Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 1 — Algebraic methods
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第1章 · 代数方法
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C01-R01 · P3 · Chapter 1

$$F(x) = \frac{x^2 - 9}{x^2 + x - 6}.$$

(a) Simplify F and state the restrictions inherited from the original expression. (3)

(b) Solve $F(x) = 2$ using the original restrictions. (3)

(Total for Question 1 is 6 marks)

$$A(x) = \frac{3}{x-1} - \frac{2}{x+2}.$$

(a) Express A as a single fraction in simplest form. (3)

(b) Find the exact real solutions of $A(x) = 1$. (3)

(Total for Question 2 is 6 marks)

3. P3-C01-R03 · P3 · Chapter 1

$P(x) = 2x^3 - 3x^2 + 4x - 5.$

(a) Find constants a, b, c, d such that $\dfrac{P(x)}{x-2} = ax^2 + bx + c + \dfrac{d}{x-2}$. (3)

(b) Find all x for which this quotient equals $2x^2 + x + 7$. (3)

(Total for Question 3 is 6 marks)

4. P3-C01-R04 · P3 · Chapter 1

For real x, $R(x) = \dfrac{x^2 - 4x + 4}{x^2 - 4}$.

(a) Simplify R and list every excluded input. (3)

(b) Find all real values of x for which $R(x) > 0$. (3)

(Total for Question 4 is 6 marks)

5. P3-C01-R05 · P3 · Chapter 1

$$E = \frac{x^2 - 1}{x^2 + 3x + 2} \div \frac{x - 1}{x + 4}.$$

(a) Simplify E. (3)

(b) State all excluded inputs in the original expression, explaining the exclusion caused by a zero divisor. (3)

(Total for Question 5 is 6 marks)

6.

P3-C01-R06 · P3 · Chapter 1

The identity $x^3 + ax^2 + bx + c = (x + 1)(x^2 + px + q) + 7$ holds for all real x. It is given that $a = 4, b = 1$.

(a) Find p, q, c. (3)

(b) Find the remainder when the cubic is divided by $x - 1$. (2)

(Total for Question 6 is 5 marks)

7.

P3-C01-R07 · P3 · Chapter 1

For $x \neq 0, 2$, $F(x) = \dfrac{x+2}{x} - \dfrac{x}{x-2}$.

(a) Show that $F(x) = -\dfrac{4}{x(x-2)}$. (3)

(b) Find all real solutions of $F(x) = -1/2$. (3)

(Total for Question 7 is 6 marks)

8. P3-C01-R08 · P3 · Chapter 1

Let $P(x) = x^4 + 2x^3 - 3x^2 - 4x + 4$.

(a) Divide P by $x^2 - 1$. (3)

(b) Find $P(1)$ and $P(-1)$ from your division identity. (2)

(Total for Question 8 is 5 marks)

$F(x) = \dfrac{x^3 - 8}{x^2 - 4}$.

(a) Write F in the form $x + \dfrac{a}{x + 2}$, stating its full original domain. (3)

(b) Find the values of x for which $F(x) = 4$. (3)

(Total for Question 9 is 6 marks)

Let $A = \dfrac{1}{x-1} + \dfrac{1}{x+1}$ and $B = \dfrac{1}{x-1} - \dfrac{1}{x+1}$, where $x \neq \pm 1$.

(a) Express A and B as single fractions. (2)

(b) Simplify A/B and state its domain. (3)

(Total for Question 10 is 5 marks)

11. P3-C01-R11 · P3 · Chapter 1

The rational expression $R(x) = \dfrac{x^2 + ax + 6}{x + 2}$ simplifies to a polynomial for every $x \neq -2$.

(a) Find a. (2)

(b) Find the polynomial and explain why the original expression is still undefined at $x = -2$. (3)

(Total for Question 11 is 5 marks)

$$F(x) = \frac{x^2 + 2x + 5}{x + 1}.$$

(a) Write F as a polynomial plus a proper fraction. (3)

(b) Find all real x for which $F(x) = 6$. (4)

(Total for Question 12 is 7 marks)

For $x \neq 0$, $E(x) = \dfrac{x^2 - 5x + 6}{x}$.

(a) Find constants a, b, c such that $E = ax + b + c/x$. (2)

(b) Solve $E(x) = 0$ and $E(x) = -5$, giving the two solution sets separately. (4)

(Total for Question 13 is 6 marks)

The quotient on dividing $P(x)$ by $(x-3)$ is $2x^2+x-4$ and the remainder is 5.

(a) Find $P(x)$ in expanded form. (3)

(b) Find the remainder on dividing $P(2x)$ by $x-1$. (3)

(Total for Question 14 is 6 marks)

$$F(x) = \frac{x^2 - 1}{x - 1} - \frac{x^2 - 4}{x - 2}.$$

(a) Simplify F and state every excluded input. (3)

(b) Find the complete real solution set of $F(x)^2 = 1$. (3)

(Total for Question 15 is 6 marks)

$P(x) = x^3 + ax^2 + bx + 6$. It is divisible by $x^2 + x - 2$.

(a) Find a, b. (4)

(b) Find all roots of $P(x) = 0$. (2)

(Total for Question 16 is 6 marks)

For real x, $R(x) = \dfrac{x^2 - 2x - 3}{x^2 - x - 6}$.

(a) Find a simplified expression and its original domain. (3)

(b) Find all x for which $R(x) < 1/2$. (4)

(Total for Question 17 is 7 marks)

$F(x) = \dfrac{2x^2 + 3x + 4}{x + 2}$.

(a) Find a, b, c in $F = ax + b + c/(x + 2)$. (3)

(b) Find the point where $F(x) = 2x + 2$. (3)

(Total for Question 18 is 6 marks)

19.

The identity $\dfrac{x^2+ax+b}{x-1} = x + 4 + \dfrac{6}{x-1}$ holds for $x \neq 1$.

(a) Find a, b. (3)

(b) Find the real zeros of the rational expression. (3)

(Total for Question 19 is 6 marks)

$A(x) = \dfrac{x}{x-1}$ and $B(x) = \dfrac{1}{x-1}$.

(a) Show that $A - B = 1$ for all allowed x. (3)

(b) Hence simplify $A^2 - B^2$ and solve $A^2 - B^2 = 3$. (4)

(Total for Question 20 is 7 marks)

$F(x) = \dfrac{(x-1)^2}{x^2-1}$.

(a) Write F in the form $1 + \dfrac{a}{x+b}$ and state its domain. (3)

(b) Find every input for which $F(x) = 1$. (3)

(Total for Question 21 is 6 marks)

Let $P(x) = x^4 + ax^3 + bx^2 + cx + d$. It is identically $(x^2 + 1)(x^2 - 3x + 2) + (2x - 5)$.

(a) Find a, b, c, d. (3)

(b) Find the quotient and remainder on division by $x^2 + 1$. (2)

(Total for Question 22 is 5 marks)

The expression $F(x) = \dfrac{x^3 + 1}{x + 1}$ is defined for $x \neq -1$.

(a) Simplify $F(x)$. (2)

(b) Find the minimum value of F on its original domain and the input attaining it. (3)

(Total for Question 23 is 5 marks)

For $x \neq 0, 1$, $F(x) = \dfrac{2x - 1}{x(x - 1)}$.

(a) Show that $F(x) = \dfrac{1}{x} + \dfrac{1}{x - 1}$ by combining the right-hand side. (2)

(b) Solve $F(x) = 0$ and state the domain. (3)

(Total for Question 24 is 5 marks)

The expression $\dfrac{x^3 - 3x^2 + kx + 4}{x - 2}$ has remainder 6.

(a) Find k. (2)

(b) For this k, find its quotient and remainder on division by $x - 2$. (3)

(Total for Question 25 is 5 marks)

$$F(x) = \frac{x^2 + 2x - 3}{x^2 - 2x + 1}.$$

(a) Simplify F. (2)

(b) Find all real solutions of $F(x)^2 = 4$. (4)

(Total for Question 26 is 6 marks)

27.

A polynomial P divided by $x^2 - 4$ leaves remainder $3x + 2$.

(a) Find $P(2)$ and $P(-2)$. (2)

(b) Find the remainder when $P(x) - x$ is divided by $x^2 - 4$. (3)

(Total for Question 27 is 5 marks)

$$E(x) = \frac{x+1}{x-1} + \frac{x-1}{x+1}.$$

(a) Express E as a single fraction and then in the form $2 + \dfrac{a}{x^2-1}$. (3)

(b) Find all real x for which $E(x) = 6$. (3)

(Total for Question 28 is 6 marks)

The function f is defined by $f(x) = \dfrac{x^3 - 3x^2 - 4x + 12}{x^2 - 4}$.

(a) Show that $f(x) = x - 3$ on its original domain, and state this domain. (3)

(b) For the original function f,

(i) sketch its graph, clearly showing both missing points and the axis intercepts. (2)

(ii) find all real solutions of $f(x) = \dfrac{6}{x + 2}$. (2)

(Total for Question 29 is 7 marks)

A polynomial $P(x)$ is defined by $P(x) = x^3 + 2x^2 - 5x + 4$.

(a) Express $\dfrac{P(x)}{x+1}$ in the form $ax^2 + bx + c + \dfrac{d}{x+1}$, stating a, b, c, d. (3)

(b) Let $g(x) = \dfrac{P(x)}{x+1} - (x^2 + x - 6)$.

(i) State the vertical asymptote and the range of g. (2)

(ii) solve $g(x) \geq 5$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C01-R01 · 6 marks

(a) · 3 marks

$F=(x-3)/(x-2), x\neq 2,-3.$

原分母分解(x+3)(x−2)，分子(x−3)(x+3)，约分得(x−3)/(x−2)，并排除−3、2。

M1	正确分解
M1	约分
A1	简式及原限制

(b) · 3 marks

$x=1.$

(x−3)/(x−2)=2得x−3=2x−4，x=1合法。

M1	正确线性方程
M1	原域复核
A1	解

Question 2

P3-C01-R02 · 6 marks

(a) · 3 marks

$A = \dfrac{x+8}{(x-1)(x+2)}, x \neq 1, -2.$

通分分子3(x+2)−2(x−1)=x+8。

M1	公分母
M1	正确分子
A1	单分式与域

(b) · 3 marks

$x = \pm\sqrt{10}.$

x+8=x^2+x−2给x^2=10，两根均非排除值。

M1	清合法分母
M1	解二次式
A1	两根

Question 3

P3-C01-R03 · 6 marks

(a) · 3 marks

$a=2, b=1, c=6, d=7.$

长除x−2得商2x²+x+6、余7，写成除法形式。

M1	多项式长除
M1	正确商
A1	常数四值

(b) · 3 marks

$x=9.$

约去相同多项式得6+7/(x−2)=7，故x−2=7；x9合法。

M1	利用分解简化
M1	解线性分式
A1	x

Question 4

P3-C01-R04 · 6 marks

(a) · 3 marks

$R = (x-2)/(x+2)$, $x \neq \pm 2$.

分子平方，分母平方差，约去x−2仍保留其原零点。

M1	因式分解约分
A1	简式
B1	两限制

(b) · 3 marks

$x < -2$ or $x > 2$.

同号乘积或符号表得外侧正，两个根均因原域排除。

M1	分子分母符号表
M1	原排除值
A1	解集

Question 5

P3-C01-R05 · 6 marks

(a) · 3 marks

$E = (x+4)/(x+2)$.

先把除法改乘倒数，再分解x^2-1和分母，约去$x\pm1$。

M1	除法倒数
M1	完全分解约分
A1	简式

(b) · 3 marks

$x \neq -4, -2, -1, 1$; $x = 1$ makes the divisor zero.

两原分母给-1、-2、-4；除数还不能为0，所以加排1。

M1	原分母排除
M1	零除数排除
A1	全部限制和原因

Question 6

P3-C01-R06 · 5 marks

(a) · 3 marks

$p = 3, q = -2, c = 5.$

比较x^2给p+1=4；x给q+p=1；常数c=q+7。

M1	二次系数
M1	一次系数
A1	三常数

(b) · 2 marks

11.

在原多项式代1得1+4+1+5=11。

M1	余式定理
A1	值

Question 7

P3-C01-R07 · 6 marks

(a) · 3 marks

Numerator $(x+2)(x-2)-x^2=-4$.

通分后平方差与x^2相消，分子恒-4。

M1	公分母
M1	正确分子
A1	目标式

(b) · 3 marks

$x=-2,4$.

$x(x-2)=8$，方程$x^2-2x-8=(x-4)(x+2)$，两根均合法。

M1	正确交叉相乘
M1	二次求解
A1	两根

Question 8

P3-C01-R08 · 5 marks

(a) · 3 marks

Quotient x^2+2x-2; remainder $-2x+2$.

长除法后次数低于2的余式必须保留一次项。

M1	正确长除
A1	商
A1	余式

(b) · 2 marks

$P(1)=0$, $P(-1)=4$.

在±1，除式项为0，分别取余式$-2x+2$。

M1	使用余式求值
A1	两值

Question 9

P3-C01-R09 · 6 marks

(a) · 3 marks

$F = x + 4/(x + 2), x \neq \pm 2.$

立方差分子(x−2)(x²+2x+4)，分母(x−2)(x+2)，再除法。

M1	立方差与平方差
M1	正确商余重写
A1	式与原域

(b) · 3 marks

$x = 1 \pm \sqrt{5}.$

x+4/(x+2)=4给x²−2x−4=0；应用二次公式得1±√5，两根均不在原排除值±2。

M1	合法清分母
M1	解二次式
A1	根与域

Question 10

P3-C01-R10 · 5 marks

(a) · 2 marks

$A = 2x/(x^2-1)$; $B = 2/(x^2-1)$.

同分母，分子相加为2x，相减为2。

M1	正确通分
A1	两式

(b) · 3 marks

$A/B = x$, $x \neq \pm 1$.

B在原域不为0，两个分式相除约去共同非零分母与2。

M1	合法相除约分
M1	检查除数非零
A1	简式与域

Question 11

P3-C01-R11 · 5 marks

(a) · 2 marks

$a = 5$.

除式成为因式，分子在−2为0：4−2a+6=0。

M1	因式条件
A1	a

(b) · 3 marks

$x + 3$; the original denominator is zero at -2.

分子(x+2)(x+3)约分，但原式0/0不定义，不能由约分后值自动补点。

M1	正确因式分解约分
A1	多项式
B1	缺失点理由

Question 12

P3-C01-R12 · 7 marks

(a) · 3 marks

$F = x + 1 + 4/(x+1),\ x \neq -1.$

分子=$(x+1)^2+4$，除以$x+1$。

M1	正确平方重写
M1	分项相除
A1	完整式

(b) · 4 marks

$x = 2 \pm \sqrt{5}.$

置$u=x+1\neq 0$，$u+4/u=6$给$u^2-6u+4=0$；$u=3\pm\sqrt{5}$，换回x。

M1	正确二次方程
M1	精确两根
M1	换回与域检查
A1	x集合

Question 13

P3-C01-R13 · 6 marks

(a) · 2 marks

$a = 1, b = -5, c = 6$.

逐项除x。

M1	分项相除
A1	三常数

(b) · 4 marks

$E = 0$: $x = 2, 3$; $E = -5$: no real solutions.

第一式分子零；第二式x+6/x=0给$x^2+6=0$，在实数无解。

M1	第一分子方程
A1	两根
M1	第二平方关系
A1	无实根

Question 14

P3-C01-R14 · 6 marks

(a) · 3 marks

$2x^3 - 5x^2 - 7x + 17.$

P=(x−3)(2x²+x−4)+5，展开。

M1	正确除法恒等式
M1	展开
A1	多项式

(b) · 3 marks

$-1.$

需算P2，由除法恒等式P2=(−1)(8+2−4)+5=−1。

M1	复合余式输入
M1	正确代2
A1	余式

Question 15

P3-C01-R15 · 6 marks

(a) · 3 marks

$F = -1,\ x \neq 1, 2.$

两商各约分为x+1、x+2，相减−1，但保留两原分母零点。

M1	两平方差分解约分
A1	常数
B1	域

(b) · 3 marks

All real x except $1, 2$.

在整个原域F恒−1，所以平方恒1；排除点不能计入。

M1	恒等式判断
M1	保留原限制
A1	完整集合

Question 16

P3-C01-R16 · 6 marks

(a) · 4 marks

$a=-2, b=-5.$

设商x+c，常数−2c=6给c−3；展开(x²+x−2)(x−3)=x³−2x²−5x+6。

M1	设正确商次数
M1	常数比较
M1	展开
A1	a与b

(b) · 2 marks

$x=-2, 1, 3.$

除式(x+2)(x−1)，再与商(x−3)列根。

M1	分解除式
A1	三根

Question 17

P3-C01-R17 · 7 marks

(a) · 3 marks

$(x+1)/(x+2),\ x \neq 3, -2.$

分子(x−3)(x+1)，分母(x−3)(x+2)，约共同因子。

M1	完全分解
M1	约分
A1	式与域

(b) · 4 marks

$-2 < x < 0.$

移项为x/[2(x+2)]<0，符号相反只在−2到0；原排除3不在此区间。

M1	合成单分式
M1	符号表
M1	原域检查
A1	区间

Question 18

P3-C01-R18 · 6 marks

(a) · 3 marks

$a = 2, b = -1, c = 6$.

除法或比较$(2x-1)(x+2)+6$。

M1	正确商
M1	余式
A1	三值

(b) · 3 marks

$x = 0$, common value 2.

$-1+6/(x+2)=2$，$x+2=2$，代得2。

M1	正确约去共同项
M1	解线性分式
A1	点

Question 19

P3-C01-R19 · 6 marks

(a) · 3 marks

$a = 3, b = 2.$

乘回分母，分子=(x−1)(x+4)+6=x^2+3x+2。

M1	乘回合法分母
M1	展开比较
A1	两系数

(b) · 3 marks

$x = -1, -2.$

分子=(x+1)(x+2)，两个零点都不等于排除1。

M1	分子分解
M1	原域核查
A1	两零点

Question 20

P3-C01-R20 · 7 marks

(a) · 3 marks

$(x-1)/(x-1)=1, x\neq 1.$

同分母相减后约分，保留x不得1。

M1	合并分子
M1	合法约分
A1	恒等式及域

(b) · 4 marks

$(x+1)/(x-1)$; solution $x=2$.

平方差(A−B)(A+B)=A+B，再通分；x+1=3x−3给x2。

M1	利用平方差和(a)
M1	单分式
M1	解方程与域
A1	解

Question 21

P3-C01-R21 · 6 marks

(a) · 3 marks

$F = 1 - 2/(x+1),\ x \neq \pm 1.$

约分为(x−1)/(x+1)，将分子写成(x+1)−2。

M1	因式约分
M1	重写分子
A1	要求形式与域

(b) · 3 marks

No real solution.

1−2/(x+1)=1要求−2/(x+1)=0，有限非零分母时不可能。

M1	正确消去常数
M1	非零分子判断
A1	无解

Question 22

P3-C01-R22 · 5 marks

(a) · 3 marks

$a=-3, b=3, c=-1, d=-3.$

先展开$x^4-3x^3+3x^2-3x+2$，再加$2x-5$。

M1	正确乘法展开
M1	合并余式
A1	四系数

(b) · 2 marks

Quotient x^2-3x+2; remainder $2x-5$.

已给恒等式的剩余项次数1小于除式次数2，因此是合法余式。

M1	识别次数条件
A1	商余

Question 23

P3-C01-R23 · 5 marks

(a) · 2 marks

$x^2 - x + 1$.

立方和$x^3+1=(x+1)(x^2-x+1)$，合法约分。

M1	立方和分解
A1	简式

(b) · 3 marks

$3/4$ at $x = 1/2$.

配方$(x-1/2)^2+3/4$，最小点$1/2$不在排除-1。

M1	正确配方
M1	检查原域
A1	最小值和位置

Question 24

P3-C01-R24 · 5 marks

(a) · 2 marks

Combined numerator $(x-1)+x=2x-1$.

只需通分验证题目给出的恒等分解。

M1	正确公分母
A1	分子一致

(b) · 3 marks

$x=1/2$, domain $x\neq 0,1$.

分母有限非零时分式为零当且仅当分子为零。

M1	分子零条件
M1	原域筛选
A1	解和域

Question 25

P3-C01-R25 · 5 marks

(a) · 2 marks

$k = 3$.

分子在2处8−12+2k+4=2k=6。

M1	余式条件
A1	k

(b) · 3 marks

Quotient $x^2 - x + 1$; remainder 6.

长除或展开$(x-2)(x^2-x+1)+6=x^3-3x^2+3x+4$。

M1	正确长除
M1	常数项及余式复核
A1	商余

Question 26

P3-C01-R26 · 6 marks

(a) · 2 marks

$F=(x+3)/(x-1),\ x\neq 1.$

分子(x+3)(x−1)，分母(x−1)²。

M1	分解并约分
A1	简式与域

(b) · 4 marks

$x=5,\ -1/3.$

先F=±2；x+3=2x−2得5；x+3=−2x+2得−1/3，两根均合法。

M1	分出正负两式
M1	解两线性式
M1	原域检查
A1	全部根

Question 27

P3-C01-R27 · 5 marks

(a) · 2 marks

8 and -4.

除式在± 2为0，函数值等于余式在这两个点的值。

M1	余式代入两个根
A1	两值

(b) · 3 marks

$2x + 2$.

线性余式减x后仍次数低于2，可以直接相减。

M1	合并余式
M1	次数合法
A1	新余式

Question 28

P3-C01-R28 · 6 marks

(a) · 3 marks

$E = (2x^2 + 2)/(x^2 - 1) = 2 + 4/(x^2 - 1).$

分子为两个平方和$2x^2+2$，再写成$2(x^2-1)+4$。

M1	通分平方和
M1	商余重写
A1	两形式

(b) · 3 marks

$x = \pm\sqrt{2}.$

$4/(x^2-1)=4$给$x^2=2$，两个根均不为±1。

M1	合法清分母
M1	正负根和域
A1	两根

Question 29

P3-C01-R29 · 7 marks

(a) · 3 marks

$f(x) = x - 3,\ x \neq -2, 2.$

分子分组为$x^2(x-3) - 4(x-3) = (x^2-4)(x-3)$，约分后仍需排除原分母零点$\pm 2$。

M1	分组因式分解
A1	化简式
B1	原定义域

(b) For the original function f,

(i) · 2 marks

Line $y = x - 3$ with holes at $(-2, -5), (2, -1)$; intercepts $(0, -3), (3, 0)$.

画直线但在原式未定义的两点画空心圆；两个坐标轴截距都在合法定义域。

B1	直线与截距
B1	两个空心点

(ii) · 2 marks

$x = -3, 4.$

保留$x \neq \pm 2$；$(x-3)(x+2) = 6$得$x^2 - x - 12 = 0$，两根$-3, 4$均合法。

M1	在合法定义域清分母
A1	两根并检验

Question 30

P3-C01-R30 · 7 marks

(a) · 3 marks

$x^2 + x - 6 + 10/(x+1)$.

多项式除法给商x^2+x-6、余数10；因此$a=1, b=1, c=-6, d=10$，且$x \neq -1$。

M1	多项式除法
A1	商
A1	余数

(b) Let $g(x) = \dfrac{P(x)}{x+1} - (x^2+x-6)$.

(i) · 2 marks

$x = -1$; $(-\infty, 0) \cup (0, \infty)$.

由(a)得$g(x) = 10/(x+1)$；分母零点为竖直渐近线，分子非零，因此值域不含0，其余实数均可取得。

B1	渐近线
B1	完整值域

(ii) · 2 marks

$-1 < x \leq 1$.

$x<-1$时$g<0$不满足；$x>-1$时乘正分母得$10 \geq 5(x+1)$，故$x \leq 1$，合并得所求区间。

M1	区分分母符号
A1	精确端点

Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 2 — Functions and graphs
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第2章 · 函数与图像
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

Figure 1 gives the complete graph of $y = f(x)$, formed from straight line segments joining $(-4, 1)$, $(-1, 4)$, $(2, -2)$ and $(5, 1)$. All four endpoints are included.

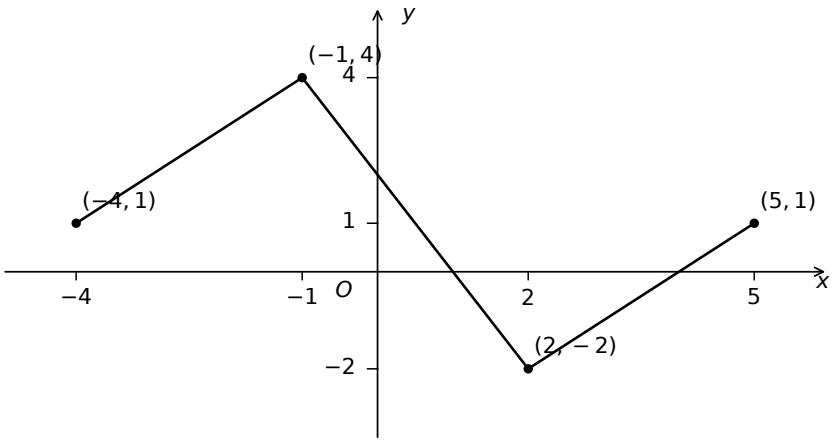


Figure 1

(a) Find $ff(2)$ and all solutions of $f(x) = 1$. (3)

(b) Let $g(x) = 2x - 1$ for real x. Find the domain and range of fg. (3)

(c) Sketch $y = |f(x)|$, marking every zero and every corner point. (3)

(Total for Question 1 is 9 marks)

The function $f(x) = 4 - (x - 1)^2$ has domain $[-1, 4]$. Define $g(x) = |f(x)|$ on the same domain, and $h(x) = f(|x|)$ wherever this is defined. Figure 2 shows the original graph.

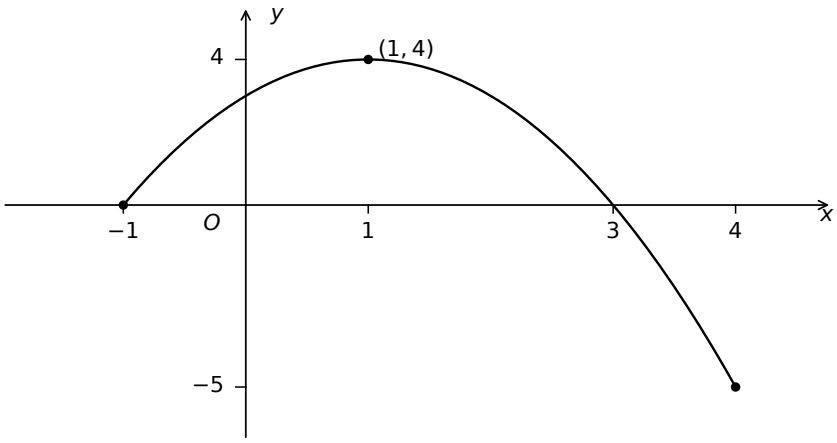


Figure 2

(a) Find the domain and range of h and sketch its graph, labelling its zeros and maximum points. (4)

(b) Find the range of g. (2)

(c) Find all k for which $y = g(x)$ and $y = k$ have exactly two distinct intersections. (3)

(Total for Question 2 is 9 marks)

An instrument maps a non-negative input x to the reading $f(x) = (ax + b)/(x + 2)$. Its calibration graph has horizontal asymptote $y = 3$ and passes through $(2, 2)$, as shown in Figure 3.

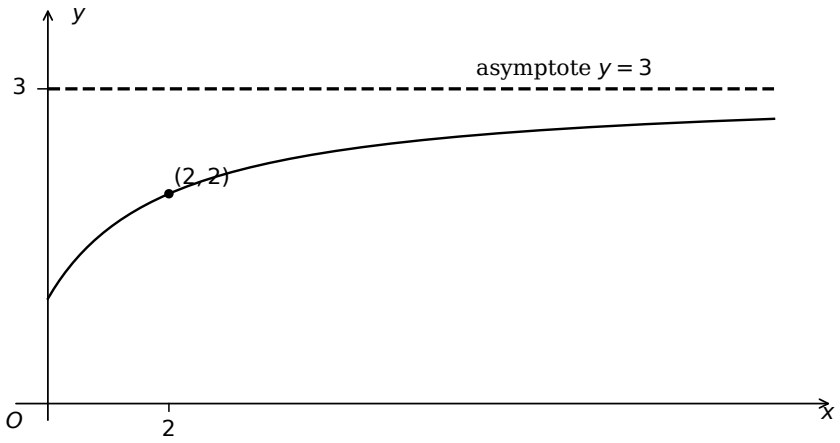


Figure 3

(a) Find a, b and write $f(x)$ in the form $3 - c/(x + 2)$. (3)

(b) Find the range of f and the inverse function, with its domain. (4)

(c) Find the input for which the reading is 0.2 below the limiting reading. Explain why the limiting reading itself is never attained. (3)

(Total for Question 3 is 10 marks)

4.

Let $f(x) = \sqrt{x - 1}$ for $x \geq 1$, and $g(x) = x^2 + 1$ for $x \geq 0$.

(a) Show that f and g are inverse functions, stating the domain of each composition. (4)

(b) The domain of g is now extended to all real numbers, while f is unchanged.

(i) Explain why g does not have an inverse function on this extended domain. (2)

(ii) Solve $fg(x) = 3$ for real x. (3)

(Total for Question 4 is 9 marks)

Figure 5 shows the complete graph of $f(x) = a|x - b| + c$ on $[-2, 6]$, where $a > 0$. Its vertex is $(2, -3)$ and its zeros are $-1, 5$. The two endpoints are included.

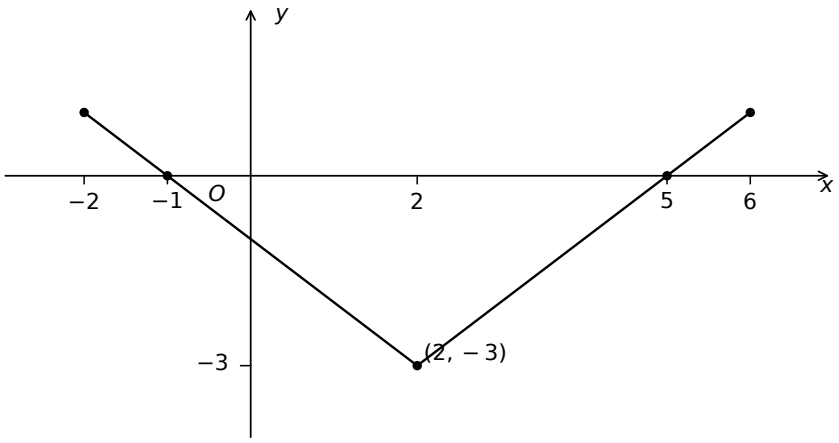


Figure 5

(a) Find a, b, c. (3)

(b) Find all k such that $f(x) + k = 0$ has two distinct solutions in the stated domain. (3)

(c) Solve $f(|x|) \geq 1$, respecting the domain of the composition. (3)

(Total for Question 5 is 9 marks)

A depth sensor in a tank of height 4 m gives reading $s = f(x) = (4 - x)^2$, where x m is the water depth and $0 \leq x \leq 4$. Figure 6 shows the calibration graph.

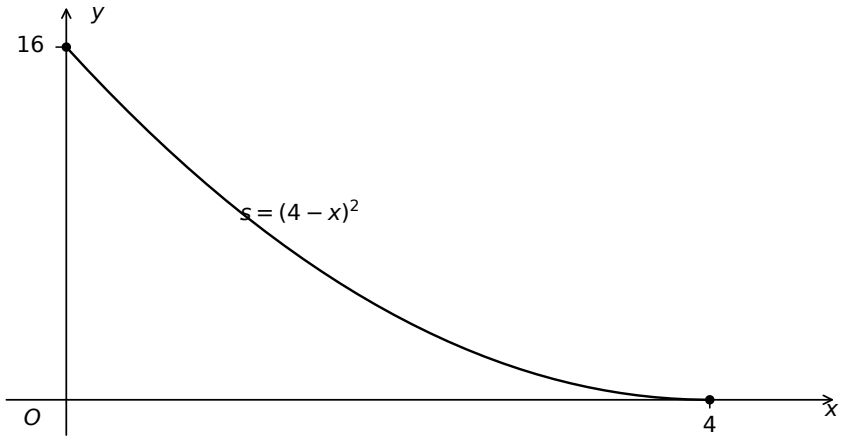


Figure 6

(a) State the range of the reading and find the depth corresponding to reading 9. (4)

(b) Find the inverse calibration function, with its domain. Explain the choice of square-root sign. (3)

(c) A second tank of height 8 m uses the sensor model $h(x) = f(x/2)$ for $0 \leq x \leq 8$.

(i) Find $h^{-1}(s)$, stating its domain. (3)

(ii) Find the depth corresponding to reading $s = 9$. (2)

(Total for Question 6 is 12 marks)

7.

Let $f(x) = 1/(x-2)$ for $x > 2$, and $g(x) = \sqrt{x+1}$ for $x \geq -1$.

(a) Find $fg(x)$ and its domain. (3)

(b) Find $gf(x)$ and its domain and range. (4)

(c) Solve $fg(x) = 1$ and $gf(x) = \sqrt{3}$, giving each answer separately. (4)

(Total for Question 7 is 11 marks)

Figure 8 shows the complete graph of f on $[-2, 6]$. Its maximum is $f(0) = 5$, its minimum is $f(3) = -1$, and all points between these output values occur. Define $h(x) = -\frac{1}{2}f(2x) + 3$.

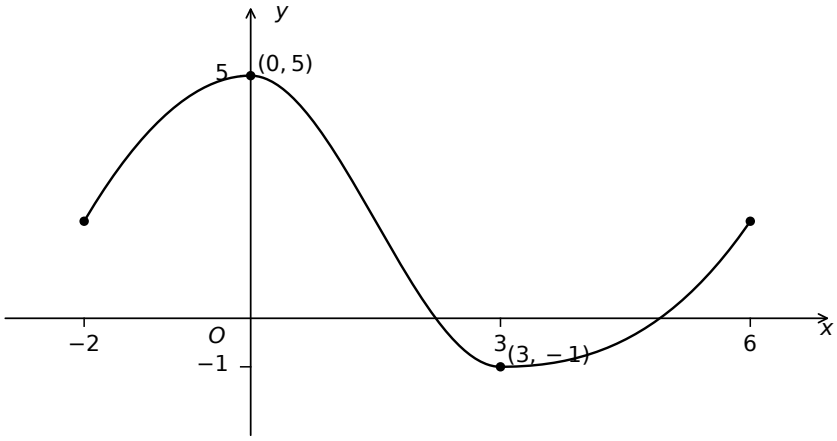


Figure 8

(a) Describe a sequence of transformations from the graph of f to that of h. (3)

(b) Find the domain, range and both extreme points of h. (3)

(c) Find the domain and range of $f(|x|)$. Explain why the range is not lost by discarding the negative-input part of f. (3)

(Total for Question 8 is 9 marks)

Let $f(x) = (x+1)/(x-1)$ for real $x \neq 1$.

(a) Find f^{-1} with its domain, and hence find $ff(x)$. (4)

(b) Using this function, give a counterexample to the statement that a function graph and its inverse graph can intersect only on $y = x$. (3)

(c) Find the points where the common graph does meet $y = x$. (2)

(Total for Question 9 is 9 marks)

The function is defined by $f(x) = x + 2$ for $-2 \leq x \leq 1$, and $f(x) = 5 - x$ for $1 < x \leq 4$. Figure 10 shows the open and closed endpoints.

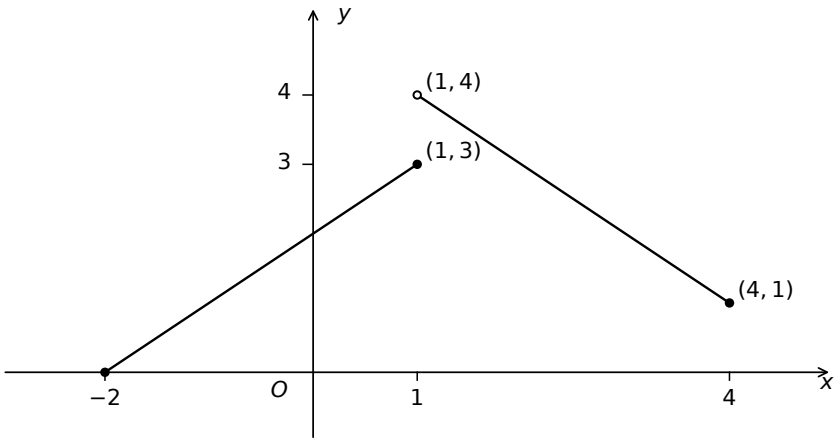


Figure 10

(a) Find the range of f and $ff(1)$. (3)

(b) Does f attain a greatest value? Explain using the endpoint at $x = 1$. (2)

(c) Restrict the domain to $[-2, 3/2]$. Find the inverse function in piecewise form, with its complete domain. (4)

(Total for Question 10 is 9 marks)

Let $f(x) = |x - 3|$ and $g(x) = |x| - 3$, both on the real numbers.

(a) Find $fg(x)$ and $gf(x)$, simplifying the outer absolute value where possible. (3)

(b) Solve $fg(x) = gf(x)$ for real x. (4)

(c) Use the input $x = 0$ to verify that the two composite functions are not identical. (2)

(Total for Question 11 is 9 marks)

Let $g(x) = 3 - \sqrt{x + 2}$ for $-2 \leq x \leq 7$. Figure 12 shows its graph.

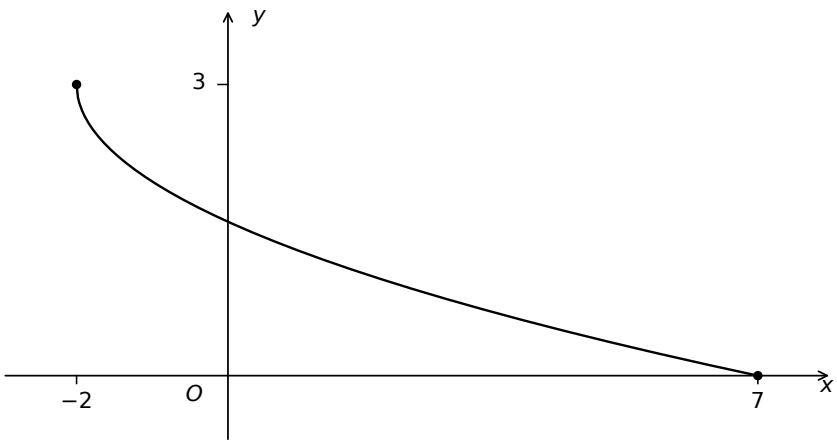


Figure 12

(a) Find its range and inverse function, with the domain of the inverse. (4)

(b) Solve $gg(x) = 1$ without expanding a nested square root. (3)

(c) Explain why gg is defined for every x in the original domain. (2)

(Total for Question 12 is 9 marks)

Let $f(x) = \sqrt{9 - x^2}$ on $[-3, 3]$, and $g(x) = x^2$ for real x.

(a) Find $fg(x)$ and its full domain and range. (4)

(b) Find $gf(x)$ and its full domain and range. (3)

(c) Solve $fg(x) = \sqrt{5}$ and $gf(x) = 5$, giving both solution sets. (2)

(Total for Question 13 is 9 marks)

The graph in Figure 14 is $y = f(x) = |x^2 - 4|$ on $[-3, 3]$. Define $h_k(x) = f(x) + k$ on this domain.

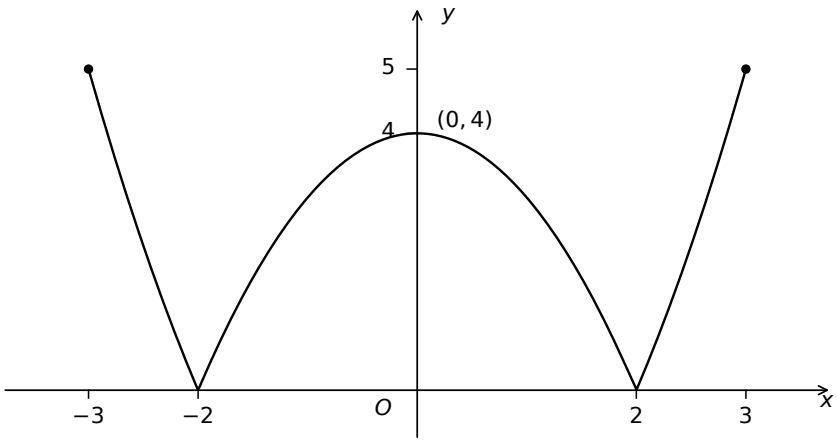


Figure 14

(a) Find the range of f and identify every point where the minimum occurs. (3)

(b) Find all k for which h_k has exactly four distinct zeros. (3)

(c) Find the value of k for which h_k has exactly three distinct zeros, and give these zeros exactly. (3)

(Total for Question 14 is 9 marks)

Let $q(x) = (x^2 - 9)/(x - 3)$ for real $x \neq 3$.

(a) Sketch the graph of q, showing the missing point. State its range. (3)

(b) Find $qq(x)$ with its complete domain and range. (4)

(c) Find q^{-1} with its domain. (2)

(Total for Question 15 is 9 marks)

Figure 16 shows $f(x) = x^3 - 3x$ on $[-2, 2]$. Its turning points are $(-1, 2)$ and $(1, -2)$, and its endpoints are $(-2, -2), (2, 2)$.

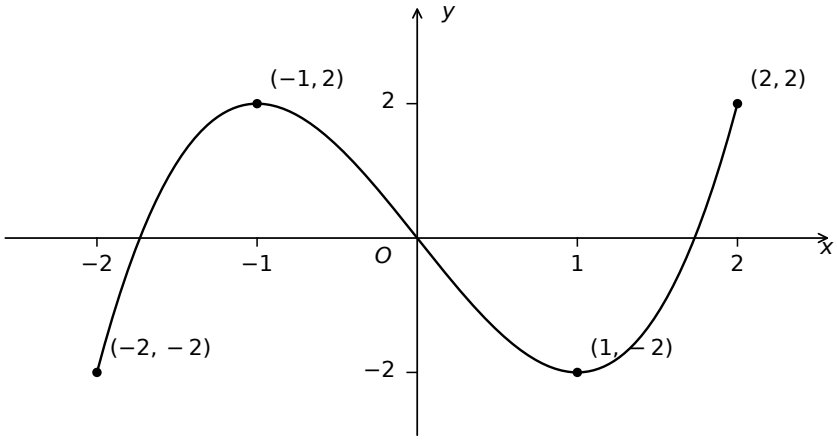


Figure 16

(a) Find all values of k for which $f(x) = k$ has exactly two distinct solutions. (3)

(b) Sketch $y = |f(x)|$, marking every zero and every point at height 2. (3)

(c) For which k does $|f(x)| = k$ have six distinct solutions? (3)

(Total for Question 16 is 9 marks)

A calibration is $f(x) = 18/(x+3)$ for $0 \le x \le 6$. A correction procedure adds 1 to a reading and then uses the inverse calibration. Its output is $H(x) = f^{-1}(f(x)+1)$ whenever this is defined.

(a) Find the range of f and f^{-1}, including its domain. (4)

(b) For the correction procedure $H(x) = f^{-1}(f(x)+1)$,

(i) Find the complete domain of H. (3)

(ii) Simplify $H(x)$ and find the input for which $H(x) = 1$. (4)

(Total for Question 17 is 11 marks)

Figure 18 gives a smooth graph $y = f(x)$ on $[-3, 4]$. Its only zeros are $-2, 1, 3$; its minimum is $(0, -2)$ and maximum is $(2, 4)$. The endpoint values are $f(-3) = 2$ and $f(4) = -1$. Define $h(x) = -2f(x/3) - 1$.

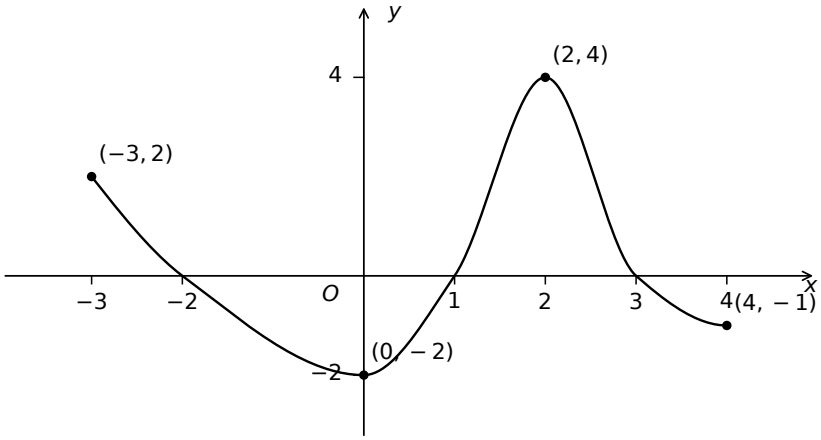


Figure 18

(a) Find the domain and range of h. (4)

(b) Find the maximum and minimum points of h. (3)

(c) Find all exact solutions of $h(x) = -1$ using the given graph information. (3)

(Total for Question 18 is 10 marks)

Two depots lie at positions 1 and 4 km on a straight road. Travelling one kilometre to the first depot has cost 1 unit; the corresponding cost to the second depot is 2 units. A delivery point at x has total cost $F(x) = |x - 1| + 2|x - 4|$.

(a) Write F in piecewise form and sketch its graph. (3)

(b) Solve $F(x) \le 6$. (2)

(c) Restrict the delivery point to $x \ge 4$. Find the inverse cost function, with its domain. (3)

(Total for Question 19 is 8 marks)

Let $f(x) = \sqrt{16 - (x-1)^2}$ for $1 \leq x \leq 5$. Figure 20 shows its graph.

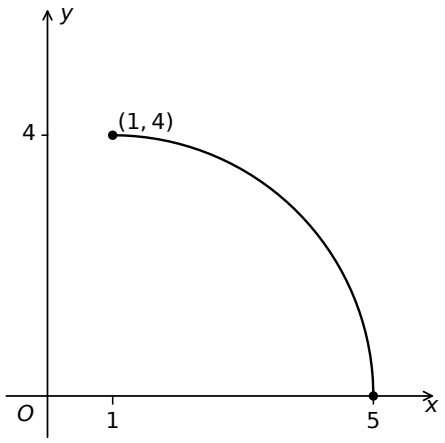


Figure 20

(a) Find f^{-1}, stating its domain and range. (4)

(b) Find all intersections of the two graphs exactly. Justify why none lies off $y = x$ for this particular pair. (4)

(Total for Question 20 is 8 marks)

The function $f(x) = |2^x - 4|$ is defined for all real x.

(a) Sketch the graph, labelling both intercepts and its horizontal asymptote, and state its range. (4)

(b) Find all values of k for which $f(x) = k$ has two distinct real solutions. (3)

(c) Solve $f(x) = 3$ exactly. (2)

(Total for Question 21 is 9 marks)

Let $f(x) = |1/x - 1|$ for real $x \neq 0$.

(a) Sketch the graph, showing all asymptotes and its zero, and state the range. (4)

(b) Find all k such that $f(x) = k$ has exactly two distinct solutions. (3)

(c) Solve $f(x) > 2$. (3)

(Total for Question 22 is 10 marks)

Let $f(x) = x^2 - 4x + 7$ for $x \geq 2$, and define $g(x) = f(x) - x$ on the same domain.

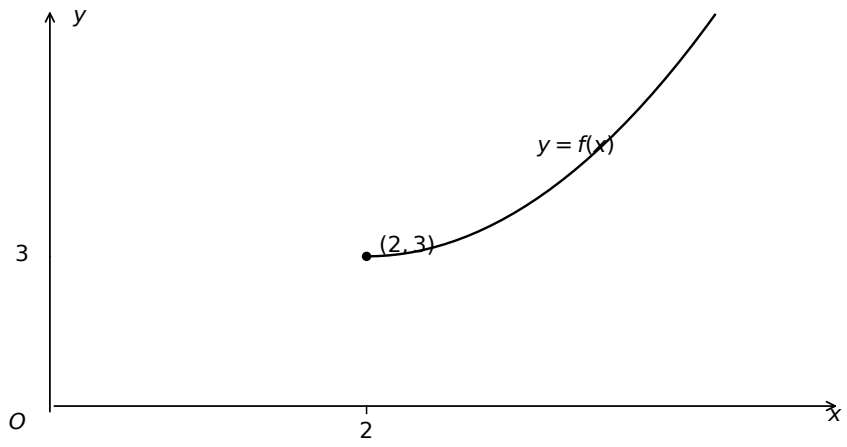


Figure 23

(a) Find f^{-1} with its domain. (3)

(b) Show that g is not one-to-one on its full domain, and find its range. (3)

(c) Find both inverses obtained by restricting g to its maximal one-to-one intervals within $[2, \infty)$. Include the domain of each inverse. (4)

(Total for Question 23 is 10 marks)

The complete graph of f is the two straight segments joining $(-2, -1)$ to $(0, 2)$ to $(2, -1)$, as shown in Figure 24.

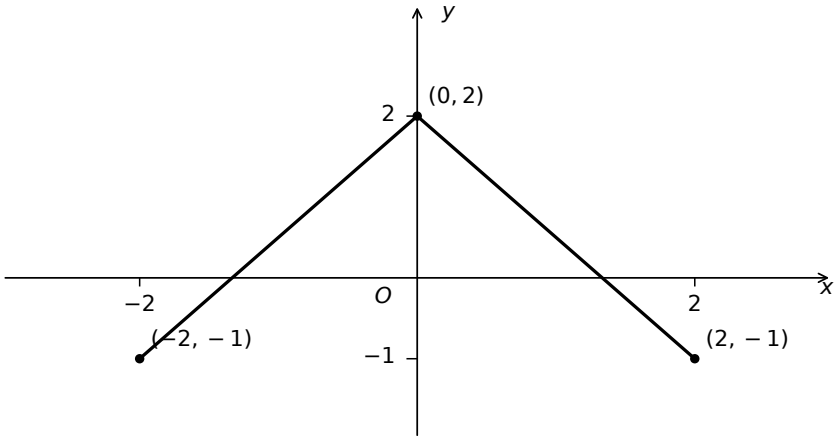


Figure 24

(a) Write an expression for f and state its range. (3)

(b) Restrict f to $[0, 2]$ and find the inverse of this restriction, including its domain. (3)

(c) Using the original full-domain f, find every solution of $ff(x) = 0$. (4)

(Total for Question 24 is 10 marks)

A two-step calibration applies $f(x) = (x+2)/(x+4)$ for $x \geq 0$, followed by $g(s) = 2s - 1$ for $1/2 \leq s < 1$. The complete calibration is $h = gf$.

(a) Find $h(x)$ and its range. (3)

(b) The complete calibration h is used to recover the original non-negative input from an output reading r.

(i) Find $h^{-1}(r)$ and state its domain. (3)

(ii) Find the input that gives reading $r = 1/4$. (2)

(c) A technician attempts to undo f first and then g. Explain the error using the reading $1/4$. (3)

(Total for Question 25 is 11 marks)

Let $f(x) = \dfrac{x-2}{x+1}$ for $x > 2$ and $g(x) = \dfrac{1}{x}$ for $x > 0$.

(a) Find the range of f and its inverse, including the domain of the inverse. (4)

(b) For the composite gf, find

(i) its expression and full domain, (3)

(ii) its range. (2)

(Total for Question 26 is 9 marks)

$f(x) = |x - 2| + 1$ is defined for $x \geq 2$. Define $h(x) = f(3 - x)$ wherever this is valid.

(a) Find f^{-1} and state its domain. (3)

(b) For h, find

(i) its domain and simplified expression, (3)

(ii) h^{-1} with its domain and range. (3)

(Total for Question 27 is 9 marks)

The function f is defined by $f(x) = x + 1$ for $0 \leq x < 2$, and $f(2) = 5$.

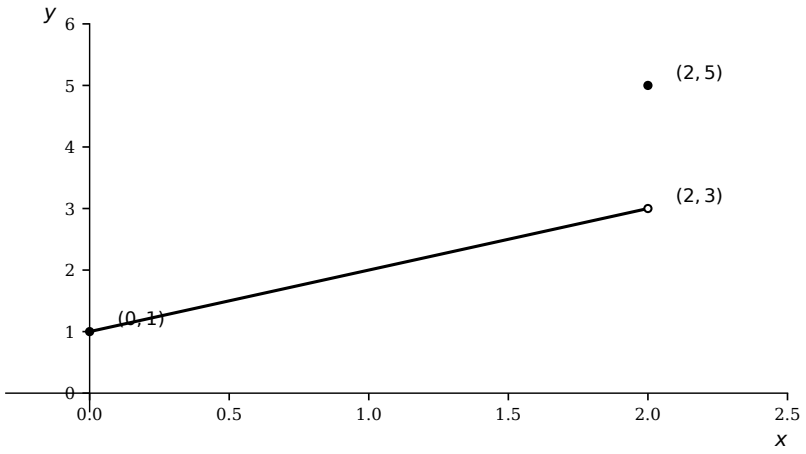


Figure 28

(a) State its range and determine whether it is one-to-one. (3)

(b) For the inverse function, find

(i) its expression on its complete domain, (3)

(ii) all x for which $ff(x)$ is defined. (3)

(Total for Question 28 is 9 marks)

The function f is defined by $f(x) = |2x - 6| + 1$ for $0 \leq x \leq 5$.

(a) Sketch the graph of f, marking the vertex and both endpoints. State its range. (3)

(b) The domain is now restricted to $k \leq x \leq 5$, where $0 \leq k < 5$, so that f has an inverse.

(i) Find the smallest possible value of k. (2)

(ii) For this restriction, find $f^{-1}(x)$ and its domain. (3)

(Total for Question 29 is 8 marks)

The functions f and g are defined by $f(x) = \sqrt{x+2}$ for $x \geq -2$, and $g(x) = \dfrac{1}{x-1}$ for $x \neq 1$.

(a) Find $gf(x)$ and state its full real domain. (3)

(b) For the composite function fg,

(i) find the set of inputs for which it is defined. (3)

(ii) solve $fg(x) = \sqrt{3}$. (2)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C02-R01 · 9 marks

(a) · 3 marks

$ff(2)=3$; $x=-4, 1/2, 5$.

f(2)=−2，在第一线段f(−2)=3。水平线y=1与首点、中央线段x=1/2以及末点相交。

M1	按顺序从图读取复合值
A1	ff(2)
A1	三个有效横坐标

(b) · 3 marks

Domain $[-3/2, 3]$; range $[-2, 4]$.

要求2x−1∈[−4,5]，解得新区间。g在此区间恰好覆盖f整个定义域，所以复合值域与f相同。

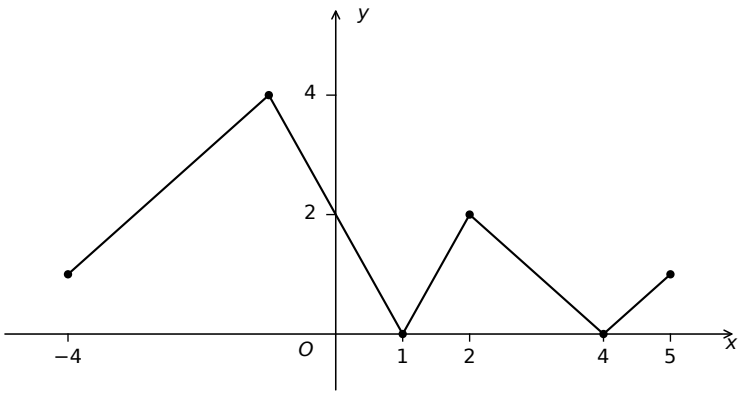
M1	内层输出必须落在外层定义域
A1	定义域
B1	完整值域及覆盖理由

(c) · 3 marks

Corner points $(-4,1), (-1,4), (1,0), (2,2), (4,0), (5,1)$; zeros $1, 4$.

保持横轴以上原图；把中央后段和最后前段的负值部分向上反射。中央线段零点1、末线段零点4，原低点(2,−2)变(2,2)。

M1	只反射原图负值部分
A1	两个零点
A1	其余全部角点和正确连线



Suggested sketch

Question 2

P3-C02-R02 · 9 marks

(a) · 4 marks

Domain $[-4,4]$, range $[-5,4]$; zeros $x=\pm 3$; maxima $(\pm 1,4)$; $h(0)=3$.

|x|只取原定义域中的$[0,4]$，再将这一半关于纵轴反射。不能把原图x<0的部分也保留。右端值−5仍是最小值。

M1	选取原图非负横坐标部分并反射
A1	定义域和值域
A1	两零点
A1	最大点及拼接点

(b) · 2 marks

$[0,5]$.

原函数最小−5、最大4且过轴；负值绝对值后能到5，故新最大值不是4。

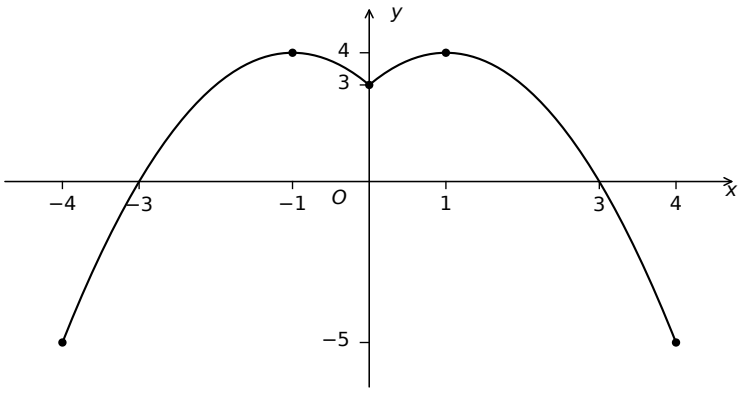
M1	比较原正最大值与负最小值绝对值
A1	完整值域

(c) · 3 marks

$k=0$ or $k=4$.

k=0时原零点−1、3两点；0

M1	按图分高度区间并处理极值
A1	k=0
A1	k=4且无额外值



Suggested sketch

Question 3

P3-C02-R03 · 10 marks

(a) · 3 marks

$a = 3, b = 2$; $f(x) = 3 - 4/(x + 2)$.

大x时分子分母首项比给a=3；点(2,2)给6+b=8。再用除法整理成3−4/(x+2)。

B1	从渐近线确定a
M1	代点求b
A1	参数与混合式

(b) · 4 marks

Range $[1, 3)$; $f^{-1}(x) = (2x - 2)/(3 - x)$, domain $1 \le x < 3$.

对x⩾0，4/(x+2)在(0,2]，故f在[1,3)。由y(x+2)=3x+2解出原输入，交换变量并用原值域作为逆函数定义域。

M1	用混合式求值域端点
A1	完整半开值域
M1	反解方程
A1	逆函数及其定义域

(c) · 3 marks

$x = 18$; $f(x) < 3$ for every finite $x \ge 0$.

与渐近线的差为4/(x+2)，令为0.2给x+2=20。对有限非负x这项严格为正，故无法正好输出3。

M1	用与极限的差建立方程
A1	输入18
B1	不能达到渐近线的代数理由

Question 4

P3-C02-R04 · 9 marks

(a) · 4 marks

$fg(x) = x$ on $[0, \infty)$; $gf(x) = x$ on $[1, \infty)$.

fg=√(x²)，因g的输入x⩾0可写x；gf=(√(x−1))²+1=x，需x⩾1。两个恒等式均必须带各自输入范围。

M1	正确计算fg并处理根号平方
A1	fg及范围
M1	正确计算gf
A1	gf及范围

(b) The domain of g is now extended to all real numbers, while f is unchanged.

(i) · 2 marks

$g(u) = g(-u)$ for every $u > 0$, so it is many-to-one.

正负同值输入输出相同，例如g(2)=g(−2)=5；逆映射将给同一输入两个输出，不能是函数。

M1	有效重复输出例子或一般关系
A1	非一一映射与无逆函数的解释

(ii) · 3 marks

$x = \pm 3$ because $fg(x) = |x|$.

√(x²)在实数域必须是|x|。扩展g允许负输入，因此−3不再受第一问域限制。

M1	识别全实数域下根号平方为绝对值
A1	两个解
B1	负解与域扩展的联系

Question 5

P3-C02-R05 · 9 marks

(a) · 3 marks

$a = 1, b = 2, c = -3.$

顶点直接给b=2、c=−3；代零点−1得3a−3=0。

B1	顶点确定b,c
M1	用一个零点定a
A1	参数

(b) · 3 marks

$-1 \le k < 3.$

解|x−2|=3−k。两不同根要求3−k>0；有限域距顶点左右最多都是4，故3−k⩽4。端点可取，顶点不能重复计为两解。

M1	用距离表达两根
M1	同时检查不同根和域端点
A1	正确半开区间

(c) · 3 marks

$x = -6$ or $x = 6.$

内层|x|必须落在[−2,6]，故|x|⩽6。再解||x|−2|⩾4：只有|x|⩾6可行，与前域结合得|x|=6。

M1	复合定义域
M1	解嵌套绝对值不等式
A1	仅两个端点

Question 6

P3-C02-R06 · 12 marks

(a) · 4 marks

$0 \le s \le 16$; depth 1 m.

4−x从4降至0，其平方取遍[0,16]；(4−x)²=9候选x=1、7，物理域只允许1。

M1	由物理域求输出范围
A1	完整值域
M1	解平方并检查物理域
A1	深度

(b) · 3 marks

$f^{-1}(s) = 4 - \sqrt{s}$, domain $[0, 16]$.

原模型中4−x≥0，因此4−x=√s。若取x=4+√s，除s=0外会超过 tank高度4；反函数域须是可实现读数范围。

M1	从原物理域选择根号正负
A1	正确逆函数
B1	逆函数定义域与解释

(c) A second tank of height 8 m uses the sensor model $h(x) = f(x/2)$ for $0 \le x \le 8$.

(i) · 3 marks

$h^{-1}(s) = 8 - 2\sqrt{s}$, $0 \le s \le 16$.

s=(4−x/2)²，由输入范围4−x/2≥0，反解x=8−2√s。

M1	正确反解及分支
A1	反函数
B1	定义域

(ii) · 2 marks

2 m.

代入逆校准8−2×3。

M1	代入同一传感器反函数
A1	深度

Question 7

P3-C02-R07 · 11 marks

(a) · 3 marks

$fg(x) = 1/(\sqrt{x+1} - 2)$; domain $x > 3$.

g输入先需x⩾−1，还要g(x)>2，故√(x+1)>2，最后得x>3。只排除分母为0还不够，因为f本身域为x>2。

M1	正确复合顺序
M1	检查外函数的完整输入域
A1	表达式和严格域

(b) · 4 marks

$gf(x) = \sqrt{1 + 1/(x-2)}$; domain $(2, \infty)$, range $(1, \infty)$.

f输入必须>2，其输出遍历正实数，全部可送入g；g对正输入的输出严格>1且无上界。

M1	正确复合
A1	域
M1	跟踪内层值域到外层
A1	值域

(c) · 4 marks

$x = 8$ and $x = 5/2$, respectively.

第一式给√(x+1)=3，x=8；第二式平方得1+1/(x−2)=3，x=5/2。分别检查其复合定义域。

M1	解第一复合并检查域
A1	第一解
M1	解第二复合并检查域
A1	第二解

Question 8

P3-C02-R08 · 9 marks

(a) · 3 marks

Horizontal scale factor $1/2$; reflection in the x-axis; vertical scale factor $1/2$; translation by $(0, 3)$.

内层2x把横坐标减半；外层−½把纵坐标反号并减半，最后纵向加3。

B1	横向尺度
B1	反射及纵向尺度
B1	最后的平移

(b) · 3 marks

Domain $[-1, 3]$; range $[1/2, 7/2]$; minimum $(0, 1/2)$, maximum $(3/2, 7/2)$.

新域由2x∈[−2,6]。负的纵向因子交换极大与极小，分别映射(0,5)和(3,−1)。

M1	正确输入范围缩放
A1	域和值域
A1	两个正确极值点

(c) · 3 marks

Domain $[-6, 6]$; range $[-1, 5]$. Both extreme outputs occur at non-negative inputs $0, 3$.

只用原f的[0,6]部分，然后反射到负横坐标；由于0和3已给最大、最小，且曲线连续，中间全部输出仍能达到。

M1	正确折叠定义域
A1	完整值域
B1	用两个极值位置解释

Question 9

P3-C02-R09 · 9 marks

(a) · 4 marks

$f^{-1}(x) = (x+1)/(x-1)$ on $x \neq 1$; $ff(x) = x$ for $x \neq 1$.

y(x−1)=x+1给x=(y+1)/(y−1)。原函数不能输出1，所以逆域排除1；它恰好是自身的逆。

M1	反解方程
A1	逆函数
B1	正确域
A1	合法复合恒等式

(b) · 3 marks

The two graphs coincide; for example $(2, 3)$ lies on both and is not on $y = x$.

第一问已得f=f逆，整幅图重合。以f(2)=3给出明确不在对角线的公共点，即构成反例。

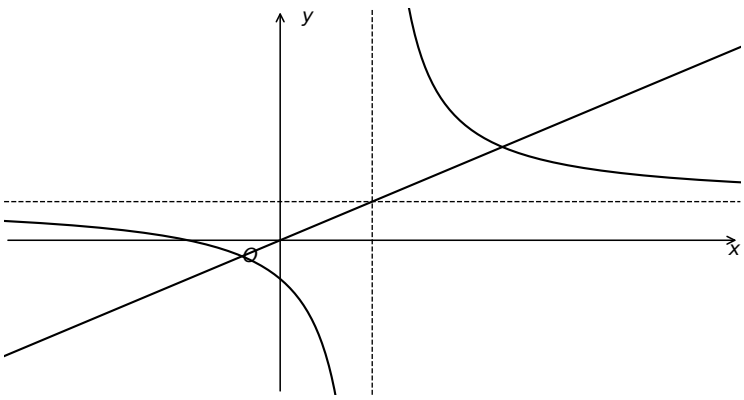
M1	由自身为逆说明两图重合
A1	有效离对角线公共点
B1	明确否定全称说法

(c) · 2 marks

$(1+\sqrt{2}, 1+\sqrt{2})$ and $(1-\sqrt{2}, 1-\sqrt{2})$.

解(x+1)/(x−1)=x得$x^2-2x-1=0$；两根均不等于1，坐标纵横相等。

M1	联立并解二次式
A1	两精确坐标且域有效



Suggested sketch

Question 10

P3-C02-R10 · 9 marks

(a) · 3 marks

Range $[0, 4)$; $ff(1) = 2$.

第一段取[0,3]；第二段取[1,4)，合并[0,4)。f(1)=3采用第一段，再算f(3)=2。

M1	正确处理两段端点
A1	完整半开值域
A1	复合值

(b) · 2 marks

No. Values approach 4 from below as $x \to 1^+$, but 4 is excluded and $f(1) = 3$.

靠近1右侧可输出比任何给定小于4的候选值更大的数，但4本身没有达到；不能把空心点当作函数值。

B1	未取得最大值
B1	空心端点及f(1)的解释

(c) · 4 marks

$f^{-1}(x) = x - 2$ for $0 \le x \le 3$; $f^{-1}(x) = 5 - x$ for $7/2 \le x < 4$.

第二段现在输入(1,1.5]、输出[3.5,4)，与第一段输出[0,3]不重叠，因此限制后是一一映射。两支分别反解并保留输出域中的缺口。

M1	检查两输出段不重叠
M1	分别反解
A1	两支公式
A1	两个区间及缺口

Question 11

P3-C02-R11 · 9 marks

(a) · 3 marks

$fg(x) = ||x| - 6|$; $gf(x) = |x - 3| - 3$.

f(g(x))=||x|−3−3|；g(f(x))=||x−3||−3，由于内层非负，可去掉外绝对值。

M1	正确两次代入顺序
A1	第一复合
A1	第二复合的化简

(b) · 4 marks

$x = -3$ or $x \geq 6$.

左边非负，故右边也须非负，先限x≤0或x≥6。x≥6时两式同为x−6，全部满足。x≤0再按−6分段：x≤−6无解，−6≤x≤0给x+6=−x，解−3。

M1	先限制右边非负的范围
M1	分别处理必要分段
A1	孤立解−3
A1	整段解x≥6

(c) · 2 marks

No: $fg(0) = 6$ while $gf(0) = 0$.

相等方程有无限多个解也不能说明两函数相同；只需在同一合法输入0处给不等输出。

B1	正确反例值
B1	非恒等的结论

Question 12

P3-C02-R12 · 9 marks

(a) · 4 marks

Range $[0,3]$; $g^{-1}(x)=(3-x)^2-2$ for $0\le x\le 3$.

根号从0增加到3，故输出从3减到0；反解√(x+2)=3−y，输出范围确保右侧非负，再平方。

M1	从原域求完整范围
A1	值域
M1	合法反解并平方
A1	逆函数及其域

(b) · 3 marks

$x=-1$.

先对外层用逆函数：g(x)=g逆(1)=2；再得x=g逆(2)=−1。核查g(−1)=2、g(2)=1。

M1	正确先撤销外层g
M1	再次用逆函数
A1	有效解

(c) · 2 marks

The range $[0,3]$ of the inner g is contained in the outer domain $[-2,7]$.

只有内层输出能全部送入外层，复合域才保持原域；此处确有[0,3]包含于[−2,7]。

B1	正确两个范围
B1	包含关系与全域可复合的解释

Question 13

P3-C02-R13 · 9 marks

(a) · 4 marks

$fg(x) = \sqrt{9 - x^4}$; domain $[-\sqrt{3}, \sqrt{3}]$; range $[0, 3]$.

g输出x^2要在f域$[-3,3]$，因非负，等价$x^2 \leqslant 3$；所得四次根式仍能由0取到3。

M1	正确复合
M1	外层域对x^2的限制
A1	定义域
A1	值域

(b) · 3 marks

$gf(x) = 9 - x^2$; domain $[-3, 3]$; range $[0, 9]$.

先做f需原域$[-3,3]$，其输出均可平方。平方抵消外根号，但不能据化简后的多项式把定义域扩展到实数。

M1	正确复合及化简
A1	保留原域
A1	完整值域

(c) · 2 marks

$x = \pm\sqrt{2}$ and $x = \pm 2$, respectively.

第一式平方得$x^4=4$，实数根$\pm\sqrt{2}$均在第一复合域；第二式得$x^2=4$，两根在第二域。

M1	分别解两个复合方程
A1	两解集及域核对

Question 14

P3-C02-R14 · 9 marks

(a) · 3 marks

Range $[0,5]$; minima $(-2,0),(2,0)$.

绝对值最小0在原二次式零点±2；端点x=±3给5，中央峰在x=0仅高4，不是全局最大。

M1	两种候选极大位置比较
A1	值域
A1	两个最小点

(b) · 3 marks

$-4<k<0$.

零点等价原图与水平y=−k相交；0<−k<4时四个交点，达到0或4时交点合并。

M1	用水平截线解释零点
M1	处理0与4两临界高度
A1	严格参数范围

(c) · 3 marks

$k=-4$; $x=0,\pm2\sqrt{2}$.

高度4触中央峰，再与两侧外支相交；解$|x^2-4|=4$给$x^2=0$或8，域内共有三个不同根。

M1	临界高度的交点合并
M1	解绝对值方程并去重
A1	k与三个精确根

Question 15

P3-C02-R15 · 9 marks

(a) · 3 marks

Line $y = x + 3$ with a hole at $(3, 6)$; range $\mathbb{R} \setminus \{6\}$.

约分得到x+3但原输入3仍不合法；在这条一一对应直线上删一个输入同时删对应输出6。

M1	正确约分但保留输入限制
A1	线与空心点
B1	值域缺6

(b) · 4 marks

$qq(x) = x + 6$, domain $\mathbb{R} \setminus \{0, 3\}$, range $\mathbb{R} \setminus \{6, 9\}$.

内层先排除x=3，还须q(x)≠3，即x+3≠3，另排除0。复合化简为x+6，所以失去输出6与9。

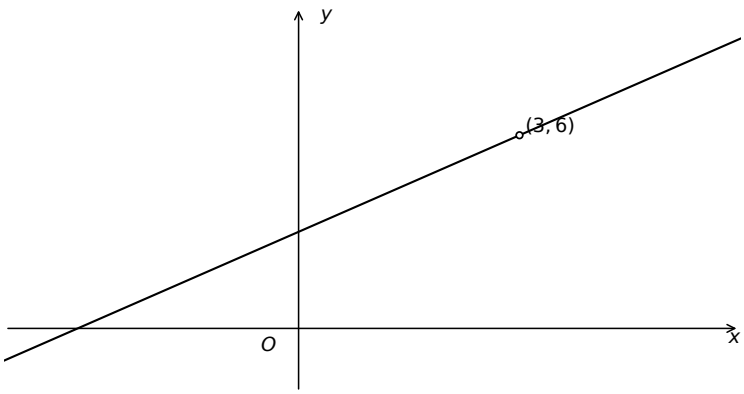
M1	同时检查两层定义域
A1	复合式
A1	完整域
A1	对应缺失值域

(c) · 2 marks

$q^{-1}(x) = x - 3,\ x \neq 6$.

反解y=x+3为原输入y−3，但逆输入不能用原函数缺失输出6。

M1	反解直线
A1	正确逆函数及其域



Suggested sketch

Question 16

P3-C02-R16 · 9 marks

(a) · 3 marks

$k = -2$ or $k = 2$.

水平高度在$(-2,2)$内给三交点；达到极大或极小时一对根合并为转折点，另有对侧端点，故两点。

M1	比较内部水平截线与临界高度
A1	两个k值
B1	不重复计相切根

(b) · 3 marks

Zeros $x = -\sqrt{3}, 0, \sqrt{3}$; height-2 points $(-2,2), (-1,2), (1,2), (2,2)$.

只把原图负值段关于横轴翻上去。零点由$x(x^2-3)=0$，端点和两转折点的绝对值都是2。

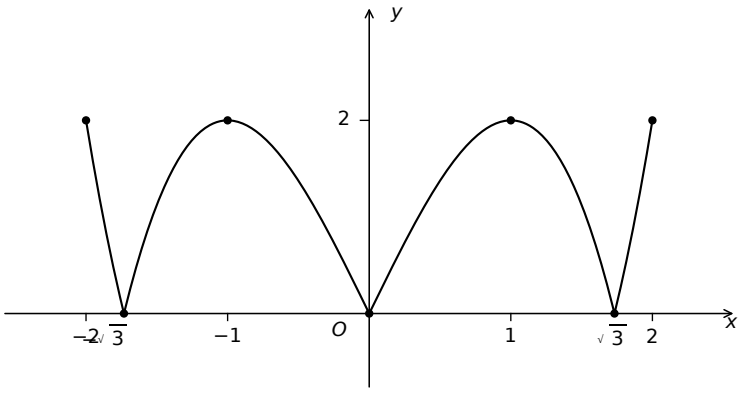
M1	正确反射负值部分
A1	三零点
A1	四个高度2的点与整体连线

(c) · 3 marks

$0 < k < 2$.

等价f=k或f=−k，每条非临界截线各三根；k>0使两组不同。k=0仅三根、k=2仅四根，故端点排除。

M1	拆成两条水平截线
M1	数清互不重合的根
A1	严格参数区间



Suggested sketch

Question 17

P3-C02-R17 · 11 marks

(a) · 4 marks

Range $[2,6]$; $f^{-1}(s)=18/s-3$ on $[2,6]$.

x+3在[3,9]内，倒数乘18得到输出[2,6]；反解18/(x+3)=s，得x=18/s−3。

M1	跟踪有限输入域到输出
A1	值域
M1	反解
A1	逆函数和域

(b) For the correction procedure $H(x)=f^{-1}(f(x)+1)$,

(i) · 3 marks

$3/5\le x\le 6$.

修正后读数必须满足2⩽f(x)+1⩽6。下限因f⩾2自动成立，上限要求18/(x+3)⩽5，故x⩾3/5，再与原域求交。

M1	检查修正读数落在逆函数定义域
M1	正确处理正分母不等式
A1	完整区间

(ii) · 4 marks

$H(x)=(15x-9)/(x+21)$; input $x=15/7$.

代逆函数得18/[18/(x+3)+1]−3，合并为(15x−9)/(x+21)。令其为1，得14x=30，且在有效域内。

M1	正确复合与通分
A1	最简式
M1	解输出方程并检查域
A1	有效输入

Question 18

P3-C02-R18 · 10 marks

(a) · 4 marks

Domain $[-9, 12]$; range $[-9, 3]$.

x/3必须在[−3,4]，所以横向放大3倍；纵输出负倍数−2交换范围两端，再减1，得[−9,3]。

M1	输入域变换
A1	定义域
M1	处理负倍数与平移
A1	值域

(b) · 3 marks

Maximum $(0, 3)$; minimum $(6, -9)$.

原最小(0,−2)映成(0,3)，原最大(2,4)映成(6,−9)；负号交换两者性质。

M1	变换两个点
A1	两坐标
B1	正确极值性质

(c) · 3 marks

$x = -6, 3, 9$.

h=−1等价f(x/3)=0，内层输入只能取原三个零点−2、1、3，故原输入各乘3。无需猜测图中其他高度的交点。

M1	目标高度转换到f的零水平
M1	正确反向横坐标映射
A1	三个解

Question 19

P3-C02-R19 · 8 marks

(a) · 3 marks

$F(x) = 9 - 3x$ for $x < 1$; $7 - x$ for $1 \le x \le 4$; $3x - 9$ for $x > 4$. Corners $(1, 6), (4, 3)$.

按1和4两地点分三段，逐个确定两个差值的符号。中段依然递减而不是水平，因为两运输权重不相同。

M1	在两个位置分段
A1	三个表达式和域
A1	两角点与正确三斜率

(b) · 2 marks

$1 \le x \le 5$.

左段要求x≥1故没有左开段内部解；中段全满足；右段3x−9≤6给x≤5，与x>4合并。

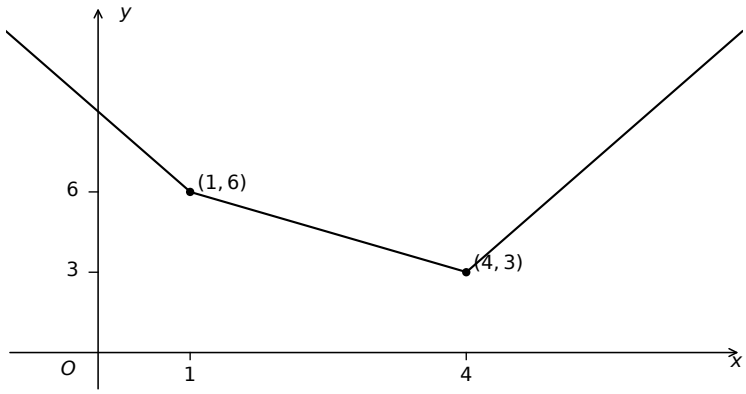
M1	各段解并结合对应范围
A1	两端与完整区间

(c) · 3 marks

$F^{-1}(c) = (c + 9)/3$ for $c \ge 3$.

该限制只留严格递增右支，成本最小3且无上界；从c=3x−9反解即可。

M1	选择一一右支并确定成本域
A1	逆函数
B1	正确逆域



Suggested sketch

Question 20

P3-C02-R20 · 8 marks

(a) · 4 marks

$f^{-1}(x) = 1 + \sqrt{16 - x^2}$ on $[0, 4]$, with range $[1, 5]$.

原横坐标≥1，所以x−1非负；反解时取正平方根。原输出从4降到0，因此逆输入域[0,4]。

M1	合法平方并解原输入
A1	选择正确分支
A1	逆函数
B1	逆域与范围

(b) · 4 marks

Only $((1 + \sqrt{31})/2, (1 + \sqrt{31})/2)$.

原图满足$(x-1)^2+y^2=16$，逆图满足$x^2+(y-1)^2=16$，相减得$y=x$。代回解$2x^2-2x-15=0$，负号根不在两个图的共同域；正号根合法。此理由来自本题两方程，不是普遍定理。

M1	写出两图方程并相减得y=x
M1	解二次方程
A1	正确精确坐标
B1	排除非法根与完整性

Question 21

P3-C02-R21 · 9 marks

(a) · 4 marks

Intercepts $(0,3),(2,0)$; asymptote $y=4$ as $x\to-\infty$; range $[0,\infty)$.

x<2时函数4−2^x，左端趋4但小于4；x≥2时2^x−4无限增长。左支的极限4不妨碍右支实际达到4。

M1	正确反射原指数负值支
A1	两截点
B1	渐近线方向
B1	完整值域

(b) · 3 marks

$0<k<4$.

拆成2^x=4+k与2^x=4−k。第一式k≥0时可解；第二式要求k<4；k=0两式同根，k=4第二式真数0无解。

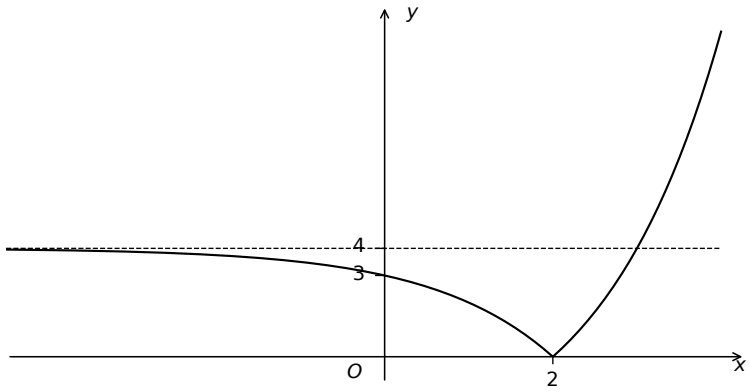
M1	两指数分支
M1	真数严格正的条件
A1	正确开区间

(c) · 2 marks

$x=0$ or $x=\log_2 7$.

分别由2^x=1或7，得到0与$\log_2 7$；两真数均正，不需P3指数微分。

M1	正确两绝对值分支
A1	两精确解



Suggested sketch

Question 22

P3-C02-R22 · 10 marks

(a) · 4 marks

Asymptotes $x = 0, y = 1$; zero $(1, 0)$; range $[0, \infty)$.

先画1/x−1，再把负值部分翻上；在x<0与x>1均为1−1/x，但分属不同分支；0

M1	正确三段形状
A1	两渐近线
B1	零点
B1	完整值域

(b) · 3 marks

$k > 0, k \neq 1$.

两候选输入为1/(1+k)与1/(1−k)。k=0时两候选重合；k=1时第二个不存在；其他正k均给两个合法非零输入。

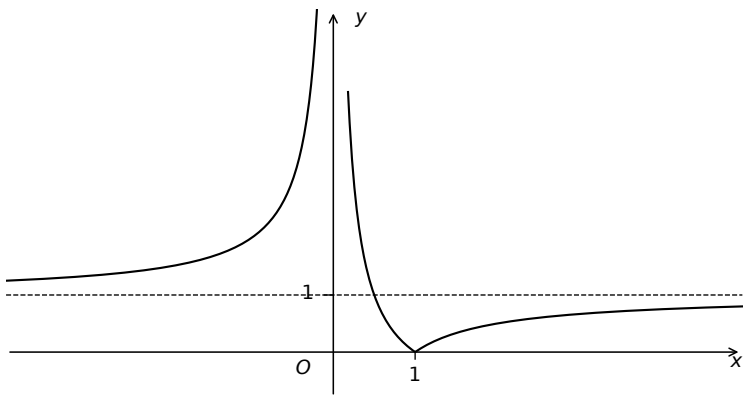
M1	解两个有理式分支
M1	检查重根和零分母例外
A1	完整参数集合

(c) · 3 marks

$-1 < x < 0$ or $0 < x < 1/3$.

按负横坐标、正横坐标小于1和大于1分段；负区1−1/x>2给−12给01时函数<1无解。

M1	完整分段并按分母符号解不等式
A1	负区
A1	正区与排除0



Suggested sketch

Question 23

P3-C02-R23 · 10 marks

(a) · 3 marks

$f^{-1}(x) = 2 + \sqrt{x-3}$ for $x \geq 3$.

f=(x−2)²+3，原横坐标在顶点右侧，输出≥3，取正分支求逆。

M1	配方并限制根号分支
A1	逆函数
B1	逆域

(b) · 3 marks

$g(2) = g(3) = 1$; range $[3/4, \infty)$.

g=x²−5x+7=(x−5/2)²+3/4。定义域包含顶点两侧的2和3，所以一一性丢失；顶点仍在域内，给最小3/4。

M1	配方或有效重复值
A1	非一一的具体证据
A1	完整值域

(c) · 4 marks

On $[2, 5/2]$: inverse $(5 - \sqrt{4x-3})/2$ for $3/4 \leq x \leq 1$. On $[5/2, \infty)$: inverse $(5 + \sqrt{4x-3})/2$ for $x \geq 3/4$.

以顶点5/2分割原输入范围。两支反解为5/2±√(x−3/4)；左支仅有原输入[2,2.5]，所以其输出域是[.75,1]，不能把两逆函数都写成同一全半轴。

M1	找两段最大单调输入区间
M1	反解两个根号分支
A1	左支公式及有限域
A1	右支公式及半轴域

Question 24

P3-C02-R24 · 10 marks

(a) · 3 marks

$f(x) = 2 - 3|x|/2$ on $[-2, 2]$; range $[-1, 2]$.

顶点在(0,2)，左右斜率大小均3/2且向下，故表达式如此；图像连续覆盖−1到2。

M1	由图恢复斜率和绝对值结构
A1	正确函数
B1	完整值域

(b) · 3 marks

$f^{-1}(x) = (4 - 2x)/3$ on $[-1, 2]$.

右支是2−3x/2，一一递减；反解并使用这条支覆盖的输出区间[−1,2]。

M1	选择正确单调支
A1	逆式
B1	逆域

(c) · 4 marks

$x = \pm 4/9$.

外层f的零点是±4/3，但内层f只能输出[−1,2]，所以−4/3不可能。只解f(x)=4/3，得|x|=4/9。

M1	找外层的两候选零输入
B1	以内层值域排除−4/3
M1	解剩下的绝对值式
A1	两根

Question 25

P3-C02-R25 · 11 marks

(a) · 3 marks

$h(x) = x/(x+4)$; range $[0, 1)$.

f可写1−2/(x+4)，所以f输出[1/2,1)，与g输入域恰匹配。代入g得2(x+2)/(x+4)−1=x/(x+4)，输出从0到趋近1。

M1	检查两步的范围匹配
A1	复合简式
A1	完整值域

(b) The complete calibration h is used to recover the original non-negative input from an output reading r.

(i) · 3 marks

$h^{-1}(r) = \dfrac{4r}{1-r}$, $0 \le r < 1$.

h(x)=x/(x+4)，r(x+4)=x给x=4r/(1−r)；原输出范围作逆函数域。

M1	正确反解
A1	反函数
B1	定义域

(ii) · 2 marks

$4/3$.

代入(i)得(4×1/4)/(3/4)=4/3。

M1	代入完整逆校准
A1	输入

(c) · 3 marks

$h^{-1} = f^{-1}g^{-1}$, so g must be undone first. Reading $1/4$ is not even in the domain $[1/2, 1)$ of f^{-1}.

输出属于最后一步g，正确先反演g。若对1/4直接用f逆，输入已不在f可实现输出范围内，算式即便能形式代入也不是合法逆校准。

M1	正确反序规则
B1	用1/4与f逆域给具体非法性
A1	完整错误解释

Question 26

P3-C02-R26 · 9 marks

(a) · 4 marks

Range $(0,1)$; $f^{-1}(x)=\dfrac{x+2}{1-x}$ for $0<x<1$.

f=1−3/(x+1)，x>2时从0上升趋近1但不取端值；反解yx+y=x−2。

M1	范围端点与开闭
M1	反解
A1	逆表达式
B1	逆定义域

(b) For the composite gf, find

(i) · 3 marks

$gf(x)=\dfrac{x+1}{x-2}$, $x>2$.

f输入先需x>2，且其输出正，均可送入g；取倒数。

M1	正确复合顺序
A1	表达式
B1	定义域

(ii) · 2 marks

$(1,\infty)$.

gf=1+3/(x−2)，x>2时分式为任意正数，1不取到。

M1	正确改写与正值
A1	完整值域

Question 27

P3-C02-R27 · 9 marks

(a) · 3 marks

$f^{-1}(x) = x + 1, x \geq 1.$

原域使f=x−1且输出⩾1，反解原输入。

M1	用限制去绝对值
M1	反解
A1	逆函数及域

(b) For h, find

(i) · 3 marks

Domain $x \leq 1$; $h(x) = 2 - x$.

内部输入3−x⩾2给x⩽1；在此域内|1−x|=1−x，所以加1为2−x。

M1	内部输入约束
M1	合法去绝对值
A1	域与式

(ii) · 3 marks

$h^{-1}(x) = 2 - x$, domain $x \geq 1$, range $x \leq 1$.

h在x⩽1上输出⩾1，反解同一线性式，交换域和值域。

M1	反解
A1	逆函数
B1	域与值域

Question 28

P3-C02-R28 · 9 marks

(a) · 3 marks

Range $[1,3)\cup\{5\}$; one-to-one.

半开线段给[1,3)，孤立端点给5；各输出只对应一个输入，3、4不在值域。

M1	正确半开输出区间
A1	加孤立值
B1	一一性结论

(b) For the inverse function, find

(i) · 3 marks

$f^{-1}(x)=x-1$ for $1\le x<3$, and $f^{-1}(5)=2$.

每一段分别反解，保留孤立输入5，不填补3到5的空缺。

M1	连续分支反解
A1	孤立分支
B1	完整定义域

(ii) · 3 marks

$0\le x\le 1$.

需要x在原域且f(x)∈[0,2]；在半开线段f=x+1⩽2给x⩽1，x=2时输出5非法。

M1	复合输入条件
M1	排除特殊端点
A1	最终区间

Question 29

P3-C02-R29 · 8 marks

(a) · 3 marks

$(0,7),(3,1),(5,5)$; $1 \le f(x) \le 7$.

分段为$7-2x$（$0 \le x \le 3$）和$2x-5$（$3 \le x \le 5$）。顶点最小值1，两端比较最大值7。

M1	正确分段图
A1	三点
B1	值域

(b) The domain is now restricted to $k \le x \le 5$, where $0 \le k < 5$, so that f has an inverse.

(i) · 2 marks

$k = 3$.

若$k < 3$，两侧支段有重复输出，非一一映射；$k = 3$开始只保留严格递增支段。

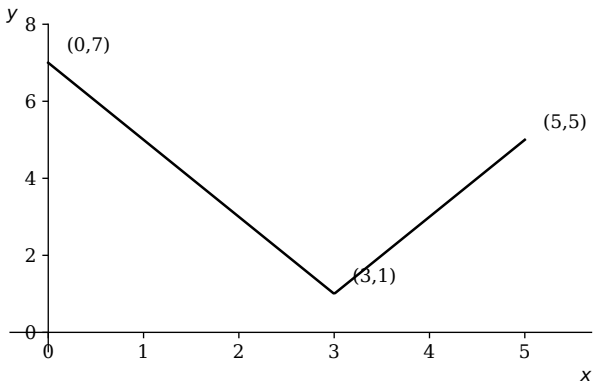
M1	用一一性判断
A1	最小k

(ii) · 3 marks

$f^{-1}(x) = (x+5)/2$, $1 \le x \le 5$.

受限函数为$y = 2x - 5$，原定义域$[3,5]$、值域$[1,5]$；解出原输入并交换字母，反函数定义域取原值域。

M1	反解正确支段
A1	表达式
B1	定义域



Suggested sketch

Question 30

P3-C02-R30 · 8 marks

(a) · 3 marks

$gf(x) = 1/(\sqrt{x+2}-1)$, $x \geq -2$, $x \neq -1$.

先要求根式$x+2 \geq 0$，再要求$f(x) \neq 1$，排除$x=-1$。两个限制均必要。

M1	正确复合顺序
A1	复合式
B1	全定义域

(b) For the composite function fg,

(i) · 3 marks

$x \leq 1/2$ or $x > 1$.

要求$1/(x-1)+2 = (2x-1)/(x-1) \geq 0$且$x \neq 1$。以$1/2, 1$分区间作符号判断。

M1	正确根式条件
M1	有理符号分析
A1	端点

(ii) · 2 marks

$x = 2$.

平方得$1/(x-1)+2 = 3$，所以$x=2$；属于前问定义域，代回根式为$\sqrt{3}$。

M1	求解复合方程
A1	合法输入

Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 3 — Trigonometric functions Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第3章 · 三角函数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C03-R01 · P3 · Chapter 3

An angle θ satisfies $\sin\theta = -5/13$ and $\pi < \theta < 3\pi/2$.

(a) Find the exact values of $\cos\theta$ and $\cot\theta$. (4)

(b) Find $\sec\theta + \operatorname{cosec}\theta$. (3)

(Total for Question 1 is 7 marks)

For $0 \leq x < 2\pi$, $\sec x = 2$.

(a) Find every exact solution. (3)

(b) Find every solution of $\sec^2 x = 4$ on the same interval. (3)

(Total for Question 2 is 6 marks)

Prove the identities on their common domains.

(a) $\sec^2 x - 1 = \tan^2 x.$ (3)

(b) $\dfrac{\sec^2 x - 1}{\sec^2 x} = \sin^2 x.$ (3)

(Total for Question 3 is 6 marks)

$\operatorname{cosec} x = -2$ for $-\pi \leq x \leq \pi$.

(a) Find all exact solutions. (3)

(b) Find the value of $\cot x$ at each solution. (3)

(Total for Question 4 is 6 marks)

5. P3-C03-R05 · P3 · Chapter 3

The curve is $y = \sec x$ for $-\pi \leq x \leq \pi$.

(a) Sketch it, showing all vertical asymptotes and points where $y = \pm 1$. (3)

(b) State the range and solve $\sec x \leq -2$ on the stated domain. (4)

(Total for Question 5 is 7 marks)

$\cot x = \sqrt{3}$ for $0 < x < 2\pi$.

(a) Find all exact solutions. (3)

(b) Find every excluded input of the function $\cot x$ on $[0, 2\pi]$. (2)

(Total for Question 6 is 5 marks)

$A = \arcsin(-\sqrt{3}/2)$ and $B = \arccos(-1/2)$.

(a) Find A, B exactly. (2)

(b) Find the exact value of $\cos A + \sin B$. (2)

(Total for Question 7 is 4 marks)

8. P3-C03-R08 · P3 · Chapter 3

$f(x) = \arcsin x$ for $-1 \leq x \leq 1$.

(a) Sketch its graph and state its range. (3)

(b) Find $\arcsin(\sin(5\pi/6))$ and explain why it is not $5\pi/6$. (3)

(Total for Question 8 is 6 marks)

9. P3-C03-R09 · P3 · Chapter 3

For $0 \leq x < 2\pi$, $2\sec^2 x - 5\sec x + 2 = 0$.

(a) Find all admissible values of $\sec x$. (3)

(b) Hence find all exact x. (3)

(Total for Question 9 is 6 marks)

On the common domain, $F(x) = \dfrac{\operatorname{cosec} x - \sin x}{\cos x}$.

(a) Show that $F(x) = \cot x$. (4)

(b) Solve $F(x) = 1$ for $0 < x < \pi$. (3)

(Total for Question 10 is 7 marks)

An angle $\theta = \arctan(5/12)$.

(a) Find $\sec\theta$ and $\operatorname{cosec}\theta$ exactly. (3)

(b) Find $\cos(\arctan(5/12))$ without a calculator. (2)

(Total for Question 11 is 5 marks)

$\operatorname{cosec}^2 x - 3\cot x - 3 = 0$ for $0 < x < \pi$.

(a) Write the equation as a quadratic in $\cot x$ and solve for this ratio. (3)

(b) Find the two solutions x in radians to 3 decimal places. (4)

(Total for Question 12 is 7 marks)

The graph is $y = \text{cosec}(2x)$ for $0 < x < \pi$.

(a) State all vertical asymptotes and the coordinates where $y = \pm 1$. (3)

(b) Solve $\text{cosec}(2x) = 2$. (3)

(Total for Question 13 is 6 marks)

For $-\pi/2 < x < \pi/2$, $\sec^2 x = 10$.

(a) Find the exact possible values of $\tan x$. (3)

(b) Given that $x < 0$, find x to 3 decimal places and $\sin x$ exactly. (4)

(Total for Question 14 is 7 marks)

Let $\alpha = \arccos(3/5)$.

(a) Find $\sin\alpha$ and $\cot\alpha$ exactly. (3)

(b) Find $\arccos(\cos(2\pi - \alpha))$ exactly in terms of α. (3)

(Total for Question 15 is 6 marks)

16. P3-C03-R16 · P3 · Chapter 3

For $0 < x < 2\pi$, $\cot^2 x = 1$.

(a) Find all exact solutions. (3)

(b) At which solutions is $\sec x$ negative? Give its exact value there. (3)

(Total for Question 16 is 6 marks)

On their common domain,

(a) prove $\dfrac{1+\cot^2 x}{\operatorname{cosec} x} = \operatorname{cosec} x$. (3)

(b) solve $\dfrac{1+\cot^2 x}{\operatorname{cosec} x} = -\sqrt{2}$ for $0 \leq x < 2\pi$. (3)

(Total for Question 17 is 6 marks)

The function $f(x) = \arccos x$ has domain $[-1, 1]$.

(a) State its range and sketch the graph, marking three exact points. (3)

(b) Solve $f(2x - 1) = \pi/3$, stating the input restriction. (3)

(Total for Question 18 is 6 marks)

For $0 \leq x < 2\pi$, $\sec x + \tan x = 3$.

(a) Use $(\sec x + \tan x)(\sec x - \tan x) = 1$ to find $\sec x, \tan x$. (3)

(b) Find the unique solution x in radians to 3 decimal places. (3)

(Total for Question 19 is 6 marks)

For $0 < x < \pi$, $\operatorname{cosec} x + \cot x = 2$.

(a) Find $\operatorname{cosec} x$ and $\cot x$ exactly. (3)

(b) Find x to 3 decimal places. (3)

(Total for Question 20 is 6 marks)

The curve is $y = \cot x$ for $0 < x < \pi$.

(a) Sketch the curve with asymptotes and intercept. (3)

(b) Find the exact interval on which $\cot x \geq 1$. (3)

(Total for Question 21 is 6 marks)

For $0 \leq x < 2\pi$, $\sec^2 x - 4\tan x = 5$.

(a) Find the possible values of $\tan x$. (3)

(b) State how many solutions occur in the interval and explain the count. (3)

(Total for Question 22 is 6 marks)

$\alpha = \arcsin(1/3)$.

(a) Find $\cos\alpha$ and $\tan\alpha$ exactly. (3)

(b) Find $\arcsin(\sin(\pi + \alpha))$ in terms of α. (3)

(Total for Question 23 is 6 marks)

For real x where defined, $F(x) = \sec^2 x + \operatorname{cosec}^2 x$.

(a) Show that $F(x) = \dfrac{1}{\sin^2 x \cos^2 x}$. (3)

(b) Find $F(\pi/6)$ exactly. (3)

(Total for Question 24 is 6 marks)

Figure 25 shows $y = a\sec(2x) + b$ for $0 \leq x \leq \pi$, where $a > 0$. The points $(0, 5)$ and $(\pi/2, -1)$ lie on the curve.

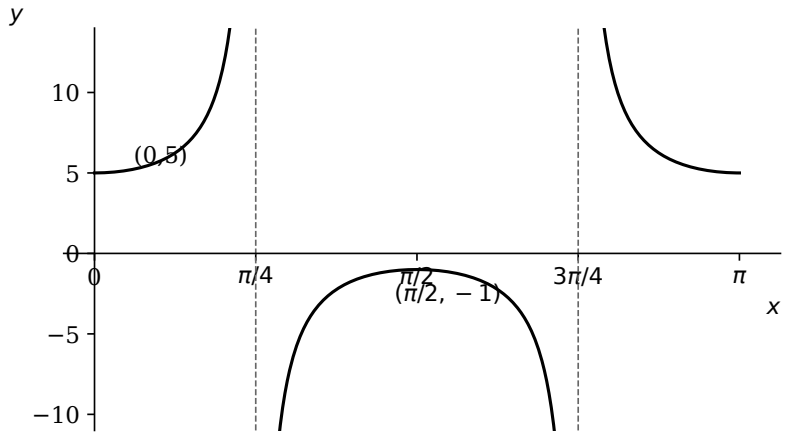


Figure 25

(a) Find a, b and state all vertical asymptotes in the interval. (3)

(b) For this curve,

(i) state its range. (2)

(ii) find the exact coordinates of all points where $y = 8$. (2)

(Total for Question 25 is 7 marks)

An angle θ lies in $(\pi/2, \pi)$ and satisfies $\tan\theta = -2$.

(a) Find the exact values of $\sec\theta$ and $\operatorname{cosec}\theta$. (3)

(b) Find $\arctan(\tan\theta)$ in terms of θ. (3)

(Total for Question 26 is 6 marks)

For $0 \leq x < 2\pi$, $\operatorname{cosec} x = \sec x$.

(a) Find all solutions. (3)

(b) Find the common reciprocal value at each solution. (3)

(Total for Question 27 is 6 marks)

Figure 28 shows the graph of $y = \arcsin x$, with principal values in $[-\pi/2, \pi/2]$. A new curve has equation $y = \arcsin(2x - 1)$.

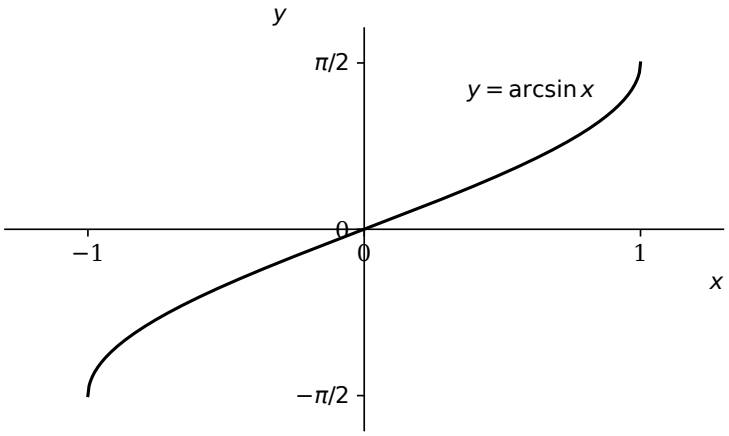


Figure 28

(a) State the domain and range of the new curve, and find its intercepts. (3)

(b) For these two graphs,

(i) find the point at which $y = \pi/6$. (2)

(ii) find $\arcsin(\sin(7\pi/6))$ and identify the corresponding point on the original graph. (2)

(Total for Question 28 is 7 marks)

For $0 < x < \pi$, $2\cot^2 x - 3\operatorname{cosec} x = 0$.

(a) Show that $\operatorname{cosec} x$ satisfies $2u^2 - 3u - 2 = 0$, where $u = \operatorname{cosec} x$. (2)

(b) Hence

(i) find all exact solutions for x in the stated interval. (2)

(ii) find the exact value of $\sec x$ at each solution. (2)

(Total for Question 29 is 6 marks)

The function $f(x) = \tan x$ is restricted to $-\pi/2 < x < \pi/2$.

(a) Sketch the graphs of f and f^{-1} on the same axes, showing their asymptotes and the line of reflection. (3)

(b) Let $\alpha = \arctan 2$.

(i) Find $\sin\alpha$ and $\cos\alpha$ exactly. (2)

(ii) find $\arctan(\tan(\pi + \alpha))$ and $\arctan(\tan(\pi - \alpha))$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C03-R01 · 7 marks

(a) · 4 marks

$-12/13$ and $12/5$.

第三象限正弦余弦均负，平方和给cos=−12/13；cot=cos/sin。

M1	平方和与象限
A1	cos
M1	倒比
A1	cot

(b) · 3 marks

$-221/60$.

倒数分别−13/12与−13/5，通分相加−221/60。

M1	两个正确倒数
M1	通分
A1	精确值

Question 2

P3-C03-R02 · 6 marks

(a) · 3 marks

$x = \pi/3, 5\pi/3.$

等价cosx=1/2，两个合法角均不使cos0。

M1	倒数转余弦
M1	象限解
A1	两角

(b) · 3 marks

$\pi/3, 2\pi/3, 4\pi/3, 5\pi/3.$

平方方程允许sec=±2，新增cos=−1/2两角。

M1	正负分支
M1	完整象限
A1	四解

Question 3

P3-C03-R03 · 6 marks

(a) · 3 marks

$\sec^2 x - 1 = (1 - \cos^2 x)/\cos^2 x = \tan^2 x$.

统一为$\cos^2$分母，再用基本恒等式。

M1	统一分母
M1	平方恒等式
A1	目标式

(b) · 3 marks

$\tan^2 x / \sec^2 x = \sin^2 x$.

用(a)再以sin/cos和1/cos代入，cos因子抵消。

M1	使用(a)
M1	倒数定义化简
A1	结果

Question 4

P3-C03-R04 · 6 marks

(a) · 3 marks

$x = -5\pi/6, -\pi/6$.

sinx=−1/2，在负半圈内两点；sin0排除无影响。

M1	转正弦
M1	负角筛选
A1	两解

(b) · 3 marks

$\sqrt{3}$ at $-5\pi/6$; $-\sqrt{3}$ at $-\pi/6$.

分别余弦负、正，而正弦都负，cot符号随之相反。

M1	两余弦值与符号
M1	cot商
A1	配对结果

Question 5

P3-C03-R05 · 7 marks

(a) · 3 marks

Asymptotes $x = \pm\pi/2$; points $(-\pi, -1), (0, 1), (\pi, -1)$.

cos零点为竖直渐近线；各段曲线符号与cos相同，绝对值至少1。

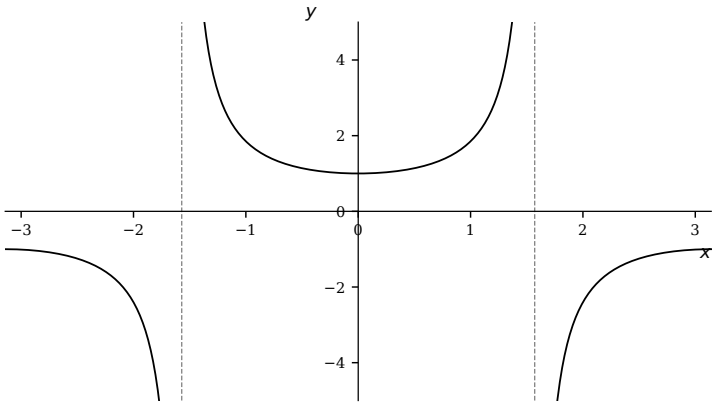
B1	渐近线
B1	三点
A1	正确三分支

(b) · 4 marks

Range $(-\infty, -1] \cup [1, \infty)$; $-2\pi/3 \le x < -\pi/2$ or $\pi/2 < x \le 2\pi/3$.

在cos负的分支，1/cos≤−2等价cos≥−1/2并保持负，按两段筛选，渐近线不取。

B1	值域
M1	负号不等式处理
M1	两分支端点
A1	完整范围



Suggested sketch

Question 6

P3-C03-R06 · 5 marks

(a) · 3 marks

$x = \pi/6, 7\pi/6.$

tanx=1/√3，周期π，两个同号象限。

M1	倒数比转换
M1	周期筛选
A1	两角

(b) · 2 marks

$0, \pi, 2\pi.$

cot=cos/sin，原分母sin零的点均无定义。

M1	原分母零点
A1	全部输入

Question 7

P3-C03-R07 · 4 marks

(a) · 2 marks

$A=-\pi/3,\ B=2\pi/3.$

arcsin主值域$[-\pi/2,\pi/2]$，arccos主值域$[0,\pi]$。

M1	正确两个主值范围
A1	两角

(b) · 2 marks

$(1+\sqrt{3})/2.$

$\cos(-\pi/3)=1/2$，$\sin(2\pi/3)=\sqrt{3}/2$。

M1	两精确三角值
A1	和

Question 8

P3-C03-R08 · 6 marks

(a) · 3 marks

Increasing from $(-1, -\pi/2)$ through $(0, 0)$ to $(1, \pi/2)$; range $[-\pi/2, \pi/2]$.

为正弦主分支关于y=x的反射，含两端点。

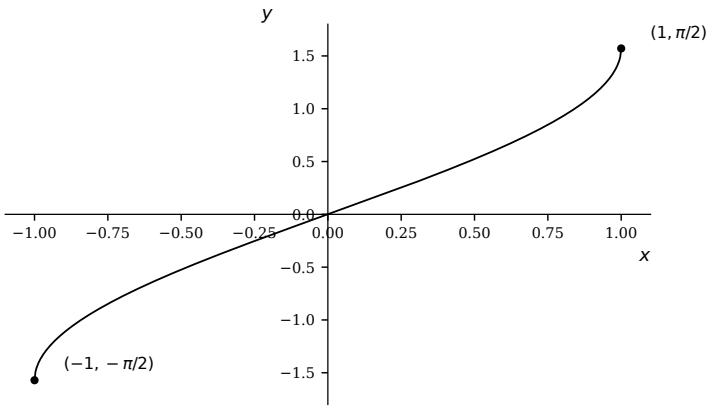
B1	三关键点
M1	递增反射形状
A1	值域

(b) · 3 marks

$\pi/6$; $5\pi/6$ lies outside the principal range of arcsine.

sin5π/6=1/2，arcsin必须取主值π/6。

M1	先算内部正弦
A1	主值
B1	主值限制理由



Suggested sketch

Question 9

P3-C03-R09 · 6 marks

(a) · 3 marks

$\sec x = 2$ only; reject $1/2$.

(2s−1)(s−2)=0，但实角正割绝对值至少1。

M1	二次分解
M1	正割值域筛选
A1	可行值

(b) · 3 marks

$\pi/3, 5\pi/3$.

cosx=1/2，取区间内两角。

M1	转余弦
M1	象限筛选
A1	两根

Question 10

P3-C03-R10 · 7 marks

(a) · 4 marks

$\operatorname{cosec} x - \sin x = (1 - \sin^2 x)/\sin x = \cos^2 x / \sin x$.

再除cosx得到cos/sin；原sin与cos都不能零。

M1	倒数与通分
M1	平方恒等式
M1	合法约分
A1	目标式

(b) · 3 marks

$x = \pi/4$.

cot=1给tan=1，主半圈仅π/4且不为原排除π/2。

M1	等价正切方程
M1	原域检查
A1	唯一角

Question 11

P3-C03-R11 · 5 marks

(a) · 3 marks

$13/12$ and $13/5$.

反正切给第一象限角，参考边5、12、13。

M1	参考三角形与主值
M1	倒数比
A1	两值

(b) · 2 marks

$12/13$.

由同一参考三角形读取余弦，不应把反三角视为倒数。

M1	正确函数嵌套解释
A1	精确值

Question 12

P3-C03-R12 · 7 marks

(a) · 3 marks

$\cot^2 x - 3\cot x - 2 = 0$; $\cot x = (3 \pm \sqrt{17})/2$.

用$\text{cosec}^2 = 1 + \cot^2$，整理并用二次公式。

M1	正确恒等替换
M1	二次公式
A1	两比值

(b) · 4 marks

$0.274, 2.082$ radians.

正cot在第一象限，负cot在第二象限；对负tan的反正切主值需加π。

M1	正确倒比
M1	负比分支加pi
M1	数值精度
A1	两角

Question 13

P3-C03-R13 · 6 marks

(a) · 3 marks

Asymptotes $x = 0, \pi/2, \pi$; points $(\pi/4, 1), (3\pi/4, -1)$.

sin2x零点给渐近线，±1处输入π/2、3π/2。

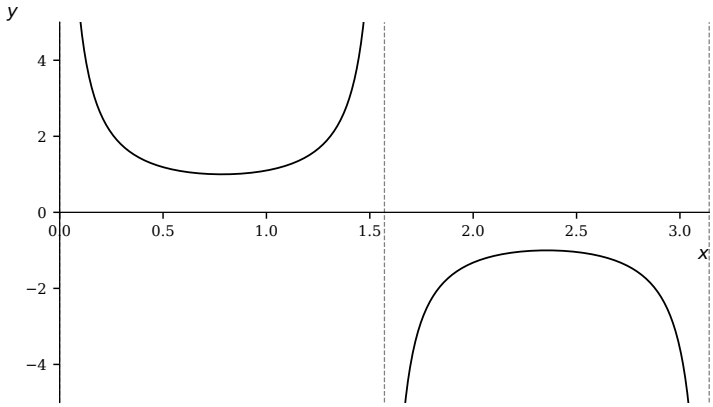
B1	三条渐近线
M1	极端输入缩放
A1	两点

(b) · 3 marks

$x = \pi/12, 5\pi/12$.

sin2x=1/2，输入在0至2π中两正半圈解，再除2。

M1	倒数转换
M1	输入范围
A1	两角



Suggested sketch

Question 14

P3-C03-R14 · 7 marks

(a) · 3 marks

$\tan x = \pm 3$.

$\tan^2=\sec^2-1=9$，两符号均在此区间可实现。

M1	平方恒等式
M1	两符号
A1	比值

(b) · 4 marks

$x = -\arctan 3 \approx -1.249$; $\sin x = -3/\sqrt{10}$.

此域cos正且x负，所以tan负；参考边−3与1，斜边$\sqrt{10}$。

M1	选负分支
A1	弧度值
M1	参考三角形符号
A1	正弦

Question 15

P3-C03-R15 · 6 marks

(a) · 3 marks

$4/5$ and $3/4$.

主值角在第一象限，勾股给sin4/5，cot比cos/sin。

M1	主值与平方和
A1	sin
A1	cot

(b) · 3 marks

α.

余弦在2π−α与α相等，而α本身位于arccos主值域。

M1	周期对称
M1	主值范围
A1	结果

Question 16

P3-C03-R16 · 6 marks

(a) · 3 marks

$\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$.

cot=±1即tan=±1，四个参考π/4角。

M1	正负分支
M1	周期筛选
A1	四解

(b) · 3 marks

$3\pi/4, 5\pi/4$; $\sec x = -\sqrt{2}$.

第二、第三象限cos负；倒数$-\sqrt{2}$。

M1	正确象限
M1	精确倒数
A1	两角及值

Question 17

P3-C03-R17 · 6 marks

(a) · 3 marks

Numerator equals $\operatorname{cosec}^2 x$, and cosecant is never zero on its domain.

使用基本倒数恒等式，合法约去非零cosec。

M1	恒等替换
M1	约分合法
A1	目标式

(b) · 3 marks

$x = 5\pi/4, 7\pi/4$.

由(a)变为sinx=−1/√2，第三第四象限两个角；sin零点原本排除。

M1	引用(a)倒数转换
M1	象限与原域
A1	两解

Question 18

P3-C03-R18 · 6 marks

(a) · 3 marks

Range $[0,\pi]$; points $(-1,\pi),(0,\pi/2),(1,0)$; decreasing.

从余弦主分支反射得到反函数图，端点均包含。

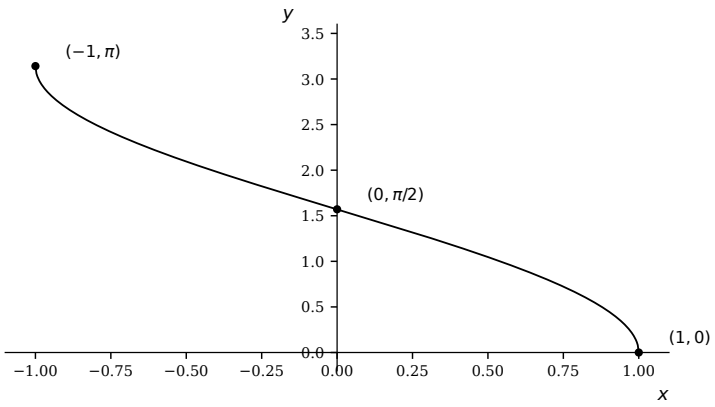
B1	三点
M1	递减形状
A1	值域

(b) · 3 marks

$0 \le x \le 1$; solution $x = 3/4$.

内部输入须−1⩽2x−1⩽1；2x−1=cosπ/3=1/2。

M1	复合定义域
M1	反解
A1	x



Suggested sketch

Question 19

P3-C03-R19 · 6 marks

(a) · 3 marks

$\sec x = 5/3$, $\tan x = 4/3$.

由恒等乘积另一因子$1/3$，两线性式相加减。

M1	找另一因子
M1	两式联立
A1	两比

(b) · 3 marks

$x = \arctan(4/3) \approx 0.927$.

sec正意味着cos正，tan正，唯一第一象限；不能加π。

M1	两比共同选象限
M1	反三角计算
A1	精度

Question 20

P3-C03-R20 · 6 marks

(a) · 3 marks

$5/4$ and $3/4$.

(cosec+cot)(cosec−cot)=1，另一因子1/2，相加减。

M1	共轭乘积
M1	联立
A1	两值

(b) · 3 marks

$x = \arctan(4/3) \approx 0.927$.

sin4/5、cot正，域内第一象限唯一。

M1	正确比值与象限
M1	计算
A1	弧度

Question 21

P3-C03-R21 · 6 marks

(a) · 3 marks

Asymptotes $x = 0, \pi$; intercept $(\pi/2, 0)$; strictly decreasing from $+\infty$ to $-\infty$.

cot=cos/sin，此半圈sin正而cos由正到负；标准图像每分支递减。

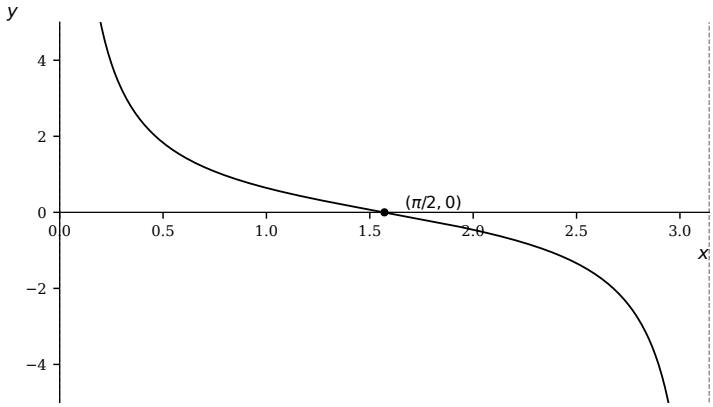
B1	渐近线
B1	零点
A1	正确分支

(b) · 3 marks

$0 < x \leq \pi/4$.

在本分支递减，等号π/4，左端为无定义渐近线。

M1	利用递减图像
M1	等号角和端点
A1	区间



Suggested sketch

Question 22

P3-C03-R22 · 6 marks

(a) · 3 marks

$\tan x = 2 \pm 2\sqrt{2}$.

以1+tan²替换sec²，得t²−4t−4=0。

M1	恒等替换
M1	二次公式
A1	两比

(b) · 3 marks

Four; each real tangent value occurs twice in a full interval of length 2π.

两不同比值，正切周期π，每一值在两个分支各出现一次，均不在渐近线。

M1	两个不同比值
M1	正切周期计数
A1	总数

Question 23

P3-C03-R23 · 6 marks

(a) · 3 marks

$2\sqrt{2}/3$ and $\sqrt{2}/4$.

主值角锐，cos=√(1−1/9)，tan为(1/3)/(2√2/3)，有理化。

M1	主值正根
M1	比值与有理化
A1	两值

(b) · 3 marks

$-\alpha$.

正弦(π+α)=−sinα，而−α位于arcsin主值区间。

M1	半周对称
M1	主值选择
A1	结果

Question 24

P3-C03-R24 · 6 marks

(a) · 3 marks

Common numerator $\sin^2 x + \cos^2 x = 1$.

以$\sin^2\cos^2$通分，原sin和cos均非零。

M1	正确公分母
M1	平方和
A1	目标式

(b) · 3 marks

$16/3$.

$\sin^2$=1/4、$\cos^2$=3/4，积3/16，取倒数。

M1	精确平方比
M1	倒数
A1	值

Question 25

P3-C03-R25 · 7 marks

(a) · 3 marks

$a = 3, b = 2$; $x = \pi/4, 3\pi/4$.

$a+b=5, -a+b=-1$给$a=3, b=2$。渐近线来自$\cos 2x = 0$。

M1	两点确定参数
A1	a,b
B1	两条渐近线

(b) For this curve,

(i) · 2 marks

$y \le -1$ or $y \ge 5$.

由$\sec 2x \le -1$或≥ 1，正伸缩3并平移2得到两支值域。

M1	基本割函数值域
A1	变换后范围

(ii) · 2 marks

$(\pi/6, 8), (5\pi/6, 8)$.

$3\sec 2x + 2 = 8$即$\cos 2x = 1/2$，$2x \in [0, 2\pi]$给$2x = \pi/3, 5\pi/3$。

M1	转换为余弦方程并匹配区间
A1	两坐标

Question 26

P3-C03-R26 · 6 marks

(a) · 3 marks

$-\sqrt{5}$ and $\sqrt{5}/2$.

参考两边2、1，斜边$\sqrt{5}$；第二象限cos负、sin正，取对应倒数。

M1	参考三角形
M1	象限符号
A1	两值

(b) · 3 marks

$\theta - \pi$.

反正切主值在$(-\pi/2, \pi/2)$，将第二象限θ减π进入负主值分支。

M1	正切周期
M1	主值区间
A1	表达式

Question 27

P3-C03-R27 · 6 marks

(a) · 3 marks

$x = \pi/4, 5\pi/4$.

原sin、cos都非零，可交叉相乘得sin=cos，tan1，两个周期解。

M1	约束下合法清分母
M1	正切与周期
A1	两角

(b) · 3 marks

$\sqrt{2}$ at $\pi/4$; $-\sqrt{2}$ at $5\pi/4$.

第一与第三象限的sin和cos分别$\pm\sqrt{2}/2$，倒数对应$\pm\sqrt{2}$。

M1	象限精确比
M1	倒数
A1	配对值

Question 28

P3-C03-R28 · 7 marks

(a) · 3 marks

$0 \le x \le 1$, $-\pi/2 \le y \le \pi/2$; $(0, -\pi/2), (1/2, 0)$.

输入满足$-1 \le 2x - 1 \le 1$，所以$x \in [0, 1]$；全体合法输入覆盖$[-1, 1]$，值域保持主值区间。令$x = 0$或$y = 0$求截距。

M1	输入不等式
A1	定义域值域
A1	截距

(b) For these two graphs,

(i) · 2 marks

$(3/4, \pi/6)$.

$2x - 1 = \sin(\pi/6) = 1/2$，所以$x = 3/4$。

M1	反函数主值关系
A1	坐标

(ii) · 2 marks

$-\pi/6$; $(-1/2, -\pi/6)$.

$\sin(7\pi/6) = -1/2$，原图在输入$-1/2$处输出主值$-\pi/6$；故对应点为$(-1/2, -\pi/6)$。$7\pi/6$不在反正弦的主值范围。

B1	正确主值
B1	用图像值域解释

Question 29

P3-C03-R29 · 6 marks

(a) · 2 marks

$(2u+1)(u-2)=0.$

用$\cot^2 x = \operatorname{cosec}^2 x - 1$代入，得到$2u^2-3u-2=0$。

M1	正确倒数三角恒等式
A1	二次式

(b) Hence

(i) · 2 marks

$x=\pi/6, 5\pi/6.$

代数根$u=2,-1/2$；由于$0<x<\pi$，$\sin x>0$且$\operatorname{cosec} x\geq 1$，只能取$u=2$。于是$\sin x=1/2$。

M1	按函数值域筛代数根
A1	两个精确角

(ii) · 2 marks

$2/\sqrt{3}$ at $\pi/6$; $-2/\sqrt{3}$ at $5\pi/6$.

两角分属第一、二象限，余弦分别为$\sqrt{3}/2,-\sqrt{3}/2$，取倒数并保留符号。

M1	象限与倒数
A1	对应值

Question 30

P3-C03-R30 · 7 marks

(a) · 3 marks

$y = \tan x$ and $y = \arctan x$, reflected in $y = x$.

正切支段过原点、竖直渐近线$x = \pm\pi/2$；反函数反射后过原点、水平渐近线$y = \pm\pi/2$。

B1	正切图及渐近线
B1	反正切图及渐近线
B1	反射直线

(b) Let $\alpha = \arctan 2$.

(i) · 2 marks

$2/\sqrt{5}$, $1/\sqrt{5}$.

主值区间内且正切为正，所以α为锐角。以对边2、邻边1构造直角三角形，斜边$\sqrt{5}$。

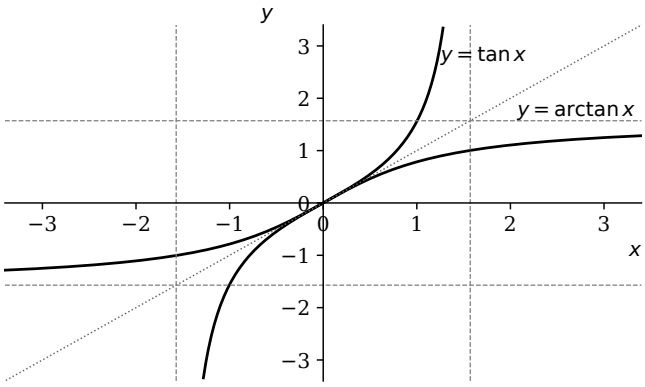
M1	用主值选象限
A1	两值

(ii) · 2 marks

α and $-\alpha$.

正切周期π使第一式内值为2；第二式正切为-2。反正切输出必须在主值支段，因此分别为$\alpha, -\alpha$。

M1	周期及符号
A1	两个主值



Suggested sketch

Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 4 — Trigonometric addition formulae Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第4章 · 三角加法公式
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C04-R01 · P3 · Chapter 4

Angles α, β are acute, with $\sin\alpha = 3/5$ and $\tan\beta = 5/12$.

(a) Find $\cos\alpha$ and $\sin\beta$ exactly. (2)

(b) Find $\sin(\alpha+\beta)$ and $\cos(\alpha-\beta)$ exactly. (3)

(Total for Question 1 is 5 marks)

$F(x) = 8\sin x + 15\cos x$.

(a) Express F as $R\sin(x + \alpha)$, where $R > 0$ and $0 < \alpha < \pi/2$. (3)

(b) Define $G(x) = 15\sin x - 8\cos x$.

(i) show that $F(x)^2 + G(x)^2 = 289$, (3)

(ii) find all possible values of $G(x)$ when $F(x) = 8$. (3)

(Total for Question 2 is 9 marks)

Without calculator technology,

(a) find the exact value of $\sin 75^\circ$, (3)

(b) find the exact value of $\tan 15^\circ$, with a rational denominator. (3)

(Total for Question 3 is 6 marks)

For $0 \leq x < 2\pi$, $\sin(2x) = \cos x$.

(a) Solve the equation exactly. (4)

(b) Find the solutions lost by dividing both sides by $\cos x$. (2)

(Total for Question 4 is 6 marks)

On their common domain,

(a) prove that $\dfrac{1-\cos 2x}{\sin 2x} = \tan x$. (3)

(b) find the exact value of this expression when $x = \pi/12$. (3)

(Total for Question 5 is 6 marks)

$\tan\alpha = 2$ and $\tan\beta = 1/3$, with both angles acute.

(a) Find $\tan(\alpha+\beta)$ exactly and determine whether $\alpha+\beta$ is acute or obtuse. (4)

(b) Find $\tan(\alpha-\beta)$ exactly. (2)

(Total for Question 6 is 6 marks)

$H(x) = 5\cos x - 12\sin x$.

(a) Write $H = R\cos(x + \alpha)$ for $R > 0$, $0 < \alpha < \pi/2$. (3)

(b) Find the total length of the intervals in $0 \le x < 2\pi$ where $H(x) > 0$. (3)

(Total for Question 7 is 6 marks)

8.

For $0 \leq x \leq \pi$, $\cos 2x + \cos x = 0$.

(a) Find all exact solutions. (4)

(b) Find $\sin 2x$ at each solution. (2)

(Total for Question 8 is 6 marks)

9. P3-C04-R09 · P3 · Chapter 4

For $0 < \theta < \pi$, $\cos\theta = -3/5$.

(a) Find $\sin(\theta/2)$ and $\cos(\theta/2)$ exactly. (3)

(b) Find $\tan(\theta/2)$ exactly. (2)

(Total for Question 9 is 5 marks)

A signal is modelled by $S(t) = 4 + 3\cos t + 3\sqrt{3}\sin t$, where t is in radians.

(a) Write S as $4 + R\cos(t - \alpha)$, with $R > 0$ and $0 < \alpha < \pi/2$. (3)

(b) On $0 \leq t < 2\pi$, find

(i) the maximum signal and the time it first occurs, (2)

(ii) all times for which the signal is 4. (3)

(Total for Question 10 is 8 marks)

Prove the identities.

(a) $\sin(x+\alpha)+\sin(x-\alpha)=2\sin x\cos\alpha.$ (3)

(b) $\cos(x-\alpha)-\cos(x+\alpha)=2\sin x\sin\alpha.$ (3)

(Total for Question 11 is 6 marks)

For $0 \leq x < 2\pi$, $\sin(x + \pi/6) = \sin(x - \pi/6)$.

(a) Use addition formulae to simplify the equation. (3)

(b) Hence solve exactly. (2)

(Total for Question 12 is 5 marks)

$\cos\theta = 5/13$ and $3\pi/2 < \theta < 2\pi$.

(a) Find $\sin 2\theta$ and $\cos 2\theta$ exactly. (4)

(b) Find $\tan 2\theta$ exactly. (2)

(Total for Question 13 is 6 marks)

For $0 < x < \pi$, $\tan 2x = \tan x$.

(a) Find all exact solutions, checking where the original tangent functions are defined. (3)

(b) Find all solutions if the domain is instead $0 \leq x \leq \pi$. (2)

(Total for Question 14 is 5 marks)

$F(x) = \cos^2 x - \sin^2 x + 2\sin x \cos x$.

(a) Express F as $R\cos(2x - \alpha)$ with exact R, α. (3)

(b) Find the least positive x at which F reaches its maximum. (3)

(Total for Question 15 is 6 marks)

A triangle has sides $AB = AC = 10$ cm and included angle 2θ, where $0 < \theta < \pi/2$.

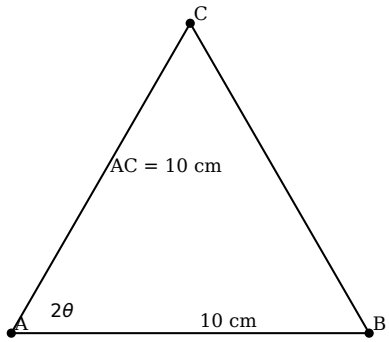


Figure 16

(a) Show that its area is $100 \sin\theta \cos\theta$. (3)

(b) Find all values of θ for which the area is $25\sqrt{3}$ cm^2. (3)

(Total for Question 16 is 6 marks)

For $0 \leq x < 2\pi$, $\sin(x + \pi/3) + \sin(x - \pi/3) = \cos 2x$.

(a) Use addition formulae to show that the left-hand side is $\sin x$. (3)

(b) Hence find every exact solution of the equation. (4)

(Total for Question 17 is 7 marks)

Let $\alpha = \arctan(1/2)$ and $\beta = \arctan(1/3)$.

(a) Find $\tan(\alpha + \beta)$ and use the principal ranges to determine $\alpha + \beta$ exactly. (4)

(b) Find $\tan(\alpha - \beta)$ and hence give the exact value of $\tan(2\alpha - \pi/4)$. (3)

(Total for Question 18 is 7 marks)

19. P3-C04-R19 · P3 · Chapter 4

For $0 \leq x < \pi$, $\cos(2x - \pi/4) = 0$.

(a) Find all exact solutions. (4)

(b) Sketch one period of $y = \cos(2x - \pi/4)$ on the stated interval, labelling the zeros and its maximum point. (3)

(Total for Question 19 is 7 marks)

Prove that $\tan(x + \pi/4) = \dfrac{1 + \tan x}{1 - \tan x}$ on the common domain.

(a) Derive the formula from the tangent addition formula. (3)

(b) Solve $\tan(x + \pi/4) = 3$ for $0 < x < \pi/2$. (3)

(Total for Question 20 is 6 marks)

$P(x) = 9\sin x + 12\cos x$.

(a) Find the greatest possible value of P and write an equation for the angles attaining it. (4)

(b) A modified signal is $Q(x) = P(x) - P(x + \pi)$. Find its maximum without re-expanding a new auxiliary angle. (3)

(Total for Question 21 is 7 marks)

For $0 \leq x \leq \pi$, $\sin(2x) = \sqrt{3}\sin x$.

(a) Find all exact solutions. (4)

(b) Find the unique interior solution and the exact value of $\cos 2x$ there. (3)

(Total for Question 22 is 7 marks)

For $0 < \theta < \pi$, $\sin\theta = 7/25$ and θ is obtuse.

(a) Find $\tan(\theta/2)$ exactly. (3)

(b) Find $\cos^2(\theta/2)$ exactly. (2)

(Total for Question 23 is 5 marks)

The identity $\sin 3x = 3\sin x - 4\sin^3 x$ is to be established.

(a) Use $\sin(2x + x)$ and double-angle formulae to prove it. (4)

(b) Hence solve $\sin 3x = \sin x$ for $0 \le x \le \pi/2$. (3)

(Total for Question 24 is 7 marks)

For $0 \leq x < 2\pi$, $\cos(x + \pi/3) = \cos(x - \pi/3)$.

(a) Simplify using addition formulae and solve. (4)

(b) Find the common cosine value at each solution. (2)

(Total for Question 25 is 6 marks)

A rotating point has height $h = 10 + 6\sin t - 8\cos t$ metres, where $0 \leq t < 2\pi$.

(a) Express h as $10 + R\sin(t - \alpha)$ for $R > 0$ and acute α. (3)

(b) Find

(i) the minimum height and the time at which it occurs, (3)

(ii) the total length of the time intervals where $h > 10$. (2)

(Total for Question 26 is 8 marks)

Prove that $\cos^4 x - \sin^4 x = \cos 2x$.

(a) Give an algebraic proof using a difference of squares. (3)

(b) Hence solve $\cos^4 x - \sin^4 x = -1/2$ for $0 \le x < \pi$. (3)

(Total for Question 27 is 6 marks)

A line segment of length 20 cm rotates through an acute angle θ. Its horizontal and vertical projections have sum 28 cm.

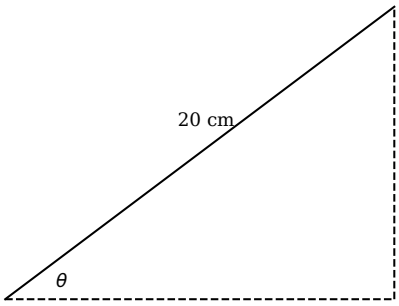


Figure 28

(a) Show that $\sin\theta + \cos\theta = 7/5$ and find $\sin 2\theta$. (3)

(b) Find the two possible angles to 1 decimal place. (3)

(Total for Question 28 is 6 marks)

Figure 29 shows the curve $y = 6\sin x + 8\cos x$ for $0 \leq x \leq 2\pi$.

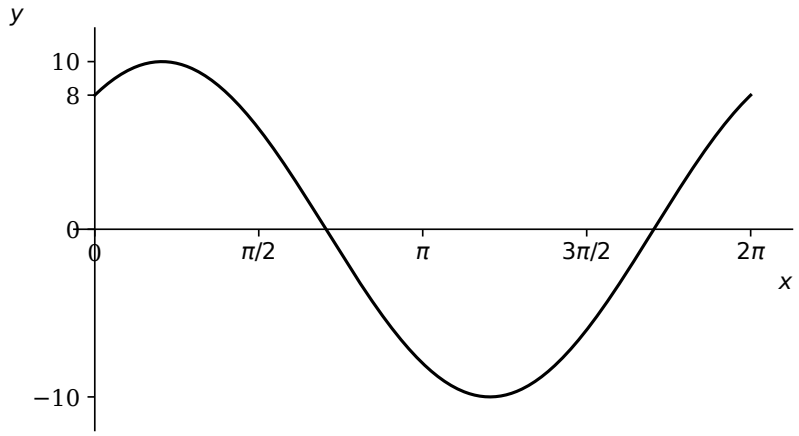


Figure 29

(a) Write $6\sin x + 8\cos x$ as $R\cos(x - \alpha)$, where $R > 0$ and $0 < \alpha < \pi/2$. Give α to 4 decimal places. (3)

(b) The horizontal line $y = k$ meets the curve at two distinct points in $0 \leq x < 2\pi$.

(i) Find the full range of k. (2)

(ii) For $k = 5$, find both x-coordinates to 3 decimal places. (2)

(Total for Question 29 is 7 marks)

Two rods of lengths 4 m and 3 m meet at a hinge. They make angles θ and $(\pi/2 - \theta)$, respectively, with the horizontal, and both extend to the right and upwards from the hinge. Here $0 < \theta < \pi/2$. The sum of their vertical projections is H m.

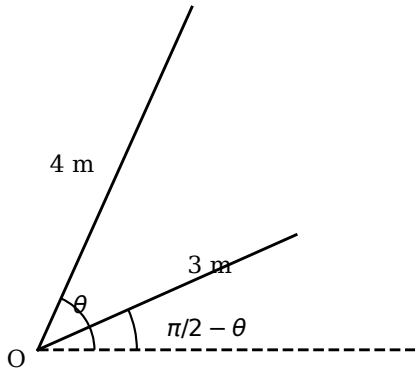


Figure 30

(a) Show that $H = 4\sin\theta + 3\cos\theta$ and express it in the form $5\sin(\theta + \beta)$, specifying β. (3)

(b) Find

(i) the greatest value of H and the exact value of θ at which it occurs. (2)

(ii) all possible values of θ for which $H = 9/2$, to 3 decimal places. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C04-R01 · 5 marks

(a) · 2 marks

$4/5$ and $5/13$.

两参考直角三角形分别3、4、5及5、12、13，锐角取正。

M1	两参考三角形
A1	两值

(b) · 3 marks

$56/65$ and $63/65$.

sin和=(3/5)(12/13)+(4/5)(5/13)=56/65；cos差=(4/5)(12/13)+(3/5)(5/13)=63/65。

M1	正弦和公式
M1	余弦差公式
A1	两精确值

Question 2

P3-C04-R02 · 9 marks

(a) · 3 marks

$R = 17$, $\alpha = \arctan(15/8)$.

比较Rcosα=8、Rsinα=15，平方和R17。

M1	正确两系数对应
M1	R
A1	精确alpha

(b) Define $G(x) = 15\sin x - 8\cos x$.

(i) · 3 marks

Cross terms cancel; sum is $289(\sin^2 x + \cos^2 x) = 289$.

展开两平方时交叉项为±240sinxcosx，抵消。

M1	展开并抵消交叉项
M1	平方和恒等式
A1	结论

(ii) · 3 marks

$G(x) = \pm 15$.

由(i)G^2=289−64=225，两符号在对应相位均能实现。

M1	代入平方和
M1	两符号
A1	值

Question 3

P3-C04-R03 · 6 marks

(a) · 3 marks

$(\sqrt{6}+\sqrt{2})/4$.

以45°+30°展开正弦和公式。

M1	选择标准角和
M1	精确三角值
A1	最简根式

(b) · 3 marks

$2-\sqrt{3}$.

用45°−30°正切差公式，再有理化(1−1/√3)/(1+1/√3)。

M1	正切差公式
M1	有理化
A1	精确值

Question 4

P3-C04-R04 · 6 marks

(a) · 4 marks

$x = \pi/6, \pi/2, 5\pi/6, 3\pi/2.$

用2sinxcosx=cosx，分解cosx(2sinx−1)=0，列两分支。

M1	双角代换
M1	零因子保留
M1	全部角
A1	四解

(b) · 2 marks

$\pi/2, 3\pi/2.$

原式在cos零点也满足，直接除法会丢掉这两根。

M1	检查零因子
A1	两角

Question 5

P3-C04-R05 · 6 marks

(a) · 3 marks

Use $1-\cos 2x = 2\sin^2 x$ and $\sin 2x = 2\sin x\cos x$.

原分母不为0保证sin与cos都非零，约分后sin/cos。

M1	两双角恒等式
M1	合法约分
A1	目标式

(b) · 3 marks

$2-\sqrt{3}$.

引用(a)变为tan15°，由角差公式得精确值。

M1	使用(a)
M1	标准角差
A1	值

Question 6

P3-C04-R06 · 6 marks

(a) · 4 marks

7; acute.

正切和(2+1/3)/(1−2/3)=7；两锐角之和在(0,π)，正切正所以仍锐角。

M1	正切和公式
M1	精确比值
M1	角和范围
A1	类型

(b) · 2 marks

1.

(2−1/3)/(1+2/3)=1。

M1	正切差公式
A1	精确值

Question 7

P3-C04-R07 · 6 marks

(a) · 3 marks

$R = 13,\ \alpha = \arctan(12/5).$

比较Rcosα=5、Rsinα=12，注意余弦和公式的负号。

M1	系数比较
M1	R
A1	alpha

(b) · 3 marks

$\pi.$

辅助角仅水平移相，完整一周期内cos正的输入总长度π；严格端点不改变总长度。

M1	化为正余弦区间
M1	完整周期长度
A1	结果

Question 8

P3-C04-R08 · 6 marks

(a) · 4 marks

$x = \pi/3, \pi$.

cos2x=2c²−1，得2c²+c−1=(2c−1)(c+1)，按0至π取每种一角。

M1	双角余弦代换
M1	分解
M1	域筛选
A1	两角

(b) · 2 marks

$\sqrt{3}/2$ and 0, respectively.

对应输入2π/3和2π。

M1	正确倍角输入
A1	两值

Question 9

P3-C04-R09 · 5 marks

(a) · 3 marks

$2/\sqrt{5}$ and $1/\sqrt{5}$.

半角在第一象限，两正平方根：$\sin^2$半角$=(1-\cos\theta)/2=4/5$，$\cos^2$半角$=1/5$。

M1	两半角平方关系
M1	正分支
A1	两值

(b) · 2 marks

2.

两半角比相除，公共根式抵消。

M1	半角比商
A1	值

Question 10

P3-C04-R10 · 8 marks

(a) · 3 marks

$S = 4 + 6\cos(t - \pi/3)$.

比较6cosα=3、6sinα=3√3得απ/3。

M1	正确系数对应
M1	R
A1	精确相位

(b) On $0 \le t < 2\pi$, find

(i) · 2 marks

Maximum 10 at $t = \pi/3$.

cos输入0时取1，检查相位落域内。

M1	余弦最大
A1	数值与位置

(ii) · 3 marks

$t = 5\pi/6, 11\pi/6$.

cos(t−π/3)=0，输入π/2+kπ，筛选两t。

M1	零余弦输入
M1	相位还原
A1	两t

Question 11

P3-C04-R11 · 6 marks

(a) · 3 marks

Expand both sine terms; the $\cos x \sin \alpha$ terms cancel.

逐项用正弦和差公式，保留sinxcosα两份。

M1	两正确展开
M1	抵消交叉项
A1	目标式

(b) · 3 marks

Expand both cosine terms; the $\cos x \cos \alpha$ terms cancel.

减去第二式时改变其两项符号，最终正的两倍sinxsinα。

M1	两余弦展开
M1	正确相减符号
A1	恒等式

Question 12

P3-C04-R12 · 5 marks

(a) · 3 marks

$\cos x = 0$.

两边展开后sinxcosπ/6抵消，余2cosxsinπ/6=0。

M1	两和差展开
M1	正确抵消
A1	简化条件

(b) · 2 marks

$x = \pi/2, 3\pi/2$.

半周两余弦零点，均在指定域内。

M1	正确零点筛选
A1	两根

Question 13

P3-C04-R13 · 6 marks

(a) · 4 marks

$-120/169$ and $-119/169$.

第四象限$\sin\theta=-12/13$；双角$\sin 2=2\sin\cos$，$\cos 2=2\cos^2-1$。

M1	象限正弦
M1	正弦双角
M1	余弦双角
A1	两值

(b) · 2 marks

$120/119$.

两双角比相除，负除负得正。

M1	比值商
A1	精确值

Question 14

P3-C04-R14 · 5 marks

(a) · 3 marks

No solutions.

同正切意味着2x=x+kπ，所以x=kπ，开区间没有；或双角公式t需检其分母且不丢cosx=0的原无定义点。

M1	正切周期相等条件
M1	原定义及开区间筛选
A1	无解

(b) · 2 marks

$x = 0, \pi.$

两个边界正切和双倍正切都定义且都为0。

M1	检查新增端点
A1	两根

Question 15

P3-C04-R15 · 6 marks

(a) · 3 marks

$F = \sqrt{2}\cos(2x - \pi/4)$.

前两项cos2x，末项sin2x，两系数1得R√2、相位π/4。

M1	双角化简
M1	辅助角系数
A1	形式

(b) · 3 marks

$x = \pi/8$.

2x−π/4=2kπ，最小正解π/8，最大值√2。

M1	最大余弦输入条件
M1	最小正支
A1	x

Question 16

P3-C04-R16 · 6 marks

(a) · 3 marks

Area $50\sin 2\theta = 100\sin\theta\cos\theta$.

两边夹角面积式，再用双角公式。

M1	正确面积50sin2theta
M1	双角转换
A1	目标式

(b) · 3 marks

$\theta = \pi/6, \pi/3$.

sin2θ=√3/2，输入2θ∈(0,π)，两角π/3、2π/3再除2。

M1	正确面积方程
M1	输入范围与两解
A1	两theta

Question 17

P3-C04-R17 · 7 marks

(a) · 3 marks

$2\sin x\cos(\pi/3)=\sin x$.

正弦和差展开后交叉项相消，余弦π/3为1/2。

M1	两加角展开
M1	抵消交叉项
A1	简式

(b) · 4 marks

$x=\pi/6,5\pi/6,3\pi/2$.

令s=sinx，用cos2x=1−2s²得(2s−1)(s+1)=0；两正弦值产生三角。

M1	双角转二次式
M1	因式分解
M1	全区间筛选
A1	三解

Question 18

P3-C04-R18 · 7 marks

(a) · 4 marks

$\tan(\alpha+\beta)=1, \alpha+\beta=\pi/4.$

和公式给(1/2+1/3)/(1−1/6)=1；两主值均锐、和在(0,π)，正切正选择第一象限π/4。

M1	正切和公式
M1	精确比1
M1	合角范围判断
A1	精确合角

(b) · 3 marks

Both equal $1/7$.

差公式(1/2−1/3)/(1+1/6)=1/7；由(a)有β=π/4−α，所以α−β=2α−π/4。

M1	正切差公式
M1	用合角关系重写
A1	两比值

Question 19

P3-C04-R19 · 7 marks

(a) · 4 marks

$x = 3\pi/8, 7\pi/8$.

输入从−π/4到7π/4，余弦零点π/2、3π/2；加π/4除2。

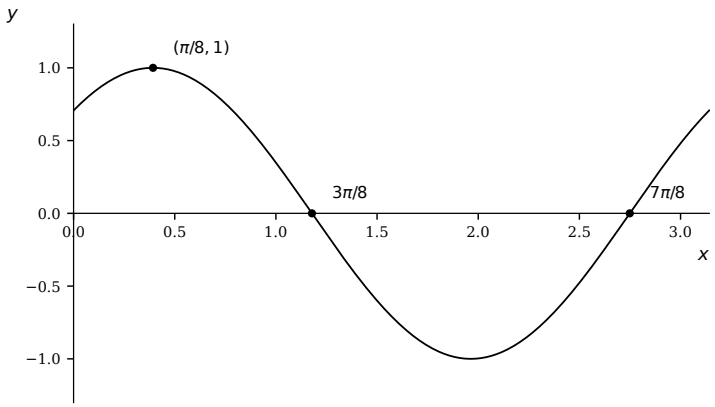
M1	输入范围
M1	零点
M1	还原x
A1	两解

(b) · 3 marks

Zeros as (a), maximum $(\pi/8, 1)$, minimum $(5\pi/8, -1)$; period π.

平移π/8、水平缩短为周期π，标出顶点及谷点，端点为$\sqrt{2}/2$。

M1	相位与周期
B1	极值点
A1	草图



Suggested sketch

Question 20

P3-C04-R20 · 6 marks

(a) · 3 marks

$\tan(\pi/4) = 1$ gives the stated expression.

正确引用加角公式，将第二角正切设1，并保留分母非零。

M1	正确和角公式
M1	精确值代入
A1	目标式

(b) · 3 marks

$x = \arctan(1/2)$.

(1+t)/(1−t)=3给t1/2；在锐域唯一且x+π/4不等于π/2。

M1	线性分式求tanx
M1	分支和域检查
A1	精确角

Question 21

P3-C04-R21 · 7 marks

(a) · 4 marks

Maximum 15; $x+\alpha=\pi/2+2k\pi$ with $\alpha=\arctan(4/3)$.

辅助角R15，Rcosα9、Rsinα12，正弦最高在π/2加整周。

M1	辅助角R
M1	正确相位
M1	最大输入
A1	值及方程

(b) · 3 marks

30.

加π使sin与cos都变号，P(x+π)=−P(x)，Q=2P，所以最高30。

M1	半周变号
M1	Q等于2P
A1	最大值

Question 22

P3-C04-R22 · 7 marks

(a) · 4 marks

$x = 0, \pi/6, \pi.$

sinx(2cosx−√3)=0，包含两端sin0，以及cos√3/2的π/6。

M1	双角分解
M1	零因子与端点
M1	余弦支
A1	三角

(b) · 3 marks

$x = \pi/6;\ \cos 2x = 1/2.$

去除0和π后唯一π/6，双倍输入π/3。

B1	内部角
M1	双角输入
A1	值

Question 23

P3-C04-R23 · 5 marks

(a) · 3 marks

7.

$\cos\theta=-24/25$，用tan半角=$\sin\theta/(1+\cos\theta)=(7/25)/(1/25)$。

M1	正确余弦负根
M1	半角关系
A1	比值

(b) · 2 marks

$1/50$.

$(1+\cos\theta)/2=(1/25)/2$。

M1	半角平方公式
A1	精确值

Question 24

P3-C04-R24 · 7 marks

(a) · 4 marks

$\sin 2x\cos x+\cos 2x\sin x=2\sin x\cos^2 x+(1-2\sin^2 x)\sin x=3\sin x-4\sin^3 x.$

依次加角、双角、平方和替换，完整显示推导，不直接把三倍角当已知。

M1	加角展开
M1	双角替换
M1	化为sin幂
A1	目标式

(b) · 3 marks

$x=0,\pi/4.$

用(a)化为$2\sin x(1-2\sin^2 x)=0$，正域内$\sin 0$或$\sqrt{2}/2$。

M1	引用(a)分解
M1	角范围筛选
A1	两角

Question 25

P3-C04-R25 · 6 marks

(a) · 4 marks

$\sin x = 0$; $x = 0, \pi$.

两式差为−2sinxsinπ/3=0，sinπ/3非零，整周期半开域两根。

M1	两余弦展开
M1	简化
M1	端点筛选
A1	两解

(b) · 2 marks

$1/2$ at 0; $-1/2$ at π.

代原相移余弦，输入π/3和4π/3。

M1	原式代入
A1	两值

Question 26

P3-C04-R26 · 8 marks

(a) · 3 marks

$R = 10$, $\alpha = \arctan(4/3)$.

比较Rcosα6和Rsinα8，减角公式提供负余弦项。

M1	正确对应
M1	R
A1	alpha

(b) Find

(i) · 3 marks

Minimum 0 at $t = 3\pi/2 + \alpha$ in the stated interval.

sin相位=−1时输入3π/2；加α<π/2仍小于2π。

M1	最低正弦
M1	选择域内相位
A1	高度和t

(ii) · 2 marks

π.

等价sin(t−α)>0，在完整周长2π内占π长度。

M1	基准线以上正弦条件
A1	总长度

Question 27

P3-C04-R27 · 6 marks

(a) · 3 marks

$(\cos^2 x - \sin^2 x)(\cos^2 x + \sin^2 x) = \cos 2x.$

先提平方差，再用平方和1和双角公式。

M1	平方差
M1	两恒等式
A1	结果

(b) · 3 marks

$x = \pi/3, 2\pi/3.$

cos2x=−1/2，输入在0至2π的2π/3、4π/3，除2。

M1	使用(a)
M1	倍角范围筛选
A1	两解

Question 28

P3-C04-R28 · 6 marks

(a) · 3 marks

Sum condition $20(\sin\theta+\cos\theta)=28$; $\sin 2\theta = 24/25$.

对和平方得1+sin2θ=49/25，所以24/25。

M1	两投影关系
M1	平方恒等式
A1	双角值

(b) · 3 marks

$36.9^\circ, 53.1^\circ$.

两锐角互余，sinθ和cosθ为3/5、4/5两种配对；也可解双角0至180°。

M1	两分支与锐域
M1	反求角
A1	两角

Question 29

P3-C04-R29 · 7 marks

(a) · 3 marks

$R = 10,\ \alpha = \arctan(3/4) = 0.6435.$

展开$R\cos(x-\alpha) = R\cos\alpha\cos x + R\sin\alpha\sin x$，比较得$R\cos\alpha = 8, R\sin\alpha = 6$。

M1	正确展开比较系数
A1	R
A1	alpha

(b) The horizontal line $y = k$ meets the curve at two distinct points in $0 \le x < 2\pi$.

(i) · 2 marks

$-10 < k < 10.$

每个内部高度在完整周期内有两个交点；在极大或极小高度仅有一个切点，超出值域无交点。半开区间避免重复周期端点。

B1	正确范围
B1	排除两端理由

(ii) · 2 marks

$x = 1.691, 5.879.$

$\cos(x-\alpha) = 1/2$，所以$x = \alpha \pm \pi/3$并调整到$[0, 2\pi)$；两解为$1.69069\ldots, 5.87949\ldots$。

M1	完整余弦解分支
A1	两个横坐标

Question 30

P3-C04-R30 · 7 marks

(a) · 3 marks

$\beta = \arctan(3/4)$.

竖直投影分别为$4\sin\theta$和$3\sin(\pi/2-\theta) = 3\cos\theta$。比较$5\cos\beta = 4, 5\sin\beta = 3$。

M1	投影与余角
M1	辅助角比较
A1	beta

(b) Find

(i) · 2 marks

$H_{\max} = 5$ m at $\theta = \arctan(4/3)$.

当$\theta + \beta = \pi/2$时正弦为1，角度位于允许区间；$\theta = \pi/2 - \beta = \arctan(4/3)$。

M1	最大值条件且合法
A1	高度与角

(ii) · 2 marks

$\theta = 0.476, 1.378$.

$\sin(\theta+\beta) = 0.9$。结合$\theta \in (0, \pi/2)$，两支为$\arcsin 0.9 - \beta$及$\pi - \arcsin 0.9 - \beta$，分别$0.47627\ldots, 1.37832\ldots$。

M1	两个正弦分支及区间
A1	两角

Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 5 — Exponentials and logarithms Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第5章 · 指数与对数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C05-R01 · P3 · Chapter 5

For $x > 1$, $\ln(x+1) - \ln(x-1) = \ln 3$.

(a) Find x exactly. (3)

(b) Find the exact value of $\ln(x^2 - 1)$ at this solution. (2)

(Total for Question 1 is 5 marks)

A cooling model is $T = 18 + 62e^{-0.2t}$, where $t \geq 0$ is in minutes and T is in °C.

(a) Find the initial temperature and limiting temperature. (2)

(b) Find the time when $T = 30$ °C, giving an exact expression and a value to 2 decimal places. (4)

(Total for Question 2 is 6 marks)

3.

$e^{2x} - 5e^{x} + 4 = 0.$

(a) Find all exact solutions. (4)

(b) Find all real solutions of $e^{2x} - 5e^{x} + 4 < 0$. (3)

(Total for Question 3 is 7 marks)

Two quantities are related by $y = Ax^n$, with $A > 0, x > 0$. A plot of $\ln y$ against $\ln x$ is a straight line through $(0, \ln 3)$ and $(\ln 2, \ln 24)$.

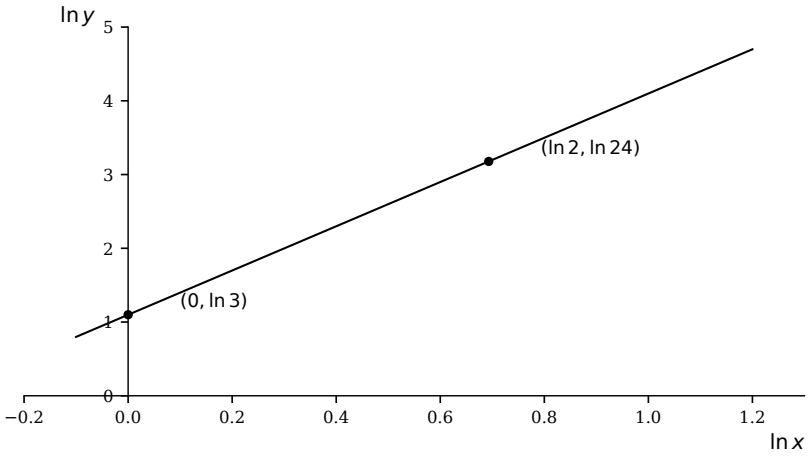


Figure 4

(a) Find A, n. (4)

(b) Find x when $y = 81$. (2)

(Total for Question 4 is 6 marks)

5.

A quantity satisfies $M = Ae^{kt}$. It has values 12 at $t = 1$ and 96 at $t = 4$.

(a) Find exact A, k. (4)

(b) Find the time when $M = 150$, to 3 decimal places. (3)

(Total for Question 5 is 7 marks)

6. P3-C05-R06 · P3 · Chapter 5

The curve is $y = \ln(4 - x)$.

(a) State its domain, vertical asymptote and axis intersections. (3)

(b) Find all x for which $y > \ln 2$. (3)

(Total for Question 6 is 6 marks)

$\ln x + \ln(x - 2) = \ln 8$, where $x > 2$.

(a) Solve exactly. (4)

(b) Find the exact value of $\ln x - \ln(x - 2)$ at the solution. (2)

(Total for Question 7 is 6 marks)

8. P3-C05-R08 · P3 · Chapter 5

A material has mass $m = 60e^{-kt}$, where $k > 0$. After 8 days its mass is 45.

(a) Find k exactly. (3)

(b) Find its half-life and the time to reach one quarter of its initial mass, in exact form. (3)

(Total for Question 8 is 6 marks)

9. P3-C05-R09 · P3 · Chapter 5

For $x > 0$, $y = Be^{ax}$. A graph of $\ln y$ against x has gradient $-3/2$ and vertical intercept 2.

(a) Find a, B. (3)

(b) Find the exact positive x at which $y = 1$. (2)

(Total for Question 9 is 5 marks)

$2e^{2x} - 7e^{x} - 4 = 0$.

(a) Find all exact real solutions, showing why one substituted root is inadmissible. (4)

(b) Find the value of e^{-x} at the solution. (2)

(Total for Question 10 is 6 marks)

The function is $f(x) = \ln(x^2 + 1)$.

(a) State its domain, range and every axis intersection. (4)

(b) Solve $f(x) = \ln 5$ and $f(x) < \ln 5$ separately. (3)

(Total for Question 11 is 7 marks)

A growth model is $P = 200e^{kt}$, with t in years. The population doubles every 3 years.

(a) Find k exactly. (2)

(b) For this model, find

(i) the population after 9 years, (2)

(ii) the least whole number of years at which the population is at least 1000. (3)

(Total for Question 12 is 7 marks)

$\ln(x+3) = 2\ln x$, where $x > 0$.

(a) Find the exact solution. (4)

(b) Find the exact value of $\ln((x+3)/x)$ at the solution. (2)

(Total for Question 13 is 6 marks)

The curve is $y = Ae^{2x} + B$. Its horizontal asymptote is $y = -3$, and it passes through $(\ln 2, 13)$.

(a) Find A, B. (3)

(b) Find its exact x-intercept and y-intercept. (3)

(Total for Question 14 is 6 marks)

For $x > 1$, $F(x) = \ln\left(\dfrac{x+1}{x-1}\right)$.

(a) Show that $F(x) > 0$. (3)

(b) Find the input at which $F(x) = \ln 5$. (3)

(Total for Question 15 is 6 marks)

A signal follows $S = 12e^{-0.4t}$ and a background signal follows $B = 3e^{-0.1t}$, for $t \geq 0$.

(a) Find exactly when $S = B$. (3)

(b) Find the interval of time when $S > B$. (3)

(Total for Question 16 is 6 marks)

A relation $y = Ae^{kx}$ has positive observations $(0, 5)$ and $(2, 20)$.

(a) Find A, k exactly. (3)

(b) Find $\ln y$ at $x = -1$ and give the corresponding exact y. (3)

(Total for Question 17 is 6 marks)

$\ln x = \ln(2 - x) + \ln 3$.

(a) State the domain restrictions and solve. (4)

(b) Find $\ln[x(2 - x)]$ at the solution. (2)

(Total for Question 18 is 6 marks)

The resistance of a component is $R = R_0 e^{kT}$. It is 10 ohms at $T = 0$ and 15 ohms at $T = 20$.

(a) Find R_0, k exactly. (4)

(b) Find the temperature change that multiplies resistance by $3/2$, and explain why it is independent of the starting temperature. (3)

(Total for Question 19 is 7 marks)

The graph of $Y = \ln(y - 2)$ against x is a straight line with gradient $-1/2$ and intercept $\ln 6$.

(a) Find y in terms of x. (3)

(b) Find the exact x at which $y = 5$ and state the horizontal asymptote of the original graph. (3)

(Total for Question 20 is 6 marks)

$e^x + e^{-x} = 5/2$.

(a) Find all exact solutions. (4)

(b) Find all possible values of $e^x - e^{-x}$ at these solutions. (3)

(Total for Question 21 is 7 marks)

A quantity is modelled by $Q = 100(1 - e^{-kt})$, where $k > 0$, $t \geq 0$. At $t = 4$, $Q = 75$.

(a) Find k exactly and the limiting value of Q. (4)

(b) Find the time at which $Q = 93.75$. (3)

(Total for Question 22 is 7 marks)

Let $f(x) = \ln(3x - 2)$.

(a) Find f^{-1} with its domain and range. (4)

(b) Solve $f(x) = 2f(1)$. (3)

(Total for Question 23 is 7 marks)

For $x > 0$, $(\ln x)^2 - \ln x - 6 = 0$.

(a) Find all exact solutions. (4)

(b) Find all x for which $(\ln x)^2 - \ln x - 6 \leq 0$. (3)

(Total for Question 24 is 7 marks)

The curves $y = e^{2x}$ and $y = 9e^{-x}$ meet.

(a) Find the intersection exactly. (3)

(b) Find the horizontal distance between the intersection and the point on $y = e^{2x}$ where $y = 1$. (3)

(Total for Question 25 is 6 marks)

A power-law model is $y = Ax^n$. A base-10 log plot has intercept 2 and gradient -3.

(a) Find A, n and the value of y at $x = 2$. (4)

(b) State the gradient and intercept if natural logarithms are used instead on both axes. (2)

(Total for Question 26 is 6 marks)

27.

For real x, $F(x) = \ln(e^x + 1) - x$.

(a) Show that $F(x) = \ln(1 + e^{-x})$. (3)

(b) Find x when $F(x) = \ln 3$. (3)

(Total for Question 27 is 6 marks)

28. P3-C05-R28 · P3 · Chapter 5

A model for concentration is $C = 8e^{-0.3t} + 2$, with $t \geq 0$.

(a) Find the concentration after 5 time units, to 3 significant figures. (2)

(b) Find the set of times for which the concentration is below 3. Give the exact boundary time and its value to 2 decimal places. (4)

(Total for Question 28 is 6 marks)

The mass M grams of a sample at time t hours is modelled by $M = Ae^{-kt}$, where $A, k > 0$. The mass is 60 g at $t = 2$ and 24 g at $t = 5$.

(a) Find k and A, giving each to 4 significant figures. (3)

(b) Using unrounded constants,

(i) find the time at which the sample mass is 10 g, to 3 significant figures. (2)

(ii) find the length of time needed for any mass in this model to fall to one half of its value, and explain why the result does not depend on that mass. (3)

(Total for Question 29 is 8 marks)

Variables x, y satisfy $x > 0$, $y > 2$ and $y = 2 + px^q$, where $p, q > 0$. A graph of $\ln(y - 2)$ against $\ln x$ passes through $(0, \ln 3)$ and $(\ln 4, \ln 24)$.

(a) Find p and q exactly. (3)

(b) For this model,

(i) find x when $y = 83$. (2)

(ii) find the percentage increase in y when x increases from 1 to 1.21. Give your answer to 3 significant figures. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C05-R01 · 5 marks

(a) · 3 marks

$x = 2$.

合并商：(x+1)/(x−1)=3，解一次式，检查x>1。

M1	对数商法则
M1	线性求解
A1	合法根

(b) · 2 marks

$\ln 3$.

代x2得$x^2-1=3$，真数为正。

M1	正确代入
A1	值

Question 2

P3-C05-R02 · 6 marks

(a) · 2 marks

80 °C and 18 °C.

t0时指数1；t增大时衰减项趋0。

B1	初温
B1	极限温度

(b) · 4 marks

$t = 5\ln(31/6) \approx 8.21$ min.

62e^{−.2t}=12，取ln得−.2t=ln(6/31)。

M1	隔离指数
M1	自然对数
A1	精确时间
A1	精度

Question 3

P3-C05-R03 · 7 marks

(a) · 4 marks

$x = 0, \ln 4.$

u=e^x>0，(u−1)(u−4)=0，取ln回解。

M1	正变量代换
M1	因式分解
M1	还原
A1	两根

(b) · 3 marks

$0 < x < \ln 4.$

二次式在1

M1	二次不等式
M1	单调映回x
A1	区间

Question 4

P3-C05-R04 · 6 marks

(a) · 4 marks

$A = 3, n = 3.$

lny=nlnx+lnA，截距ln3；斜率ln(24/3)/ln2=3。

M1	正确线性化
M1	截距求A
M1	斜率求n
A1	参数

(b) · 2 marks

$x = 3.$

$81=3x^3$，x正给3。

M1	代回原模型
A1	x

Question 5

P3-C05-R05 · 7 marks

(a) · 4 marks

$k = \ln 2, A = 6.$

相除e^{3k}=8给kln2，再由Ae^k12求A6。

M1	两条件相除
M1	对数求k
M1	回求A
A1	精确参数

(b) · 3 marks

$t = \ln 25 / \ln 2 \approx 4.644.$

6e^{tln2}=150，tln2=ln25。

M1	正确阈值方程
M1	对数
A1	时间

Question 6

P3-C05-R06 · 6 marks

(a) · 3 marks

$x < 4$; $x = 4$; $(3, 0)$ and $(0, \ln 4)$.

真数4−x正；y0给4−x1；x0给ln4。

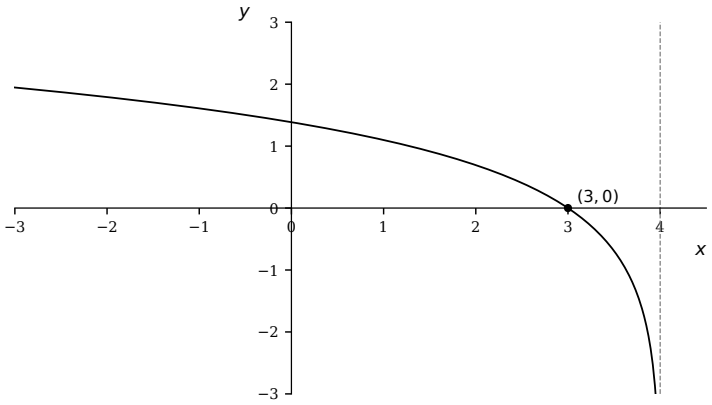
B1	域与渐近线
M1	横截距
A1	两点

(b) · 3 marks

$x < 2$.

ln严格递增，4−x>2，结果已包含在原域中。

M1	合法对数比较
M1	原域检查
A1	范围



Suggested sketch

Question 7

P3-C05-R07 · 6 marks

(a) · 4 marks

$x = 4$.

x(x−2)=8，方程(x−4)(x+2)=0，只留4。

M1	合并对数转乘积
M1	二次求根
M1	原域筛选
A1	x

(b) · 2 marks

$\ln 2$.

ln[4/2]=ln2，不能相减真数后直接取ln。

M1	对数商法则
A1	值

Question 8

P3-C05-R08 · 6 marks

(a) · 3 marks

$k = \frac{1}{8}\ln(4/3)$.

e^{-8k}=3/4，ln后求k。

M1	正确比值
M1	自然对数
A1	k

(b) · 3 marks

Half-life $8\ln 2/\ln(4/3)$; quarter time $16\ln 2/\ln(4/3)$ days.

e^{-kt}=1/2或1/4，对数指数分别ln2与2ln2，所以时间加倍。

M1	半衰关系
M1	四分之一关系
A1	两时间

Question 9

P3-C05-R09 · 5 marks

(a) · 3 marks

$a=-3/2, B=e^{2}.$

lny=ax+lnB，直读斜率和对数截距，再指数还原B。

M1	正确线性形式
A1	a
A1	B

(b) · 2 marks

$x=4/3.$

lny=0给−3x/2+2=0。

M1	使用对数直线求零值
A1	x

Question 10

P3-C05-R10 · 6 marks

(a) · 4 marks

$x = \ln 4$; reject $e^x = -1/2$.

(2u+1)(u−4)=0，u=e^x严格正。

M1	指数代换分解
M1	排负u
M1	ln还原
A1	唯一根

(b) · 2 marks

$1/4$.

e^{-x}=1/e^x=1/4。

M1	指数倒数关系
A1	值

Question 11

P3-C05-R11 · 7 marks

(a) · 4 marks

Domain $\mathbb{R}$; range $[0, \infty)$; only intercept $(0, 0)$.

$x^2+1 \geqslant 1$，ln输出$\geqslant 0$，在x0取0，任意更大真数可取。

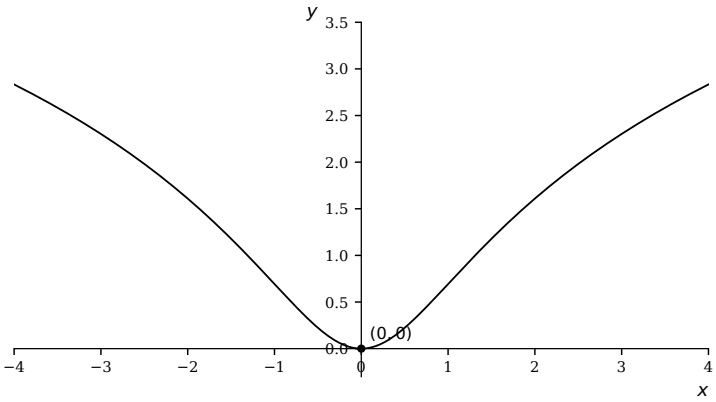
M1	真数恒正
M1	最小输出
A1	域与范围
B1	截距

(b) · 3 marks

Equality $x = \pm 2$; inequality $-2 < x < 2$.

ln单调，分别化为$x^2+1=5$或<5。

M1	合法去对数
M1	平方方程与不等式
A1	两个解集



Suggested sketch

Question 12

P3-C05-R12 · 7 marks

(a) · 2 marks

$k = \ln 2/3.$

e^{3k}=2，取ln。

M1	倍增条件
A1	k

(b) For this model, find

(i) · 2 marks

1600.

9年等于3个倍增周期，所以200×2^{3}。

M1	周期倍数
A1	数值

(ii) · 3 marks

7 years.

$t \geqslant 3\ln 5/\ln 2 \approx 6.966$，最小整数7，等号条件为至少。

M1	正确阈值不等式
M1	整数取上界
A1	年数

Question 13

P3-C05-R13 · 6 marks

(a) · 4 marks

$x = (1 + \sqrt{13})/2.$

$\ln(x+3)=\ln x^2$给$x^2-x-3=0$，正根唯一满足$x>0$。

M1	对数幂法则
M1	二次公式
M1	正域筛选
A1	根

(b) · 2 marks

$\ln x.$

由$x+3=x^2$，商为x，真数正。

M1	使用根关系化简商
A1	值

Question 14

P3-C05-R14 · 6 marks

(a) · 3 marks

$A = 4, B = -3$.

渐近线给B，e^{2ln2}=4，13=4A−3。

B1	B
M1	精确指数还原
A1	A

(b) · 3 marks

x-intercept $(\frac{1}{2}\ln(3/4), 0)$; y-intercept $(0, 1)$.

令4e^{2x}=3取ln；x0时4−3=1。

M1	指数零值方程
M1	对数求x
A1	两截距

Question 15

P3-C05-R15 · 6 marks

(a) · 3 marks

$(x+1)/(x-1) = 1 + 2/(x-1) > 1$, so its natural logarithm is positive.

利用x>1使分母正，真数严格大于1，ln单调且ln1=0。

M1	真数改写
M1	正值比较
A1	严格结论

(b) · 3 marks

$x = 3/2$.

(x+1)/(x−1)=5给x+1=5x−5，x1.5合法。

M1	去对数并清合法分母
M1	求解检查
A1	x

Question 16

P3-C05-R16 · 6 marks

(a) · 3 marks

$t = \frac{10}{3}\ln 4$.

相除4e^{-0.3t}=1，对数后t=ln4/.3。

M1	相比隔离指数
M1	ln运算
A1	精确时间

(b) · 3 marks

$0 \le t < \frac{10}{3}\ln 4$.

4e^{-0.3t}>1，随t严格减，所以在首次等值之前；包括初始时点。

M1	正确不等式方向
M1	原时间域
A1	区间

Question 17

P3-C05-R17 · 6 marks

(a) · 3 marks

$A = 5, k = \ln 2.$

首点A5，e^{2k}=4所以kln2。

B1	A
M1	对数求k
A1	k

(b) · 3 marks

$\ln y = \ln(5/2);\ y = 5/2.$

用lny=ln5+xln2，在−1处相减；也可代指数模型。

M1	正确对数线性值
M1	还原
A1	两值

Question 18

P3-C05-R18 · 6 marks

(a) · 4 marks

$0 < x < 2$; solution $x = 3/2$.

两真数均正；合并成ln[3(2−x)]，得到x=6−3x。

M1	两真数正域
M1	合并对数
M1	解并查域
A1	解

(b) · 2 marks

$\ln(3/4)$.

代1.5×.5=3/4。

M1	正确乘积代值
A1	对数值

Question 19

P3-C05-R19 · 7 marks

(a) · 4 marks

$R_0 = 10$, $k = \ln(3/2)/20$.

首条件读R0，第二e^{20k}=3/2后取ln。

B1	R0
M1	观测比例
M1	对数
A1	k

(b) · 3 marks

20 degrees; $R(T+\Delta)/R(T) = e^{k\Delta}$.

相除时初值和e^{kT}抵消，固定比值只由温差决定。

M1	正确比值式
M1	消去起始温度
A1	温差和理由

Question 20

P3-C05-R20 · 6 marks

(a) · 3 marks

$y = 2 + 6e^{-x/2}$.

ln(y−2)=−x/2+ln6，指数还原再加2。

M1	写线性式
M1	指数还原
A1	模型

(b) · 3 marks

$x = 2\ln 2$; asymptote $y = 2$.

6e^{-x/2}=3给e^{-x/2}=1/2，求x；指数衰减极限加回2。

M1	对数求x
A1	精确x
B1	原坐标渐近线

Question 21

P3-C05-R21 · 7 marks

(a) · 4 marks

$x = \pm \ln 2$.

设u=e^x>0，u+1/u=5/2，2u²−5u+2=0给u2或1/2。

M1	正变量代换
M1	二次因式
M1	ln还原
A1	两x

(b) · 3 marks

$\pm 3/2$.

可分别代u，或平方差关系：(sum)²−(difference)²=4，差平方9/4。

M1	代值或平方恒等式
M1	两符号
A1	值

Question 22

P3-C05-R22 · 7 marks

(a) · 4 marks

$k = \ln 4/4$; limit 100.

1−e^{-4k}=.75给e^{-4k}=.25；t大时指数趋0。

M1	校准方程
M1	对数
A1	k
B1	极限

(b) · 3 marks

$t = 8$.

此时指数=1/16，为t4处1/4的平方，所以时间8；或用ln求。

M1	正确指数比例
M1	求时间
A1	精确值

Question 23

P3-C05-R23 · 7 marks

(a) · 4 marks

$f^{-1}(x) = (e^x + 2)/3$, domain $\mathbb{R}$, range $(2/3, \infty)$.

原域x>2/3，原值域全实；反解e^y=3x−2，交换域范围。

M1	正确指数反解
A1	反函数
B1	逆域
B1	逆范围

(b) · 3 marks

$x = 1$.

f1=ln1=0，所以右边0，3x−2=1。

M1	先求f1
M1	反解零值
A1	x

Question 24

P3-C05-R24 · 7 marks

(a) · 4 marks

$x = e^{3}, e^{-2}$.

令u=lnx，(u−3)(u+2)=0；两实u都可指数还原为正x。

M1	对数代换
M1	因式分解
M1	指数回代
A1	两x

(b) · 3 marks

$e^{-2} \leq x \leq e^{3}$.

二次条件−2⩽lnx⩽3，指数单调，闭端点保留。

M1	二次范围
M1	单调指数化
A1	闭区间

Question 25

P3-C05-R25 · 6 marks

(a) · 3 marks

$(\frac{2}{3}\ln 3, 3^{4/3})$.

e^{3x}=9给x=ln9/3，纵坐标e^{2x}=9^{2/3}。

M1	合并指数
M1	ln求x
A1	精确交点

(b) · 3 marks

$\frac{2}{3}\ln 3$.

第二点x0，第一点横坐标正，差即该值。

M1	找零对数位置x0
M1	横差
A1	距离

Question 26

P3-C05-R26 · 6 marks

(a) · 4 marks

$A = 100, n = -3$; $y = 25/2$.

logy=logA+nlogx，截距2意味着A100而不是2；代x2。

M1	对数截距还原
B1	n
M1	代原模型
A1	y

(b) · 2 marks

Gradient -3; intercept $\ln 100 = 2\ln 10$.

lny=nlnx+lnA，换底不改变幂指数斜率，但截距依底而变。

M1	自然对数线性式
A1	两参数

Question 27

P3-C05-R27 · 6 marks

(a) · 3 marks

$x = \ln(e^x)$, so combine as a quotient $(e^x + 1)/e^x = 1 + e^{-x}$.

分母e^x严格正，日志商法则始终合法。

M1	将x改写为lne^x
M1	商化简
A1	目标式

(b) · 3 marks

$x = -\ln 2$.

1+e^{-x}=3给e^{-x}=2，求x。

M1	使用化简式
M1	指数解
A1	精确根

Question 28

P3-C05-R28 · 6 marks

(a) · 2 marks

3.79.

C5=8e^{-1.5}+2≈3.78504。

M1	正确代值
A1	精度

(b) · 4 marks

$t > \frac{10}{3}\ln 8 \approx 6.93$; boundary $(10/3)\ln 8$.

8e^{−.3t}<1，−.3t<−ln8，除负数方向翻转。

M1	正确严格指数条件
M1	不等号翻转
A1	精确边界
A1	数值

Question 29

P3-C05-R29 · 8 marks

(a) · 3 marks

$k = 0.3054$, $A = 110.5$.

两条件相除$e^{-3k} = 2/5$，所以$k = \frac{1}{3}\ln(5/2)$；再由$A = 60e^{2k} = 110.52\ldots$。

M1	比值消去A
A1	k
A1	A

(b) Using unrounded constants,

(i) · 2 marks

$t = 7.87$ hours.

用$t = 2 + \ln(60/10)/k = 2 + \ln 6/k = 7.866\ldots$，保持起算时刻为$t = 0$。

M1	对数时间方程
A1	绝对时刻

(ii) · 3 marks

$\ln 2/k = 2.27$ hours.

比值$M(t+T)/M(t) = e^{-kT} = 1/2$，所以$T = \ln 2/k$；$A$和起始时间都约去，因此与起始质量无关。

M1	比值及对数
A1	时间
B1	独立性理由

Question 30

P3-C05-R30 · 7 marks

(a) · 3 marks

$p = 3,\ q = 3/2.$

线性化为$\ln(y-2) = \ln p + q\ln x$。截距给$p = 3$；斜率$\ln(24/3)/\ln 4 = 3/2$，所以$y = 2 + 3x^{3/2}$。

M1	对数线性化
A1	p
A1	q

(b) For this model,

(i) · 2 marks

$x = 9.$

$2 + 3x^{3/2} = 83$给$x^{3/2} = 27$，所以$x = 9$。

M1	使用已求模型
A1	x

(ii) · 2 marks

$19.9\%.$

$y(1) = 5$，$y(1.21) = 2 + 3(1.21)^{3/2} = 5.993$。增幅$(5.993-5)/5 \times 100\% = 19.86\%$，即$19.9\%$。常数平移$2$使得$y$的百分比变化不同于$y-2$。

M1	输入倍率的幂
A1	百分比

Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 6 — Differentiation
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第6章 · 微分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C06-R01 · P3 · Chapter 6

The curve is $y = (x^2 + 1)e^{2x}$.

(a) Find dy/dx in factorised form. (3)

(b) Explain why the curve is strictly increasing for all real x. (3)

(Total for Question 1 is 6 marks)

$f(x) = \dfrac{\ln x}{x}$ for $x > 0$.

(a) Find $f'(x)$. (3)

(b) Find and classify its stationary point. (4)

(Total for Question 2 is 7 marks)

3.

P3-C06-R03 · P3 · Chapter 6

$y = \sin(3x^2)$, where x is in radians.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the tangent at the origin. (3)

(Total for Question 3 is 7 marks)

$C : y = \ln(2x + 1)$, where $x > -1/2$.

(a) Find the equation of the normal at $x = 0$. (3)

(b) Find the point on C where the tangent is parallel to $y = x + 3$. (3)

(Total for Question 4 is 6 marks)

$f(x) = x^2 e^{-x}$ for $x \geq 0$.

(a) Find $f'(x)$ in factorised form. (3)

(b) Find the greatest value on $[0, 4]$ and all points where it is attained. (3)

(Total for Question 5 is 6 marks)

$y = \dfrac{x+1}{x^2+1}$.

(a) Find and simplify dy/dx. (3)

(b) Find the exact stationary x-coordinates and classify them. (4)

(Total for Question 6 is 7 marks)

For $0 < x < \pi/2$, $y = \ln(\sin x)$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the tangent at $x = \pi/6$. (3)

(Total for Question 7 is 7 marks)

8.

A curve has $y = (ax + b)e^{x}$. At $(0, 3)$ its tangent has gradient 5.

(a) Find a, b. (3)

(b) Find and classify its stationary point. (4)

(Total for Question 8 is 7 marks)

9. P3-C06-R09 · P3 · Chapter 6

$f(x) = \tan(2x)$ for $-\pi/4 < x < \pi/4$.

(a) Find $f'(x)$ and $f''(x)$. (4)

(b) Find the exact point where the tangent has gradient 8 and $x > 0$. (4)

(Total for Question 9 is 8 marks)

$y = \sqrt{4x^2 + 1}$.

(a) Find dy/dx. (3)

(b) Find the exact gradient of the tangent and the normal at $x = 1$. (3)

(Total for Question 10 is 6 marks)

$f(x) = \dfrac{e^x}{x+1}$, with $x \neq -1$.

(a) Show that $f'(x) = xe^x/(x+1)^2$. (3)

(b) Find and classify every stationary point, taking account of the excluded input. (4)

(Total for Question 11 is 7 marks)

$y = x \ln x - x$ for $x > 0$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Find the stationary point and show it is a minimum. (4)

(Total for Question 12 is 8 marks)

A model is $P(t) = 30te^{-0.5t}$ for $t \geq 0$.

(a) Find the time and value of its maximum. (4)

(b) Find the rate of change at $t = 4$ and state whether P is increasing or decreasing then. (3)

(Total for Question 13 is 7 marks)

$f(x) = \sin x \cos x$, with x in radians.

(a) Find $f'(x)$ using the product rule, and simplify using a double-angle formula. (3)

(b) Find and classify the stationary points on $[0, \pi]$. (4)

(Total for Question 14 is 7 marks)

$C : y = \ln(x^2 - 4)$ with $x > 2$.

(a) Find dy/dx and d^2y/dx^2. (4)

(b) Show that the gradient is positive and decreasing throughout this domain. (3)

(Total for Question 15 is 7 marks)

$y = (3x - 1)^5$.

(a) Find the first two derivatives. (4)

(b) Find the normal at $x = 0$. (3)

(Total for Question 16 is 7 marks)

$f(x) = \dfrac{x^2}{2x+1}$, with $x \neq -1/2$.

(a) Find $f'(x)$ in factorised form. (3)

(b) Find all stationary points and classify them. (4)

(Total for Question 17 is 7 marks)

$C : y = e^{3x} - 6e^{x}$.

(a) Find its stationary point exactly. (4)

(b) Use the second derivative to classify it. (3)

(Total for Question 18 is 7 marks)

$y = \ln\left(\dfrac{x+1}{x-1}\right)$ for $x > 1$.

(a) Find dy/dx in simplest form. (3)

(b) Find the exact point where the tangent has gradient $-1/4$. (3)

(Total for Question 19 is 6 marks)

$f(x) = \cos^2(2x)$ for $0 \le x \le \pi/2$.

(a) Find $f'(x)$ and express it using a single sine function. (3)

(b) Find all points on the interval with horizontal tangents, and state their function values. (3)

(Total for Question 20 is 6 marks)

21.

A curve is $y = (x - a)e^{2x}$ and has a stationary point at $x = 1$.

(a) Find a. (3)

(b) Find the coordinates of the stationary point and prove it is a minimum. (4)

(Total for Question 21 is 7 marks)

$f(x) = \dfrac{\sin x}{1 + \cos x}$ wherever defined.

(a) Show that $f'(x) = 1/(1 + \cos x)$. (3)

(b) Find the tangent at $x = \pi/2$. (3)

(Total for Question 22 is 6 marks)

$C : y = \ln(1 + e^x)$.

(a) Find the first two derivatives. (4)

(b) Find the tangent at $x = 0$ and explain why the gradient is increasing for every real x. (4)

(Total for Question 23 is 8 marks)

$y = xe^{x^2}$.

(a) Find dy/dx in factorised form. (3)

(b) Find d^2y/dx^2 and state all real inputs where this derivative is zero. (4)

(Total for Question 24 is 7 marks)

For $x > 0$, $f(x) = \ln x + 2/x$.

(a) Find and classify the stationary point. (4)

(b) Find the greatest and least values of f on $[1, 4]$. (3)

(Total for Question 25 is 7 marks)

$C : y = \sin x + \cos x$ on $0 \leq x < 2\pi$.

(a) Find its stationary points exactly. (4)

(b) Use the second derivative to classify them and compare with the auxiliary-angle form. (3)

(Total for Question 26 is 7 marks)

For $-\pi/2 < x < \pi/2$, $y = \dfrac{1 + \sin x}{\cos x}$.

(a) Show that $\dfrac{dy}{dx} = \dfrac{1}{1 - \sin x}$. (3)

(b) For this curve, find

(i) the equation of the tangent at $x = 0$. (2)

(ii) the exact coordinates of the point where the gradient is 2. (3)

(Total for Question 27 is 8 marks)

A point P moves on a semicircular track of radius 3 m. Its horizontal coordinate relative to the centre is $x = 3\cos t$ and its height above the diameter is $y = 3\sin t$, where $0 < t < \pi$. Time t is measured in seconds.

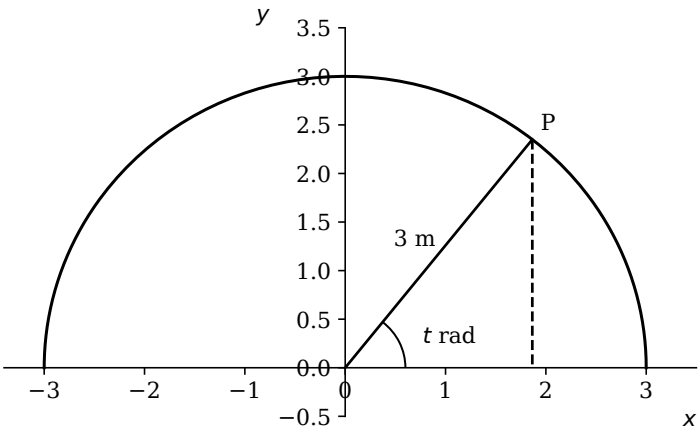


Figure 28

(a) Find dx/dt and dy/dt. (2)

(b) Let $D = x + y$.

(i) find the greatest value of D and the time at which it occurs. (2)

(ii) find the exact rate of change of D at $t = \pi/3$. (2)

(Total for Question 28 is 6 marks)

For $0 \leq x \leq \pi/2$, $f(x) = \sin x \cos x + \sin x$.

(a) Show that $f\prime(x) = 2\cos^2 x + \cos x - 1$. (3)

(b) For this function,

(i) find its greatest value on the stated closed interval. (3)

(ii) find $\int_0^{\pi/2} f(x)dx$ exactly. (2)

(Total for Question 29 is 8 marks)

The curve C has equation $y = xe^{-x}$ for $x \geq 0$. The point P on C has $x = a$, where $a > 0$.

(a) Find an equation of the tangent at P in terms of a. (3)

(b) The tangent meets the y-axis at N. Let O be the origin.

(i) Show that $ON = a^2e^{-a}$. (2)

(ii) Find the greatest possible value of ON, justifying your answer. (3)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C06-R01 · 6 marks

(a) · 3 marks

$2e^{2x}(x^2+x+1)$.

乘积法则给2xe^{2x}+2(x²+1)e^{2x}，提出2e^{2x}。

M1	乘积法则
M1	指数链式因子2
A1	因式形式

(b) · 3 marks

$x^2+x+1=(x+1/2)^2+3/4>0$ and $e^{2x}>0$, so $y'>0$.

配方证明二次因子恒正，结合指数恒正，导数严格正。

M1	二次配方
M1	两因子正性
A1	单调结论

Question 2

P3-C06-R02 · 7 marks

(a) · 3 marks

$(1-\ln x)/x^2$.

商法则[(1/x)x−lnx]/x²；也可乘x^{-1}。

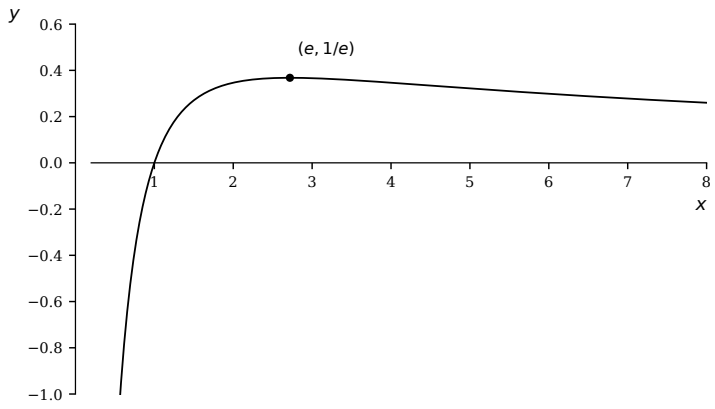
M1	商法则
M1	正确ln导数
A1	简式

(b) · 4 marks

$(e, 1/e)$ is a maximum.

导数分母正，分子1−lnx在e前正、后负，因此由增转减。

M1	驻点方程
M1	正确纵坐标
M1	导数符号改变
A1	分类



Suggested sketch

Question 3

P3-C06-R03 · 7 marks

(a) · 4 marks

$6x\cos(3x^2);\ 6\cos(3x^2) - 36x^2\sin(3x^2).$

一阶用链式，二阶对6x和余弦乘积求导，内层导数6x再次出现。

M1	一阶链式
A1	一阶导
M1	二阶乘积链式
A1	二阶导

(b) · 3 marks

$y = 0.$

原函数在0为0，导式在0也为0，所以切线为x轴。

B1	点
M1	梯度代值
A1	切线

Question 4

P3-C06-R04 · 6 marks

(a) · 3 marks

$y = -x/2$.

原点在曲线上，切线斜率2/(2x+1)在0为2，法线−1/2。

M1	链式对数导数
M1	法线负倒数
A1	方程

(b) · 3 marks

$(1/2, \ln 2)$.

令2/(2x+1)=1得x1/2，回代。

M1	梯度条件
M1	域及回代
A1	点

Question 5

P3-C06-R05 · 6 marks

(a) · 3 marks

$e^{-x}x(2-x)$.

乘积法则，e^{-x}导数带负号，提出e^{-x}x。

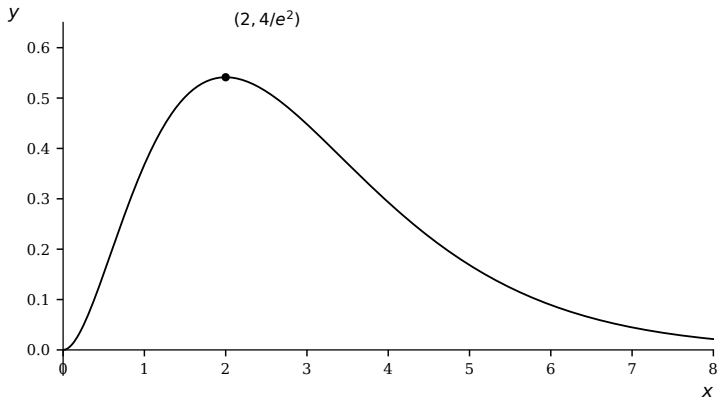
M1	乘积法则
M1	负链式因子
A1	因式式

(b) · 3 marks

$4e^{-2}$, only at $x=2$.

导式在0至2正、2至4负；比较端点0与16e^{-4}，驻点最大。

M1	候选点和端点
M1	符号或比较
A1	最大值和唯一位置



Suggested sketch

Question 6

P3-C06-R06 · 7 marks

(a) · 3 marks

$(-x^2-2x+1)/(x^2+1)^2$.

商法则分子$(x^2+1)-2x(x+1)$，分母平方始终正。

M1	正确商法则
M1	分子化简
A1	导式

(b) · 4 marks

$x=-1-\sqrt{2}$ minimum; $x=-1+\sqrt{2}$ maximum.

分子$2-(x+1)^2$，在两根间正，外侧负，所以左根由负转正，右根由正转负。

M1	两根
M1	导数符号
M1	类型变化
A1	分类

Question 7

P3-C06-R07 · 7 marks

(a) · 4 marks

$\cot x;\ -\operatorname{cosec}^2 x.$

ln链式导数cos/sin=cot，再用余切标准导数或商法则。

M1	对数链式
A1	一阶
M1	二阶求导
A1	二阶

(b) · 3 marks

$y=\sqrt{3}(x-\pi/6)-\ln 2.$

y=ln(1/2)=−ln2，梯度cotπ/6=√3。

B1	纵坐标
M1	梯度
A1	切线

Question 8

P3-C06-R08 · 7 marks

(a) · 3 marks

$b = 3, a = 2.$

点给b3；y'=(ax+a+b)e^x，在0给a+b5。

B1	b
M1	乘积求导与梯度条件
A1	a

(b) · 4 marks

$(-5/2, -2e^{-5/2})$ is a minimum.

导式(2x+5)e^x，零点−5/2，分子线性由负转正；原线性因子2x+3在此为−2。

M1	导数零点
M1	正确纵坐标
M1	符号变化
A1	分类

Question 9

P3-C06-R09 · 8 marks

(a) · 4 marks

$2\sec^2(2x)$; $8\sec^2(2x)\tan(2x)$.

一阶有输入导数2；二阶对$\sec^2$复合求导再乘2。

M1	一阶链式
A1	一阶
M1	二阶链式
A1	二阶

(b) · 4 marks

$(\pi/6, \sqrt{3})$.

$2\sec^2 2x=8$给$\cos^2 2x\,1/4$；输入$2x\in(0,\pi/2)$，取$\cos 1/2$即$2x\pi/3$。

M1	梯度方程
M1	输入范围选支
M1	原函数回代
A1	点

Question 10

P3-C06-R10 · 6 marks

(a) · 3 marks

$4x/\sqrt{4x^2+1}$.

写成(4x^2+1)^{1/2}，外导1/2乘内导8x。

M1	外幂导数
M1	内导链式
A1	简式

(b) · 3 marks

Tangent $4/\sqrt{5}$; normal $-\sqrt{5}/4$.

代导式x1，再取负倒数。

M1	导数代值
M1	法线负倒数
A1	两比

Question 11

P3-C06-R11 · 7 marks

(a) · 3 marks

Quotient numerator $e^x(x+1) - e^x = xe^x$.

商法则后公因子提取，分母必须保留平方。

M1	正确商法则
M1	分子化简
A1	目标式

(b) · 4 marks

Only $(0, 1)$, a minimum on the branch $x > -1$.

指数不为零，分母正，导数符号由x；在0负转正。x=−1未定义，不能算驻点或连接两支比较全局。

M1	找唯一导数零点
M1	域分支说明
M1	符号改变
A1	分类

Question 12

P3-C06-R12 · 8 marks

(a) · 4 marks

$\ln x$; $1/x$.

xlnx乘积导数lnx+1，减去x导数1后抵消。

M1	乘积导数
A1	一阶
M1	再求导
A1	二阶

(b) · 4 marks

$(1,-1)$; $y''(1)=1>0$.

lnx0给x1，回代−1；二阶正。

M1	解驻点
M1	纵坐标
M1	二阶判别
A1	点与类型

Question 13

P3-C06-R13 · 7 marks

(a) · 4 marks

$t = 2$; maximum $60/e$.

P'=30e^{−.5t}(1−.5t)，在2由正转负，代P2。

M1	乘积链式
M1	驻点
M1	最大性
A1	时间和值

(b) · 3 marks

$P'(4) = -30e^{-2}$; decreasing.

代导式得负值，负号说明此时下降。

M1	正确导式代值
A1	精确变化率
B1	方向解释

Question 14

P3-C06-R14 · 7 marks

(a) · 3 marks

$\cos^2 x - \sin^2 x = \cos 2x$.

分别求导sin与cos，第二项带负；再用双角恒等式。

M1	乘积法则
M1	正确负号
A1	双角式

(b) · 4 marks

$(\pi/4, 1/2)$ maximum; $(3\pi/4, -1/2)$ minimum.

cos2x0得两根，f″=−2sin2x在两处负、正。

M1	区间内导数零点
M1	纵坐标
M1	二阶符号
A1	分类

Question 15

P3-C06-R15 · 7 marks

(a) · 4 marks

$2x/(x^2-4)$; $-2(x^2+4)/(x^2-4)^2$.

一阶ln链式，二阶用商法则，分子$2(x^2-4)-4x^2$。

M1	一阶链式
A1	一阶
M1	二阶商法则
A1	简式

(b) · 3 marks

$y'>0$ and $y''<0$ for $x>2$.

一阶分子分母都正；二阶分子严格负、分母正，分别说明曲线上升且其斜率下降。

M1	一阶符号
M1	二阶符号
A1	区分两结论

Question 16

P3-C06-R16 · 7 marks

(a) · 4 marks

$15(3x-1)^4$; $180(3x-1)^3$.

两次外幂求导都乘输入导数3，第二次15×4×3=180。

M1	一阶链式
A1	一阶
M1	二阶链式
A1	二阶

(b) · 3 marks

$y=-x/15-1$.

点(0,−1)，切线梯度15，法线−1/15。

B1	点
M1	法线斜率
A1	方程

Question 17

P3-C06-R17 · 7 marks

(a) · 3 marks

$2x(x+1)/(2x+1)^2$.

商法则分子2x(2x+1)−2x²=2x(x+1)。

M1	商法则和内导2
M1	因式分解
A1	导式

(b) · 4 marks

$(-1,-1)$ maximum; $(0,0)$ minimum.

导数分母正，x(x+1)在−1两侧正转负，在0负转正；注意−1/2将域分开。

M1	两导数零点
M1	纵坐标
M1	各分支符号
A1	分类

Question 18

P3-C06-R18 · 7 marks

(a) · 4 marks

$\left(\frac{1}{2}\ln 2, -4\sqrt{2}\right)$.

y'=3e^x(e^{2x}−2)，零点e^{2x}=2，原式e^x(2−6)。

M1	因式化导数
M1	对数求x
M1	精确y
A1	点

(b) · 3 marks

Minimum, since $y'' = 9e^{3x} - 6e^{x}$ has value $12\sqrt{2} > 0$ there.

e^{3x}=2√2，代9×2√2−6√2=12√2。

M1	正确二阶导
M1	精确代值
A1	最小性

Question 19

P3-C06-R19 · 6 marks

(a) · 3 marks

$-2/(x^2-1)$.

先写ln(x+1)−ln(x−1)，导数1/(x+1)−1/(x−1)再通分。

M1	合法拆对数
M1	两项求导
A1	简式

(b) · 3 marks

$(3, \ln 2)$.

−2/(x²−1)=−1/4给x²9，域x>1只取3；回代ln(4/2)。

M1	梯度方程
M1	域筛选
A1	点

Question 20

P3-C06-R20 · 6 marks

(a) · 3 marks

$-2\sin 4x$.

链式给−4cos2xsin2x，用sin4x=2sin2xcos2x。

M1	外平方与内角链式
M1	双角化简
A1	导式

(b) · 3 marks

$x = 0, \pi/4, \pi/2$; values $1, 0, 1$.

4x在0到2π，sin4x0的输入0、π、2π；端点的导函数也为0。

M1	输入范围零点
M1	包括端点
A1	三x及值

Question 21

P3-C06-R21 · 7 marks

(a) · 3 marks

$a = 3/2$.

y'=e^{2x}[1+2(x−a)]，在1零给3−2a=0。

M1	乘积链式
M1	驻点条件
A1	a

(b) · 4 marks

$(1, -e^2/2)$; the derivative factor is $2(x-1)$ and changes from negative to positive.

代a得原函数(1−1.5)e^2；指数正，导数由线性因子定号。

M1	纵坐标
M1	因式符号
M1	负转正
A1	类型

Question 22

P3-C06-R22 · 6 marks

(a) · 3 marks

Quotient numerator $\cos x(1+\cos x)+\sin^2 x = 1+\cos x$.

分母导数为$-\sin$，商法则中的相减再变加；用平方和约分。

M1	正确商法则符号
M1	平方和化简
A1	导式

(b) · 3 marks

$y = x + 1 - \pi/2$.

点$(\pi/2, 1)$，导数1，点斜式。

B1	点
M1	梯度
A1	方程

Question 23

P3-C06-R23 · 8 marks

(a) · 4 marks

$e^x/(1+e^x)$; $e^x/(1+e^x)^2$.

一阶对数链式，二阶商法则两e^{2x}项抵消。

M1	一阶链式
A1	一阶
M1	二阶商法则
A1	二阶

(b) · 4 marks

$y = x/2 + \ln 2$; second derivative is strictly positive.

点(0,ln2)，一阶1/2；二阶指数正、分母正，梯度递增。

M1	点与斜率
A1	方程
M1	二阶符号
A1	梯度说明

Question 24

P3-C06-R24 · 7 marks

(a) · 3 marks

$e^{x^2}(1+2x^2)$.

乘积导数e^{x^2}+x e^{x^2}2x，提出指数。

M1	乘积法则
M1	内层2x
A1	因式形式

(b) · 4 marks

$2xe^{x^2}(3+2x^2)$; only $x=0$.

再乘积求导，合并为e^{x^2}(6x+4x^3)；指数和3+2x^2恒正，所以仅x0。

M1	二阶乘积链式
M1	因式整理
M1	正因子判零
A1	导式及根

Question 25

P3-C06-R25 · 7 marks

(a) · 4 marks

$(2, \ln 2 + 1)$ minimum.

f′=(x−2)/x²，在2负转正，原值ln2+1。

M1	正确导数
M1	零点与原值
M1	符号变化
A1	类型

(b) · 3 marks

Least $1 + \ln 2$ at 2; greatest 2 at 1.

f1=2，f4=ln4+1/2≈1.8863，比较端点和驻点。

M1	两端点数值
M1	与驻点比较
A1	两全局极值

Question 26

P3-C06-R26 · 7 marks

(a) · 4 marks

$(\pi/4, \sqrt{2})$ and $(5\pi/4, -\sqrt{2})$.

y'=cosx−sinx=0，tan1但需cos非零，得到两角并代原式。

M1	正确导数
M1	全部角
M1	精确y
A1	两点

(b) · 3 marks

Maximum then minimum; $y'' = -y$ and $y = \sqrt{2}\sin(x + \pi/4)$.

二阶在两个点分别−√2和√2；辅助角幅值√2，与最高最低一致。

M1	二阶符号分类
M1	辅助角式
A1	一致说明

Question 27

P3-C06-R27 · 8 marks

(a) · 3 marks

$y\prime = 1/(1-\sin x)$.

商法则分子为$\cos^2 x + (1+\sin x)\sin x = 1+\sin x$。再由$\cos^2 x = (1-\sin x)(1+\sin x)$约分；该开区间内所除因子非零。

M1	正确商法则
M1	平方恒等式化简
A1	目标式

(b) For this curve, find

(i) · 2 marks

$y = x + 1$.

$y(0)=1$，$y\prime(0)=1$，所以切线$y-1=x$。

M1	点与梯度
A1	方程

(ii) · 3 marks

$(\pi/6, \sqrt{3})$.

$1/(1-\sin x)=2$给$\sin x = 1/2$；规定开区间内唯一解$x=\pi/6$，代入$y=(3/2)/(\sqrt{3}/2)=\sqrt{3}$。

M1	导数方程及区间
A1	横坐标
A1	纵坐标

Question 28

P3-C06-R28 · 6 marks

(a) · 2 marks

$dx/dt = -3\sin t$, $dy/dt = 3\cos t$.

分别对给定的两个时间函数求导。角量以弧度计。

B1	水平变化率
B1	竖直变化率

(b) Let $D = x + y$.

(i) · 2 marks

$3\sqrt{2}$ m at $t = \pi/4$ s.

$D = 3(\cos t + \sin t) = 3\sqrt{2}\sin(t + \pi/4)$；区间内最大值在$t = \pi/4$。也可求$D\prime = 3(\cos t - \sin t)$并检查两侧符号。

M1	辅助角或导数条件
A1	合法最大值与时刻

(ii) · 2 marks

$\frac{3}{2}(1 - \sqrt{3})\ \mathrm{m\,s^{-1}}$.

$D\prime = 3(\cos t - \sin t)$，代入$t = \pi/3$得$3(1/2 - \sqrt{3}/2)$。

M1	对和求导并代值
A1	精确变化率

Question 29

P3-C06-R29 · 8 marks

(a) · 3 marks

$f\prime(x) = 2\cos^2 x + \cos x - 1.$

乘积法则给$\cos^2 x - \sin^2 x + \cos x$，再用$\sin^2 x = 1 - \cos^2 x$化成所需形式。

M1	正确乘积法则
M1	三角恒等式
A1	导数

(b) For this function,

(i) · 3 marks

$3\sqrt{3}/4$ at $x = \pi/3$.

$f\prime = (2\cos x - 1)(\cos x + 1)$。区间内唯一驻点$\cos x = 1/2$即$x = \pi/3$；函数先增后减，且端点值$0, 1$均小于$3\sqrt{3}/4$。

M1	因式分解与合法驻点
M1	比较端点或符号
A1	最大值

(ii) · 2 marks

$3/2$.

$\sin x \cos x = \frac{1}{2}\sin 2x$；积分为$[-\frac{1}{4}\cos 2x - \cos x]_0^{\pi/2} = 3/2$。只使用标准三角积分与恒等式。

M1	用二倍角识别标准积分
A1	精确值

Question 30

P3-C06-R30 · 8 marks

(a) · 3 marks

$y - ae^{-a} = e^{-a}(1-a)(x-a)$.

乘积法则给$y\prime = e^{-x}(1-x)$；以$P=(a, ae^{-a})$写点斜式。

M1	正确乘积法则
A1	梯度
A1	参数切线

(b) The tangent meets the y-axis at N. Let O be the origin.

(i) · 2 marks

$ON = a^2e^{-a}$.

在切线方程中令$x=0$，得$y_N = ae^{-a} - ae^{-a}(1-a) = a^2e^{-a} > 0$，所以该纵坐标就是长度$ON$。

M1	求纵截距
A1	正确正长度

(ii) · 3 marks

$4e^{-2}$ at $a=2$.

对a求导，$d(ON)/da = e^{-a}(2a-a^2) = ae^{-a}(2-a)$。$a>0$时在$2$左侧为正、右侧为负，故全局最大值在$a=2$，值为$4e^{-2}$。

M1	对截距函数求导
M1	两侧符号证明最大
A1	最大值

Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 7 — Integration
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第7章 · 积分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

For $x > 0$,

(a) find $\int \left(3e^{2x} - \frac{4}{x}\right) dx$, (4)

(b) hence evaluate $\int_1^2 \left(3e^{2x} - \frac{4}{x}\right) dx$ exactly. (3)

(Total for Question 1 is 7 marks)

2. P3-C07-R02 · P3 · Chapter 7

The curve C has gradient $dy/dx = 6\cos 3x - 2\sin x$ and passes through $(0, 5)$.

(a) Find the equation of C. (3)

(b) Find the exact value of y at $x = \pi/6$. (2)

(Total for Question 2 is 5 marks)

For $x > -1/2$,

(a) find $\displaystyle\int \frac{6}{2x+1}\,dx$, (4)

(b) evaluate $\displaystyle\int_0^1 \frac{6}{2x+1}\,dx$ exactly. (2)

(Total for Question 3 is 6 marks)

4.

P3-C07-R04 · P3 · Chapter 7

For real x, let $F(x) = (x^2 + 1)^4$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_0^1 x(x^2 + 1)^3 \, dx$ exactly. (3)

(Total for Question 4 is 5 marks)

R is bounded by $y = e^x$, the x-axis and the lines $x = 0, x = \ln 3$.

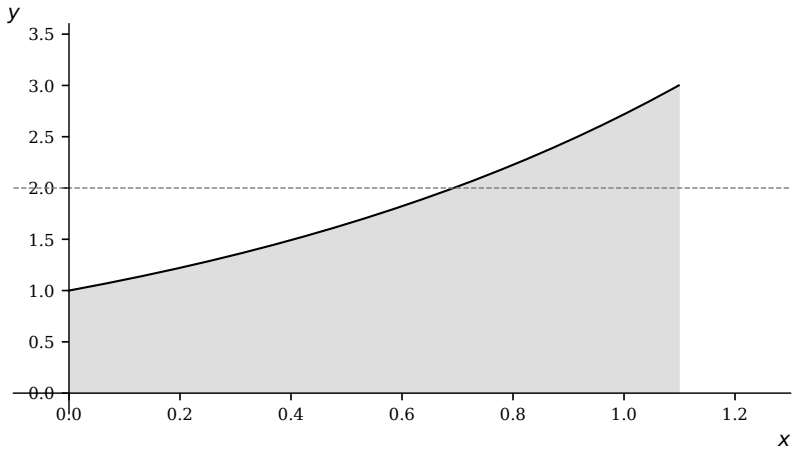


Figure 5

(a) Find its exact area. (3)

(b) The horizontal line $y = 2$ is drawn.

(i) find its intersection with $y = e^x$, (2)

(ii) find the exact area of the part of R above $y = 2$. (3)

(Total for Question 5 is 8 marks)

Find each indefinite integral.

(a) $\int \cos(5x - 2)\, dx.$ (4)

(b) $\int \sec^2(3x)\, dx.$ (4)

(Total for Question 6 is 8 marks)

For $x \geq 0$, $f(x) = \ln(x^2 + 4)$.

(a) Find $f'(x)$. (2)

(b) Hence find $\displaystyle\int_0^2 \frac{3x}{x^2 + 4}\,dx$ exactly. (3)

(Total for Question 7 is 5 marks)

The curve $C : y = \sin x$ on $[0, \pi]$ is cut by $y = 1/2$.

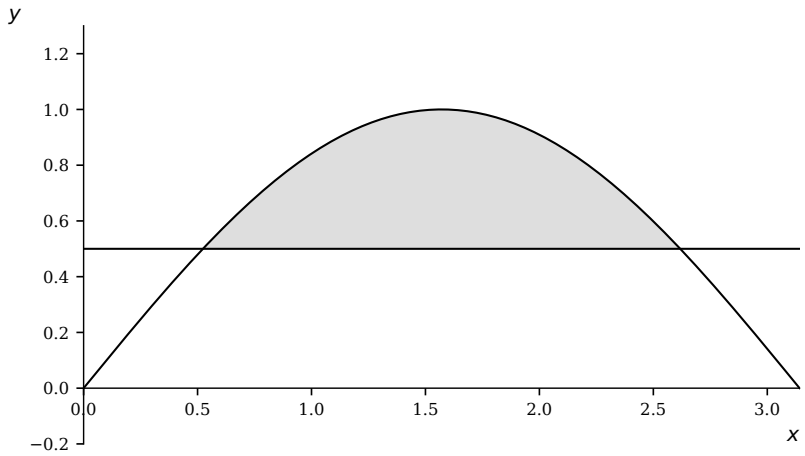


Figure 8

(a) Find the two intersection points. (2)

(b) Find the exact area of the region enclosed by the curve and the line $y = 1/2$. (4)

(Total for Question 8 is 6 marks)

For $x > 0$,

(a) find $\int \left(\frac{x^2 + 3x + 2}{x} \right) dx$, (4)

(b) evaluate this integral between $x = 1$ and $x = 3$. (3)

(Total for Question 9 is 7 marks)

$F(x) = e^{x^2}$.

(a) Find $F'(x)$. (2)

(b) Hence find $\int_0^{\sqrt{\ln 3}} 4xe^{x^2}\,dx$ exactly. (3)

(Total for Question 10 is 5 marks)

For $0 < x < \pi$,

(a) show that $\dfrac{d}{dx}\ln(\sin x) = \cot x$, (3)

(b) hence evaluate $\displaystyle\int_{\pi/6}^{\pi/2} \cot x \, dx$. (3)

(Total for Question 11 is 6 marks)

12. P3-C07-R12 · P3 · Chapter 7

The curve C satisfies $dy/dx = 4e^{-2x} + 3$ and passes through $(0, 1)$.

(a) Find the equation of C. (3)

(b) Find the exact y-coordinate when $x = \ln 2$. (3)

(Total for Question 12 is 6 marks)

Use double-angle formulae to evaluate exactly:

(a) $\displaystyle\int_0^{\pi/2} \sin^2 x \, dx,$ (4)

(b) $\displaystyle\int_0^{\pi/4} \cos^2 x \, dx.$ (4)

(Total for Question 13 is 8 marks)

Let $F(x) = \sin^3 x$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_0^{\pi/2} \sin^2 x \cos x \, dx$. (3)

(Total for Question 14 is 5 marks)

The line $y = 3$ meets $C : y = e^{2x} + 1$ at P. Region R lies between the line, curve and the y-axis.

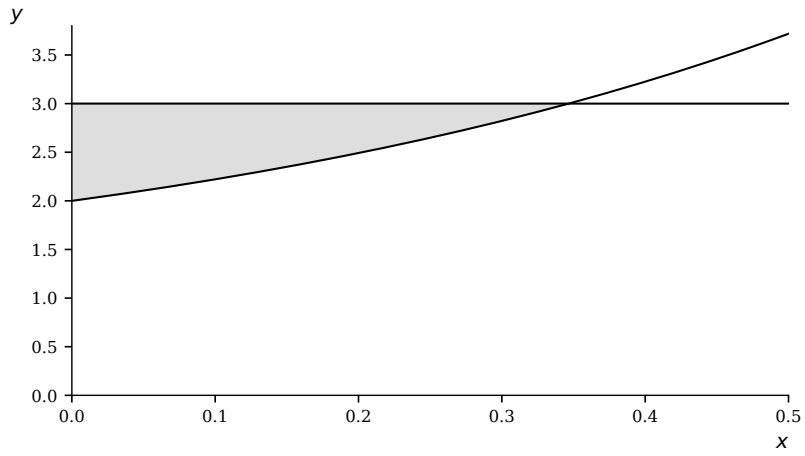


Figure 15

(a) Find P exactly. (3)

(b) Find the exact area of R. (4)

(Total for Question 15 is 7 marks)

For $x > -1$,

(a) find $\int \left(\frac{2}{x+1} + 3(x+1)^2 \right) dx$, (4)

(b) evaluate the integral from 0 to 1 exactly. (2)

(Total for Question 16 is 6 marks)

$F(x) = \dfrac{1}{2x+3}$ for $x > -3/2$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_0^1 \frac{5}{(2x+3)^2}\,dx$. (3)

(Total for Question 17 is 5 marks)

The gradient of a curve is $dy/dx = 2\sec^2 x - \sin x$ on $-\pi/2 < x < \pi/2$. The curve passes through $(0, 4)$.

(a) Find its equation. (3)

(b) Find the exact value of y at $x = \pi/3$. (3)

(Total for Question 18 is 6 marks)

For real x, $F(x) = \ln(1 + e^{2x})$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_0^{\ln 2} \frac{e^{2x}}{1 + e^{2x}}\,dx$ exactly. (3)

(Total for Question 19 is 5 marks)

The graph of $f(x) = \cos x - \sin x$ on $[0, \pi/2]$ crosses the x-axis once.

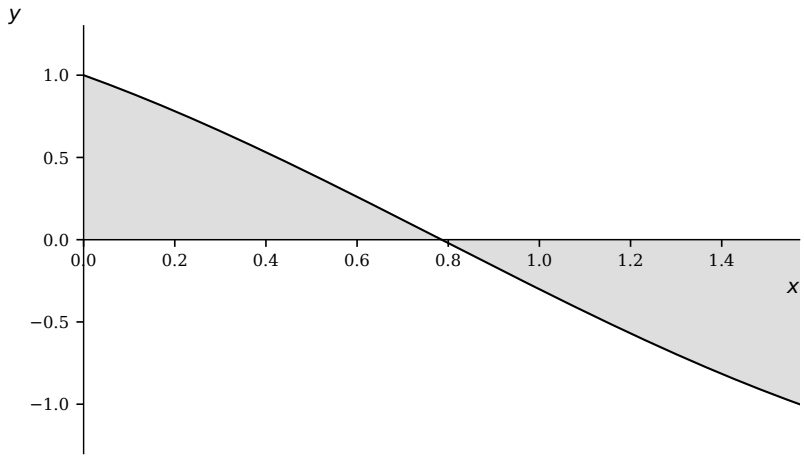


Figure 20

(a) Find the zero exactly. (2)

(b) Find the total positive area between the curve and the x-axis. (4)

(Total for Question 20 is 6 marks)

21.

For $x > 0$, $F(x) = x^2 \ln x - \frac{1}{2}x^2$.

(a) Show that $F'(x) = 2x \ln x$. (3)

(b) Hence evaluate $\displaystyle\int_1^e x \ln x \, dx$. (3)

(Total for Question 21 is 6 marks)

A rate is modelled by $R(t) = 6e^{-0.5t}$ units per minute for $t \geq 0$. The total amount accumulated by time t is defined as $\int_0^t R(s)\,ds$.

(a) Find the exact accumulated amount at time t. (3)

(b) Find the time when 9 units have accumulated. (3)

(Total for Question 22 is 6 marks)

Use the identity $\cos 2x = 2\cos^2 x - 1$.

(a) Find $\displaystyle\int (4\cos^2 x - 1)\,dx$. (4)

(b) Find the exact integral from 0 to $\pi/3$. (2)

(Total for Question 23 is 6 marks)

$f(x) = (3x + 2)^4$.

(a) Find an antiderivative of f. (3)

(b) Find $a > 0$ such that $\displaystyle\int_0^a f(x)\,dx = 992/15$. (4)

(Total for Question 24 is 7 marks)

Let $F(x) = \cos(\ln x)$ for $x > 0$.

(a) Find $F'(x)$. (2)

(b) Hence evaluate $\displaystyle\int_1^{e^{\pi/2}} \frac{\sin(\ln x)}{x}\,dx$. (3)

(Total for Question 25 is 5 marks)

The curve $C : y = e^{x} - e^{-x}$ is considered on $[-\ln 2, \ln 2]$.

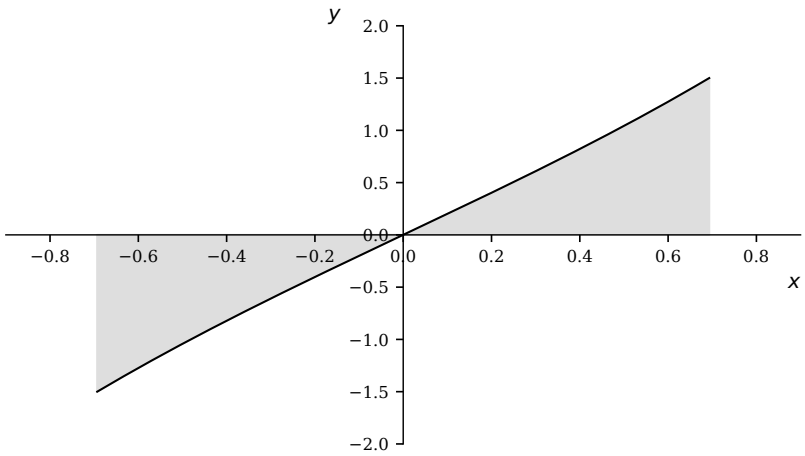


Figure 26

(a) Find its zero and show that its signed integral on this interval is zero. (4)

(b) Find the total area between the curve and the axis on this interval. (4)

(Total for Question 26 is 8 marks)

27. P3-C07-R27 · P3 · Chapter 7

The curve has gradient $dy/dx = 5/(5x + 1)$ for $x > -1/5$, and passes through $(0, 2)$.

(a) Find its equation. (3)

(b) Find where this curve crosses the x-axis. (3)

(Total for Question 27 is 6 marks)

For $0 < x < \pi/2$, $F(x) = \ln(\tan x)$.

(a) Show that $F'(x) = 1/(\sin x \cos x)$. (3)

(b) Hence evaluate $\displaystyle\int_{\pi/6}^{\pi/4} \frac{1}{\sin x \cos x}\, dx$. (3)

(Total for Question 28 is 6 marks)

A function F is given by $F(x) = (x - 2)e^{2x}$.

(a) Show that $F\prime(x) = (2x - 3)e^{2x}$. (2)

(b) Hence, using your answer to (a),

(i) find $\int_0^1 (2x - 3)e^{2x}dx$ exactly. (2)

(ii) find the positive area bounded by $y = (2x - 3)e^{2x}$, the x-axis and the lines $x = 0, x = 1$. (2)

(Total for Question 29 is 6 marks)

For $0 \leq x \leq \pi/4$, the curve C has equation $y = \sec^2 x + 2\cos 2x$.

(a) Find the exact area between C, the x-axis and the lines $x = 0, x = \pi/4$. (3)

(b) The line $y = k$ is drawn across the same vertical strip.

(i) Find k if the signed integral $\int_0^{\pi/4}(y-k)dx$ is zero. (2)

(ii) Explain why this does not mean that the curve is the horizontal line $y = k$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C07-R01 · 7 marks

(a) · 4 marks

$\frac{3}{2}e^{2x} - 4\ln x + C.$

指数积分除内层常数2；1/x积分为lnx，原域x正。

M1	指数积分
M1	对数积分
A1	两项
B1	常数

(b) · 3 marks

$\frac{3}{2}(e^4 - e^2) - 4\ln 2.$

使用(a)原函数在上限减下限，ln1=0。

M1	正确代上下限
M1	对数化简
A1	精确值

Question 2

P3-C07-R02 · 5 marks

(a) · 3 marks

$y = 2\sin 3x + 2\cos x + 3.$

积分cos3x除3，积分−2sinx为2cosx；点条件给常数3。

M1	三角积分及内层因子
M1	点条件
A1	常数与曲线

(b) · 2 marks

$5 + \sqrt{3}.$

2sinπ/2=2，2cosπ/6=√3，另加3。

M1	正确输入精确值
A1	y

Question 3

P3-C07-R03 · 6 marks

(a) · 4 marks

$3\ln(2x+1)+C$.

对数真数导数2，外系数需为6/2=3，域内真数正。

M1	识别对数导数
M1	内层因子调整
A1	原函数
B1	常数

(b) · 2 marks

$3\ln 3$.

代上下限$3\ln 3-3\ln 1$。

M1	正确上下限
A1	值

Question 4

P3-C07-R04 · 5 marks

(a) · 2 marks

$8x(x^2+1)^3$.

外幂4乘内导2x。

M1	链式
A1	导式

(b) · 3 marks

$15/8$.

被积式为F′/8，原函数F/8，上下限(16−1)/8。

M1	导数识别及比例
M1	上下限
A1	精确值

Question 5

P3-C07-R05 · 8 marks

(a) · 3 marks

2.

e^x恒正，积分[e^x]0至ln3=3−1。

M1	正确积分与界
M1	原函数代值
A1	面积

(b) The horizontal line $y = 2$ is drawn.

(i) · 2 marks

$(\ln 2, 2)$.

e^x2，取ln。

M1	对数反解
A1	点

(ii) · 3 marks

$1 - 2\ln(3/2)$.

只在ln2到ln3，积分(e^x−2)，结果(3−2ln3)−(2−2ln2)。

M1	正确上下界和上减下
M1	积分
A1	精确面积

Question 6

P3-C07-R06 · 8 marks

(a) · 4 marks

$\frac{1}{5}\sin(5x-2)+C$.

正弦导数给余弦，再除输入导数5。

M1	标准三角积分
M1	系数1/5
A1	原函数
B1	常数

(b) · 4 marks

$\frac{1}{3}\tan 3x+C$.

tan3x导数3sec²3x，因此积分除3，取同一连续定义区间。

M1	识别正切导数
M1	输入因子
A1	原函数
B1	常数

Question 7

P3-C07-R07 · 5 marks

(a) · 2 marks

$2x/(x^2+4)$.

对数链式，分子为内部二次式导数。

M1	对数链式
A1	导式

(b) · 3 marks

$\frac{3}{2}\ln 2$.

原函数$(3/2)\ln(x^2+4)$，上下限给$(3/2)\ln(8/4)$。

M1	正确识别比例
M1	上下限及对数商
A1	值

Question 8

P3-C07-R08 · 6 marks

(a) · 2 marks

$(\pi/6, 1/2), (5\pi/6, 1/2).$

sin在半圈取1/2于参考角π/6及补角。

M1	合法角范围
A1	两点

(b) · 4 marks

$\sqrt{3} - \pi/3.$

积分π/6至5π/6的(sinx−1/2)，原函数−cosx−x/2。

M1	正确界和差
M1	原函数
M1	精确代值
A1	面积

Question 9

P3-C07-R09 · 7 marks

(a) · 4 marks

$x^2/2 + 3x + 2\ln x + C$.

先逐项除x得x+3+2/x，最后一项用对数积分。

M1	正确化简
M1	混合积分
A1	三项
B1	常数

(b) · 3 marks

$10 + 2\ln 3$.

多项式部分差(9−1)/2+3(3−1)=10，另加2ln3。

M1	上下限
M1	对数项
A1	值

Question 10

P3-C07-R10 · 5 marks

(a) · 2 marks

$2xe^{x^2}$.

指数链式内导2x。

M1	链式
A1	导式

(b) · 3 marks

4.

原函数2e^{x^2}，上端x^2ln3给6，下端给2。

M1	识别比例
M1	正确根式上限
A1	精确值

Question 11

P3-C07-R11 · 6 marks

(a) · 3 marks

Derivative $\cos x / \sin x = \cot x$.

域内sin正，ln合法；链式化为余切。

M1	对数链式
M1	倒数比化简
A1	目标式

(b) · 3 marks

$\ln 2$.

[lnsinx]上下限=ln1−ln(1/2)。

M1	识别原函数
M1	精确三角值代限
A1	值

Question 12

P3-C07-R12 · 6 marks

(a) · 3 marks

$y = -2e^{-2x} + 3x + 3$.

指数积分除−2；初值−2+C=1给C3。

M1	负指数积分
M1	积分常数条件
A1	方程

(b) · 3 marks

$5/2 + 3\ln 2$.

e^{-2ln2}=1/4，故−1/2+3ln2+3。

M1	指数对数还原
M1	代完整曲线
A1	值

Question 13

P3-C07-R13 · 8 marks

(a) · 4 marks

$\pi/4$.

$\sin^2=(1-\cos2x)/2$，原函数$x/2-\sin2x/4$。

M1	正确半角平方重写
M1	标准积分
M1	上下限
A1	值

(b) · 4 marks

$\pi/8+1/4$.

$\cos^2=(1+\cos2x)/2$，原函数$x/2+\sin2x/4$。

M1	双角重写
M1	积分
M1	精确代限
A1	值

Question 14

P3-C07-R14 · 5 marks

(a) · 2 marks

$3\sin^2 x\cos x$.

三次外幂乘sin的导数cos。

M1	链式
A1	导式

(b) · 3 marks

$1/3$.

原函数$\sin^3 x/3$，上端1下端0。

M1	识别导数比例
M1	上下限
A1	值

Question 15

P3-C07-R15 · 7 marks

(a) · 3 marks

$P = (\frac{1}{2}\ln 2, 3)$.

e^{2x}=2，取ln后除2。

M1	指数方程
M1	自然对数
A1	点

(b) · 4 marks

$\ln 2 - 1/2$.

在0至ln2/2，直线在上，积分2−e^{2x}，原函数2x−e^{2x}/2。

M1	正确差和界
M1	指数积分因子
M1	上下限
A1	面积

Question 16

P3-C07-R16 · 6 marks

(a) · 4 marks

$2\ln(x+1)+(x+1)^3+C$.

两项分别识别ln与三次幂导数；输入导数1。

M1	对数积分
M1	幂识别
A1	两项
B1	常数

(b) · 2 marks

$7+2\ln 2$.

原函数差：2ln2+8−1。

M1	正确代限
A1	精确值

Question 17

P3-C07-R17 · 5 marks

(a) · 2 marks

$-2/(2x+3)^2$.

写负一次幂，乘内导2。

M1	负幂链式
A1	导式

(b) · 3 marks

$1/3$.

原函数−5/[2(2x+3)]，上下限−1/2+5/6=1/3。

M1	识别比例与负号
M1	上下限
A1	值

Question 18

P3-C07-R18 · 6 marks

(a) · 3 marks

$y = 2\tan x + \cos x + 3$.

积分$\sec^2$得tan，−sin得cos，初值给C3。

M1	两项标准积分
M1	初值
A1	曲线

(b) · 3 marks

$7/2 + 2\sqrt{3}$.

域内π/3合法；tan √3、cos1/2，另加3。

M1	精确三角值
M1	完整式代入
A1	y

Question 19

P3-C07-R19 · 5 marks

(a) · 2 marks

$2e^{2x}/(1+e^{2x})$.

对数链式内导2e^{2x}。

M1	链式
A1	导式

(b) · 3 marks

$\frac{1}{2}\ln(5/2)$.

原函数F/2；上端e^{2ln2}=4，差半ln5−半ln2。

M1	识别半倍原函数
M1	指数代限
A1	对数精确值

Question 20

P3-C07-R20 · 6 marks

(a) · 2 marks

$x = \pi/4$.

cos=sin，两者均非负，锐域唯一tan1。

M1	正确方程与域
A1	零点

(b) · 4 marks

$2(\sqrt{2}-1)$.

在0至π/4积分f，在π/4至π/2积分−f，原函数sinx+cosx；两块均√2−1。

M1	在零点分段
M1	正负面积正确
M1	原函数代限
A1	总面积

Question 21

P3-C07-R21 · 6 marks

(a) · 3 marks

Product derivative $2x\ln x + x - x = 2x\ln x$.

乘积法则产生的x与后项导数x抵消。

M1	乘积求导
M1	正确抵消
A1	导式

(b) · 3 marks

$(e^2+1)/4$.

使用题目已给原函数的一半；F(e)=$e^2/2$，F1=$-1/2$。

M1	识别一半F
M1	两限值
A1	精确值

Question 22

P3-C07-R22 · 6 marks

(a) · 3 marks

$12(1-e^{-t/2})$.

原函数−12e^{-s/2}，代t和0。

M1	指数积分含负因子
M1	上下限
A1	累积式

(b) · 3 marks

$t=2\ln 4$.

12(1−e^{-t/2})=9给指数1/4，取ln求时间。

M1	累积阈值方程
M1	对数解
A1	精确时间

Question 23

P3-C07-R23 · 6 marks

(a) · 4 marks

$x + \sin 2x + C$.

被积式=1+2cos2x，积分为x+sin2x。

M1	正确双角重写
M1	标准积分
A1	两项
B1	常数

(b) · 2 marks

$\pi/3 + \sqrt{3}/2$.

代原函数，sin2π/3=√3/2。

M1	精确代限
A1	值

Question 24

P3-C07-R24 · 7 marks

(a) · 3 marks

$(3x+2)^5/15$.

五次幂求导带15倍，所以除15。

M1	幂加1
M1	内层因子3
A1	原函数

(b) · 4 marks

$a=2/3$.

[(3a+2)^5−32]/15=992/15给(3a+2)^5=1024=4^5，得3a+2=4，所以a=2/3。

M1	正确定积分方程
M1	五次正根
M1	正域筛选
A1	a

Question 25

P3-C07-R25 · 5 marks

(a) · 2 marks

$-\sin(\ln x)/x$.

外余弦导数负正弦，内ln导数1/x。

M1	复合链式
A1	导式

(b) · 3 marks

1.

原函数−cosln⁡x，上端−cosπ/2=0，下端−cos0=−1。

M1	识别负F
M1	对数上限
A1	值

Question 26

P3-C07-R26 · 8 marks

(a) · 4 marks

Zero $x = 0$; antiderivative $e^{x} + e^{-x}$ has equal values at opposite endpoints.

原式0给e^{2x}=1；原函数两端都2+1/2，所以差0。

M1	零点
M1	正确原函数
M1	两端同值
A1	零积分

(b) · 4 marks

1.

左右面积相同，在0至ln2积分原式为(2+.5)−2=.5，再乘2。

M1	分段或对称
M1	正确半面积
M1	乘2
A1	总面积

Question 27

P3-C07-R27 · 6 marks

(a) · 3 marks

$y = \ln(5x + 1) + 2$.

分子正好是真数导数5，ln原函数；初值C2。

M1	对数识别
M1	常数条件
A1	曲线

(b) · 3 marks

$((e^{-2} - 1)/5, 0)$.

ln(5x+1)=−2给5x+1=e^{-2}，所得x大于−1/5合法。

M1	指数反解
M1	域检验
A1	坐标

Question 28

P3-C07-R28 · 6 marks

(a) · 3 marks

$F' = \sec^2 x / \tan x = 1/(\sin x \cos x)$.

ln链式导数，倒数定义化简，域内所有比正。

M1	链式正切导数
M1	三角比化简
A1	目标式

(b) · 3 marks

$\frac{1}{2}\ln 3$.

[ln tanx]=ln1−ln(1/√3)=半ln3。

M1	原函数识别
M1	精确三角代限
A1	值

Question 29

P3-C07-R29 · 6 marks

(a) · 2 marks

$F\prime = (2x-3)e^{2x}$.

乘积法则给$e^{2x} + 2(x-2)e^{2x} = (2x-3)e^{2x}$。

M1	乘积与链式法则
A1	正确整理

(b) Hence, using your answer to (a),

(i) · 2 marks

$2 - e^2$.

直接识别被积函数为$F\prime$，所以积分$F(1) - F(0) = -e^2 - (-2) = 2 - e^2$；不需要分部积分。

M1	由已证明导数识别原函数
A1	精确值

(ii) · 2 marks

$e^2 - 2$.

区间内$2x - 3 < 0$且$e^{2x} > 0$，整条曲线在轴下方，几何面积等于前问有向积分的相反数。

B1	判断曲线符号
A1	正面积

Question 30

P3-C07-R30 · 7 marks

(a) · 3 marks

2.

区间内两项均非负。原函数$\tan x + \sin 2x$，代入端点为$1 + 1 = 2$。

M1	两项标准积分
M1	正确上下限
A1	面积

(b) The line $y = k$ is drawn across the same vertical strip.

(i) · 2 marks

$k = 8/\pi$.

由(a)得$2 - k\pi/4 = 0$，所以$k = 8/\pi$。

M1	用已得积分形成方程
A1	k

(ii) · 2 marks

The areas above and below the line cancel in the signed integral.

零有向积分仅说明正负部分抵消，并不说明每一点的函数差为零；例如$y(0) = 3$而$k = 8/\pi \neq 3$。

B1	有向面积抵消
B1	用给定曲线举例

Pearson Edexcel International A Level · Pure Mathematics 3

Chapter 8 — Numerical methods
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P3 第8章 · 数值方法
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P3-C08-R01 · P3 · Chapter 8

Let $f(x) = x^3 + 2x - 7$.

(a) Show that $f(x) = 0$ has exactly one real root, and that it lies between 1 and 2. (4)

(b) The iteration is $x_{n+1} = \sqrt[3]{7 - 2x_n}$, starting with $x_1 = 1.5$.

(i) Find x_2, x_3, x_4 to 6 decimal places. (3)

(ii) Continue to obtain the root to 3 decimal places and verify the rounding using a sign-change interval. (3)

(Total for Question 1 is 10 marks)

For $x > 0$, $f(x) = x + \ln x - 3$.

(a) Show that it has exactly one root in $(2, 3)$. (4)

(b) Use $x_{n+1} = 3 - \ln x_n$ with $x_1 = 2$ to find x_2, x_3, x_4, and then the root to 3 decimal places. (4)

(Total for Question 2 is 8 marks)

The curves $y = e^{-x}$ and $y = x$ meet at $x = \alpha$.

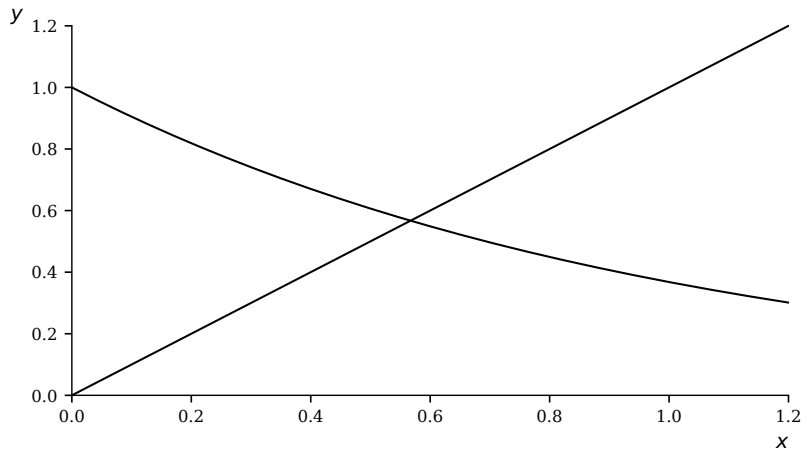


Figure 3

(a) Explain why there is exactly one intersection for real x and locate it in $(0, 1)$. (4)

(b) Use $x_{n+1} = e^{-x_n}$, $x_1 = 0.5$, to calculate x_2 to x_5 to 6 decimal places and give α to 4 decimal places. (4)

(Total for Question 3 is 8 marks)

$f(x) = x\cos x - 0.4$ on $0 \le x \le 0.6$.

(a) Show that a root lies in $(0.4, 0.5)$. (3)

(b) Derive a fixed-point formula $x_{n+1} = 0.4/\cos x_n$. Starting with $x_1 = 0.45$, find x_2, x_3, x_4 to 6 decimal places. (4)

(Total for Question 4 is 7 marks)

A decaying response is $R(t) = (t+1)e^{-t}$ for $t \geq 0$. A detector switches off when $R = 0.2$.

(a) Show that the switching equation can be written $t = \ln(5(t+1))$ and explain why it has one positive solution. (4)

(b) Use this iteration from $t_1 = 3$ to find t_2, t_3, t_4 to 6 decimal places and the switching time to 3 decimal places. (4)

(Total for Question 5 is 8 marks)

$f(x) = (x^2 - 2)^2$.

(a) Explain why $f(1)f(2) > 0$ does not prove there is no root in $(1, 2)$. Find the root exactly. (3)

(b) Use $x_{n+1} = \frac{1}{2}(x_n + 2/x_n)$ with $x_1 = 1$ to find x_2, x_3, x_4 to 6 decimal places. (3)

(Total for Question 6 is 6 marks)

$f(x) = 1/(x - 1)$.

(a) Calculate $f(0.9)$ and $f(1.1)$ and explain why their opposite signs do not establish a root between them. (3)

(b) Show algebraically that f has no real zero. (3)

(Total for Question 7 is 6 marks)

The equation $x^2 - x - 2 = 0$ has a positive root α.

(a) Find both exact roots and derive $x = \sqrt{x + 2}$ for the positive root. (3)

(b) Use initial value $x_1 = 1$.

(i) Find x_2, x_3, x_4 for $x_{n+1} = \sqrt{x_n + 2}$. (2)

(ii) Find x_2, x_3 for $x_{n+1} = x_n^2 - 2$ and explain why this does not approximate the requested root. (3)

(Total for Question 8 is 8 marks)

$f(x) = x^3 - 4x + 1$.

(a) Show that there is a root in each interval $(-3, -2)$, $(0, 1)$ and $(1, 2)$. (3)

(b) Explain why these account for every real root, and calculate each to 3 decimal places. (3)

(Total for Question 9 is 6 marks)

On $x \geq 0$, the equation $\sin x = \ln(x+1)$ has the root $x = 0$ and another root $\alpha \in (1.6, 1.8)$.

(a) Verify the stated bracket for the non-zero root. (3)

(b) Use $x_{n+1} = e^{\sin x_n} - 1$ from $x_1 = 1.7$ to obtain x_2, x_3, x_4 to 6 decimal places. Explain why starting at 0 would not find α. (4)

(Total for Question 10 is 7 marks)

$f(x) = x^3 + x - 5$. It is known to have a unique real root.

(a) Perform two bisections starting from $[1, 2]$, stating the retained interval after each. (4)

(b) Find the root to 3 decimal places and verify it using the half-unit rounding boundaries. (3)

(Total for Question 11 is 7 marks)

The area beneath $y = 1 + e^x$ from 0 to $a > 0$ is 5.

(a) Show that $a = \ln(6 - a)$ and state the restriction needed for this iteration. (3)

(b) From $a_1 = 1.5$, find a_2, a_3, a_4 to 6 decimal places and the area-splitting input to 3 decimal places. (4)

(Total for Question 12 is 7 marks)

$f(x) = x^3 - 2x - 4$. A proposed root is 1.999.

(a) Calculate the residual $f(1.999)$ and explain why a small residual does not mean that the decimal is an exact root. (3)

(b) Find the exact root in $[1, 3]$ and show that it is unique on this interval. (3)

(Total for Question 13 is 6 marks)

The iteration $x_{n+1} = \ln(4 - x_n)$ is used to solve $e^x + x - 4 = 0$.

(a) State the real-input restriction and explain why $x_1 = 5$ is invalid. (3)

(b) Using $x_1 = 1$, find x_2, x_3, x_4 to 6 decimal places and the root to 3 decimal places. (4)

(Total for Question 14 is 7 marks)

$f(x) = x^3 - 2x - 2$.

(a) Show that $f(x) = 0$ can be written $x = \sqrt[3]{2x + 2}$, and verify a root in $(1, 2)$. (3)

(b) Use the derived iteration with $x_1 = 1.5$ to find x_2, x_3, x_4 and then the root to 4 decimal places. (4)

(Total for Question 15 is 7 marks)

16. P3-C08-R16 · P3 · Chapter 8

An iteration is $x_{n+1} = (a + \ln x_n)/2$. Its fixed point is known to be 2.

(a) Find a exactly and the equation solved by this iteration. (3)

(b) For this a, start at $x_1 = 1.8$ and find x_2, x_3, x_4 to 6 decimal places. (3)

(Total for Question 16 is 6 marks)

$f(x) = (x^3 - 7)^3$.

(a) Find its exact real root and determine whether f changes sign there. (3)

(b) Calculate $f(1.9125)$ and $f(1.9135)$, and use them to give the root correct to 3 decimal places. (3)

(Total for Question 17 is 6 marks)

18.

A cost model is $C(q) = q + \ln q$ for positive integer output q. A project requires $C(q) \geq 7$.

(a) Show that the equation $C(x) = 7$ has one real solution for $x > 0$. (3)

(b) Approximate the real threshold to 3 decimal places and find the least permitted integer output. (4)

(Total for Question 18 is 7 marks)

For $0 < x < \pi/2$, $\cot x = x$.

(a) Derive the iteration $x_{n+1} = \arctan(1/x_n)$ and verify a root in $(0.8, 0.9)$. (4)

(b) From $x_1 = 0.85$, find x_2, x_3, x_4 and then the root to 4 decimal places. (4)

(Total for Question 19 is 8 marks)

$f(x) = 3x\cos x - 1$ on $(0, \pi/2)$.

(a) Show that there is one root in $(0.3, 0.4)$ and another in $(1.2, 1.4)$. (4)

(b) To approximate the larger root use $x_{n+1} = \arccos(1/(3x_n))$.

(i) state why $x_1 = 0.2$ is invalid, (2)

(ii) from $x_1 = 1.3$, find x_2, x_3, x_4 to 6 decimal places. (3)

(Total for Question 20 is 9 marks)

On $x \geq 0$, $\ln(2x+1) = x^2$ has the root 0 and a positive root α.

(a) Verify that $\alpha \in (1, 1.2)$. (3)

(b) Use $x_{n+1} = \sqrt{\ln(2x_n + 1)}$, $x_1 = 1.1$, to find x_2, x_3, x_4 and α to 3 decimal places. (4)

(Total for Question 21 is 7 marks)

The iteration $x_{n+1} = \frac{1}{2}\cos x_n$ starts at $x_1 = 0.5$.

(a) Find x_2 to x_6, using 6 decimal places. (3)

(b) Let $f(x) = x - \frac{1}{2}\cos x$. Verify a sign change across the last two computed values and obtain the fixed point to 4 decimal places. (4)

(Total for Question 22 is 7 marks)

$f(x) = \ln x - (x-1)/2$, with $x > 0$. A calculator gives a proposed root $x = 3.5129$.

(a) Calculate $f(3.5129)$ to at least 6 significant figures and state whether this decimal is an exact root. (3)

(b) Find the other exact root and verify that a second root lies in $(3.5, 3.6)$. (4)

(Total for Question 23 is 7 marks)

A time t, measured in minutes, satisfies $t = \ln(t + 3)$. An iteration starts from $t_1 = 1.5$.

(a) Find t_2, t_3, t_4 to 6 decimal places. (3)

(b) If $s = 60t$ is measured in seconds, derive an equivalent iteration for s and state the corresponding initial value. (3)

(Total for Question 24 is 6 marks)

The model equation is $x^3 + x = p$ for $x > 0$. The measured parameter satisfies $4.99 \le p \le 5.01$.

(a) Show that the positive solution increases as p increases, using the monotonicity of $x^3 + x$. (3)

(b) Find the smallest and largest possible solutions to 4 decimal places and determine whether the model fixes x to 3 decimal places. (4)

(Total for Question 25 is 7 marks)

The equation is $x = 1 + \sqrt{3 - x}$.

(a) Find the exact real solution, checking roots introduced by squaring. (4)

(b) Use $x_{n+1} = 1 + \sqrt{3 - x_n}$ with $x_1 = 1.5$ to find x_2, x_3, x_4 and their absolute errors relative to the exact root. (3)

(Total for Question 26 is 7 marks)

For $0 < x < \pi/2$, $\cos x = x/2$.

(a) Derive both fixed-point forms $x = 2\cos x$ and $x = \arccos(x/2)$. (3)

(b) Use $x_1 = 1$ in each iteration.

(i) Calculate x_2, x_3, x_4 for $x_{n+1} = 2\cos x_n$. (3)

(ii) Calculate x_2, x_3, x_4 for $x_{n+1} = \arccos(x_n/2)$ and find the root to 4 decimal places. (4)

(Total for Question 27 is 10 marks)

$f(x) = \ln(x - 1) + x$ for $x > 1$.

(a) Find a sign-change interval of width 0.1 containing a root, and prove this root is unique. (4)

(b) Use $x_{n+1} = 1 + e^{-x_n}$ from $x_1 = 1.25$ to obtain x_2, x_3, x_4 and the root to 4 decimal places. (4)

(Total for Question 28 is 8 marks)

The function $f(x) = x + \ln x - 2$ is defined for $x > 0$.

(a) Show that $f(x) = 0$ has exactly one root in $1 < x < 2$. (3)

(b) The iteration $x_{n+1} = 2 - \ln x_n$ is used with $x_0 = 1.5$.

(i) Calculate x_1, x_2, x_3, giving each to 5 decimal places. (2)

(ii) Show by a sign change that the root rounded to 2 decimal places is 1.56. (3)

(Total for Question 29 is 8 marks)

The equation $\cos x = x$ has a root α in $0 < x < 1$. Angles are in radians.

(a) Show that this root is unique. (3)

(b) Two rearrangements are proposed: $x_{n+1} = \cos x_n$ and $x_{n+1} = \arccos x_n$. Both start at $x_0 = 0.7$.

(i) Calculate the next two iterates for each rearrangement to 4 decimal places. (2)

(ii) For the inverse-cosine iteration, the next iterate is approximately 0.8617 and the following one is approximately 0.5321. State why these four iterates do not justify reporting α to 3 decimal places. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P3-C08-R01 · 10 marks

(a) · 4 marks

$f(1)=-4$, $f(2)=5$; $f'(x)=3x^2+2>0$.

多项式连续、端值异号给存在；导数恒正给至多一根。

M1	两异号值
M1	连续性
M1	导数恒正
A1	唯一根

(b) The iteration is $x_{n+1}=\sqrt[3]{7-2x_n}$, starting with $x_1=1.5$.

(i) · 3 marks

$1.587401, 1.563932, 1.570303$

逐次代入，不将中间值提前截为显示精度。

M1	正确首迭代
M1	顺序代入
A1	三值

(ii) · 3 marks

1.569; $f(1.5685)=-0.0041884559<0$, $f(1.5695)=0.0051968274>0$.

多次迭代近似根$1.568946\cdots$，用两个舍入边界代f所得异号，并结合(a)唯一性。

M1	迭代求近似
M1	舍入边界异号
A1	三位小数

Question 2

P3-C08-R02 · 8 marks

(a) · 4 marks

$f(2) = \ln 2 - 1 < 0$, $f(3) = \ln 3 > 0$; $f' = 1 + 1/x > 0$.

域内连续、异号且严格递增。

M1	异号端值
M1	连续性
M1	正导数
A1	唯一性

(b) · 4 marks

$2.306853, 2.164116, 2.227988$; root 2.208.

所有实际迭代输入保持正，逐项用未舍入值；继续到要求精度。

M1	正确公式代入
M1	三值
M1	持续迭代
A1	根

Question 3

P3-C08-R03 · 8 marks

(a) · 4 marks

$e^{-x} - x$ is strictly decreasing, with values 1 and $e^{-1} - 1$ at $0, 1$.

导数-e^{-x}-1恒负，多项函数连续且0、1异号。

M1	差函数
M1	负导数
M1	异号
A1	唯一性

(b) · 4 marks

$0.606531, 0.545239, 0.579703, 0.560065$; $\alpha = 0.5671$.

先列四次迭代，给四位小数根时需继续计算，不能仅用前四次值。

M1	正确迭代
M1	四值
M1	继续计算
A1	根

Question 4

P3-C08-R04 · 7 marks

(a) · 3 marks

$f(0.4) = -0.031575602,\ f(0.5) = 0.038791281$

两值异号，函数在该闭区间连续，故存在根。

M1	两端计算
M1	连续性
A1	存在根

(b) · 4 marks

$0.444224, 0.442995, 0.442737$

原式移项除cosx，在该区间cos正不为0；按式计算。

M1	合法等价重排
M1	正确首步
M1	后续计算
A1	三值

Question 5

P3-C08-R05 · 8 marks

(a) · 4 marks

$e^{t}=5(t+1)$; $R'=-te^{-t}<0$ for $t>0$, with $R(0)=1$ and $R(4)=5e^{-4}<0.2$.

正量取ln合法；响应在正域严格下降，连续端值跨过门槛，所以唯一。

M1	等价对数重排
M1	单调导数
M1	阈值跨越
A1	唯一性

(b) · 4 marks

$2.995732, 2.994665, 2.994398$; $t=2.994$.

逐项迭代，继续至指定精度，时间域为非负。

M1	正确迭代
M1	三值
M1	继续
A1	时间

Question 6

P3-C08-R06 · 6 marks

(a) · 3 marks

$f(1)=1, f(2)=4$, but $f(\sqrt{2})=0$; the curve touches the axis without a sign change.

变号是存在根的充分检验，不变号不构成无根证明；本题偶重根只接触。

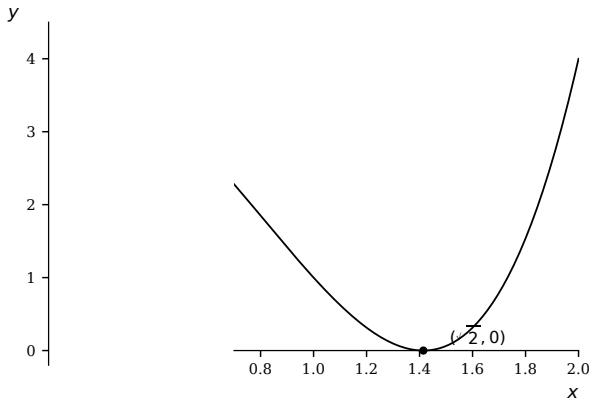
M1	端值与精确根
M1	接触不变号
A1	逻辑说明

(b) · 3 marks

$1.500000, 1.416667, 1.414216$

给定的是固定点迭代；每一步输入均非零。

M1	正确首项
M1	递推值
A1	三值



Suggested sketch

Question 7

P3-C08-R07 · 6 marks

(a) · 3 marks

-10 and 10; f is not continuous on the interval because it is undefined at 1.

变号定理的连续前提失效，不能把渐近线穿越当作过轴。

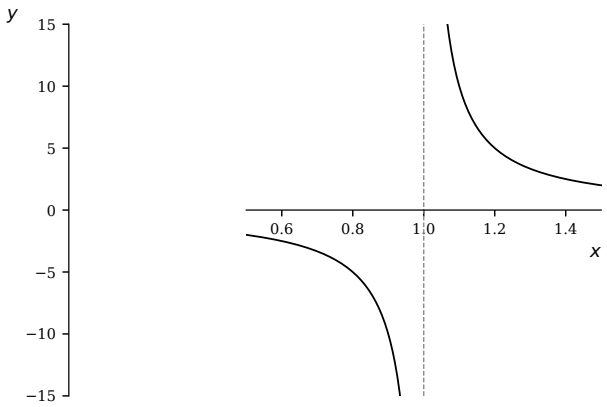
M1	两端值
M1	指出x1未定义
A1	连续前提说明

(b) · 3 marks

A non-zero numerator 1 divided by a finite non-zero denominator cannot be 0.

所有合法输入分子恒1，分式不可能为0。

M1	非零分子
M1	原域
A1	无根结论



Suggested sketch

Question 8

P3-C08-R08 · 8 marks

(a) · 3 marks

Roots $2, -1$; positive branch gives $x = \sqrt{x+2}$.

原方程(x−2)(x+1)=0；平方根重排只保留非负分支。

M1	两根
M1	合法正分支
A1	迭代形式

(b) Use initial value $x_1 = 1$.

(i) · 2 marks

$1.732051, 1.931852, 1.982890$

逐步取正平方根。

M1	迭代
A1	三值

(ii) · 3 marks

$x_2 = -1, x_3 = -1$; it reaches the negative root instead.

等价固定点方程可有多个根，不同重排和起点可落入不同根；本题需求明确是正根。

M1	两步计算
M1	对照根符号
A1	失败原因

Question 9

P3-C08-R09 · 6 marks

(a) · 3 marks

Values at $-3, -2, 0, 1, 2$ are $-14, 1, 1, -2, 1$, respectively.

三个不相交区间均连续且端值异号，分别至少一根。

M1	五个端值
M1	三次连续变号判断
A1	三根存在

(b) · 3 marks

Roots $-2.115,\ 0.254,\ 1.861$; a cubic has at most three distinct roots.

三个不同根已由互不相交区间保证，三次多项式不可能有第四个不同实根；用区间缩小或适当迭代计算。

M1	次数上界理由
M1	对三段数值求根
A1	三个近似

Question 10

P3-C08-R10 · 7 marks

(a) · 3 marks

$f(1.6) = 0.044062158$, $f(1.8) = -0.055771786$

函数在1.6至1.8连续，两端差值异号。

M1	两差值
M1	连续性
A1	区间

(b) · 4 marks

$1.695719, 1.697182, 1.696687$; starting at 0 stays at 0.

对数方程指数还原得递推；0是该递推的另一固定点，起于0将永远不动。

M1	正确迭代
M1	三值
M1	检查零起点
A1	说明

Question 11

P3-C08-R11 · 7 marks

(a) · 4 marks

$[1.5, 2]$, then $[1.5, 1.75]$.

f1.5=−.125、f1.75=2.109375；每次取异号的半区间。

M1	第一中点值
M1	第一保留区间
M1	第二中点及区间
A1	两区间

(b) · 3 marks

Root 1.516; check 1.5155 and 1.5165.

缩小到同一舍入格内，并检查两格边界端值异号；唯一性已给。

M1	数值求根
M1	正确舍入格
A1	根

Question 12

P3-C08-R12 · 7 marks

(a) · 3 marks

$a + e^a - 1 = 5 \Rightarrow e^a = 6 - a$; need $a < 6$.

先定积分得到面积方程，指数为正，取ln要求6−a正。

M1	正确积分
M1	等价重排
A1	对数限制

(b) · 4 marks

$1.504077, 1.503171, 1.503373$; $a = 1.503$.

逐步用未舍入值，解的语义是面积上界而非函数高度。

M1	正确公式
M1	三值
M1	继续计算
A1	a

Question 13

P3-C08-R13 · 6 marks

(a) · 3 marks

$f(1.999) = -0.009994001$, $f(2) = 0$

确根要求f恰为0；1.999代入非零，只是近似；本题f2恰零。

M1	正确残差
M1	非零与确根区别
A1	说明

(b) · 3 marks

$x = 2$; $f' = 3x^2 - 2 > 0$ on $[1, 3]$.

验证f2零，区间导数正给最多一根。

M1	验证根
M1	正导数
A1	唯一性

Question 14

P3-C08-R14 · 7 marks

(a) · 3 marks

$x_n < 4$ is required; at 5 the logarithm has negative argument.

限制针对每一步送入递推的当前值，不能只检查最终根。

M1	真数正条件
M1	起值代入
A1	原因

(b) · 4 marks

$1.098612, 1.065189, 1.076643$; root 1.074.

该起点的后续输入都小于4，逐次计算并继续。

M1	正确首步
M1	后续迭代
M1	根求值
A1	结果

Question 15

P3-C08-R15 · 7 marks

(a) · 3 marks

$x^3 = 2x + 2$; $f(1) = -3$, $f(2) = 2$.

实立方根重排不增加或丢失实根，端值异号且连续。

M1	等价重排
M1	两端异号
A1	存在根

(b) · 4 marks

$1.709976, 1.756569, 1.766578$; root 1.7693.

逐步迭代，四位小数根须继续而不是把x4直接当根。

M1	正确首步
M1	三值
M1	继续
A1	四位根

Question 16

P3-C08-R16 · 6 marks

(a) · 3 marks

$a = 4 - \ln 2$; $2x - \ln x - a = 0$ for $x > 0$.

固定点代2= (a+ln2)/2，反解参数；移项给原方程。

M1	固定点条件
M1	参数
A1	方程和域

(b) · 3 marks

$1.947320, 1.986653, 1.996652$

参数保持精确4−ln2，不将它提前舍入。

M1	正确精确参数代入
M1	顺序迭代
A1	三值

Question 17

P3-C08-R17 · 6 marks

(a) · 3 marks

$x = \sqrt[3]{7}$; yes, because both the inner cubic and its third power preserve sign.

三次外幂只在内部零点为0且保持正负号，所以穿越而非接触。

M1	精确零点
M1	奇次幂符号
A1	变号结论

(b) · 3 marks

$f(1.9125) = -1.0598645e-07$, $f(1.9135) = 2.4370558e-07$; root 1.913.

两个数值虽很小但仍异号；这一区间是舍入到1.913的格，非仅按残差大小决定精度。

M1	边界准确代值
M1	异号与唯一根
A1	舍入值

Question 18

P3-C08-R18 · 7 marks

(a) · 3 marks

$C' = 1 + 1/x > 0$; $C(5) < 7 < C(6)$.

正域连续严格递增，5和6跨门槛，故唯一临界值。

M1	正导数
M1	端值跨越
A1	唯一性

(b) · 4 marks

Threshold 5.327; least $q = 6$.

可用x=7−lnx迭代，也可持续缩小区间；最终输出要满足整数模型而不是直接使用实根。

M1	正确数值方法
M1	根精度
M1	整数筛选
A1	q

Question 19

P3-C08-R19 · 8 marks

(a) · 4 marks

$f(0.8) = 0.1712146,\ f(0.9) = -0.10644885$

正域内取倒比tanx=1/x，主值反正切落在原锐域，重排合法；cot差连续异号。

M1	合法反三角重排
M1	两端值
M1	连续域
A1	区间

(b) · 4 marks

$0.866302, 0.856914, 0.862302$; root 0.8603.

按弧度计算，继续至要求精度。

M1	弧度迭代
M1	三值
M1	继续
A1	根

Question 20

P3-C08-R20 · 9 marks

(a) · 4 marks

$f(0.3) = -0.14019716$, $f(0.4) = 0.10527319$; $f(1.2) = 0.30448792$, $f(1.4) = -0.286138$

两个不相交且连续的区间分别端值异号，至少各一根。

M1	第一对异号
M1	第二对异号
M1	连续性
A1	两根存在

(b) To approximate the larger root use $x_{n+1} = \arccos(1/(3x_n))$.

(i) · 2 marks

$1/(3x_1) = 5/3 > 1$, outside the domain of arccosine.

反余弦真数必须在$[-1,1]$，该起点超界。

M1	代起点检查真数
A1	域理由

(ii) · 3 marks

$1.311490, 1.313813, 1.314278$

每步用主值$\arccos$，迭代输入均为合法正数。

M1	正确首步
M1	顺序递推
A1	三值

Question 21

P3-C08-R21 · 7 marks

(a) · 3 marks

$f(1)=0.098612289,\ f(1.2)=-0.21622457$

差函数连续，两个值异号。

M1	两端值
M1	连续性
A1	存在

(b) · 4 marks

$1.078495, 1.072203, 1.070339;\ \alpha=1.070.$

选择非负平方根与正根任务一致，起点不取另一个固定点0。

M1	正确正分支
M1	三值
M1	继续计算
A1	正根

Question 22

P3-C08-R22 · 7 marks

(a) · 3 marks

$0.438791, 0.452633, 0.449649, 0.450300, 0.450158$

所有角用弧度，显示舍入但内部保留更多位。

M1	正确首步
M1	五次递推
A1	五值

(b) · 4 marks

$f(x_5) = 0.00014144376 > 0$, $f(x_6) = -3.0776166e-05 < 0$; root 0.4502.

不能只因相邻迭代接近就说两者必为界；本题实际代入差函数核查符号后才构成变号区间。

M1	用未舍入相邻值检查
M1	连续变号
M1	继续至精度
A1	根

Question 23

P3-C08-R23 · 7 marks

(a) · 3 marks

$f(3.5129) = -8.0928203e-06$, not zero; it is not an exact root.

直接代值检验，微小但非零的残差不能称确解。

M1	准确计算残差
M1	与零比较
A1	结论

(b) · 4 marks

$x = 1$; $f(3.5) = 0.0027629685$, $f(3.6) = -0.019066155$

原式代1为0；第二区间函数连续且端值异号，两个根不能混同。

B1	精确根1
M1	区间两端值
M1	连续性
A1	第二根存在

Question 24

P3-C08-R24 · 6 marks

(a) · 3 marks

$1.504077, 1.504983, 1.505184$

以分钟作为递推变量，按给定公式计算。

M1	首步
M1	连续迭代
A1	三值

(b) · 3 marks

$s_{n+1} = 60\ln(s_n/60 + 3)$; $s_1 = 90$.

将t_n=s_n/60代入分钟公式，再两边乘60；不能只把ln外的单位改名。

M1	正确内部单位换算
M1	外部乘60
A1	公式和起值

Question 25

P3-C08-R25 · 7 marks

(a) · 3 marks

Derivative $3x^2 + 1 > 0$, so larger output p requires larger input x.

正域函数严格递增，一一对应，比较不同p对应x的顺序，无须引入隐函数求导。

M1	正导数
M1	一一单调对应
A1	顺序结论

(b) · 4 marks

Range approximately $[1.5147, 1.5172]$; no unique 3-decimal value.

分别数值求两个参数端值的根；它们舍入到三位小数不同，故测量不确定性已超过要求，更多迭代不能消除参数误差。

M1	两端参数求根
M1	根精度
M1	三位舍入比较
A1	结论

Question 26

P3-C08-R26 · 7 marks

(a) · 4 marks

$x = 2$.

原式要求$1 \leqslant x \leqslant 3$；平方$(x-1)^2 = 3-x$得$(x-2)(x+1)=0$，$-1$不满足原式。

M1	原限制
M1	平方求两根
M1	排增根
A1	唯一根

(b) · 3 marks

$2.224745, 1.880486, 2.058071$; absolute errors $0.224745, 0.119514, 0.058071$.

分别将三个未舍入迭代值减2取绝对值，最终显示六位小数。

M1	正确三迭代
M1	绝对误差定义
A1	三误差

Question 27

P3-C08-R27 · 10 marks

(a) · 3 marks

Both follow algebraically; the second uses the principal arccosine range containing the required root.

第一式乘2；第二式取主值反余弦，必须说明原根在其主值域。

M1	第一重排
M1	第二反三角重排
A1	分支说明

(b) Use $x_1 = 1$ in each iteration.

(i) · 3 marks

$1.080605, 0.941590, 1.177006$

分别递推，不因两式同解就假定迭代速度或行为相同。

M1	首步
M1	递推
A1	三值

(ii) · 4 marks

$1.047198, 1.019727, 1.035770$; root 1.0299.

该序列继续计算可逼近所需根，使用弧度并保留内部精度。

M1	三值
M1	继续迭代
M1	原式核查
A1	根

Question 28

P3-C08-R28 · 8 marks

(a) · 4 marks

$[1.2, 1.3]$; $f' = 1/(x-1) + 1 > 0$.

f1.2=ln.2+1.2<0，f1.3=ln.3+1.3>0，正域连续严格递增。

M1	两端异号
M1	连续性
M1	正导数
A1	唯一性

(b) · 4 marks

$1.286505, 1.276235, 1.279086$; root 1.2785.

对数式指数还原得到迭代，后继值自动大于1，保持原定义域。

M1	合法重排
M1	三值
M1	继续及域
A1	根

Question 29

P3-C08-R29 · 8 marks

(a) · 3 marks

Exactly one root.

$f(1) = -1 < 0$，$f(2) = \ln 2 > 0$，连续性给出一根；$f\prime(x) = 1 + 1/x > 0$说明严格递增，至多一根。

M1	两端异号及连续
M1	严格递增
A1	唯一性

(b) The iteration $x_{n+1} = 2 - \ln x_n$ is used with $x_0 = 1.5$.

(i) · 2 marks

$1.59453, 1.53342, 1.57250.$

逐次把未舍入前值代入$2 - \ln x_n$，只在报告数值时保留五位小数。

M1	正确迭代公式
A1	前三项

(ii) · 3 marks

$f(1.555) < 0 < f(1.565)$; root rounds to 1.56.

计算$f(1.555) \approx -0.00352445$，$f(1.565) \approx 0.01288582$；唯一根位于舍入边界之间，所以保留两位小数为$1.56$。仅凭前几项迭代不能替代该精度检验。

M1	用舍入区间端点检验
A1	正确异号
B1	两位小数结论

Question 30

P3-C08-R30 · 7 marks

(a) · 3 marks

Unique root in $(0,1)$.

令$f(x)=\cos x-x$，$f(0)=1>0$、$f(1)=\cos 1-1<0$；且$f\prime=-\sin x-1<0$，所以连续且严格递减，恰有一根。

M1	异号存在性
M1	导数符号
A1	唯一结论

(b) Two rearrangements are proposed: $x_{n+1}=\cos x_n$ and $x_{n+1}=\arccos x_n$. Both start at $x_0=0.7$.

(i) · 2 marks

Cosine: $0.7648, 0.7215$; inverse cosine: $0.7954, 0.6511$.

两种迭代都用弧度，分别连续代入两次。使用未舍入值计算第二步。

M1	正确执行两个不同迭代
A1	四个数值

(ii) · 2 marks

The iterates are not stable to three decimal places.

已列项仍有显著振荡，且差距远大于0.001，不能作为三位小数精度证据。题目未要求引用收敛判别定理。

B1	振荡或未稳定
B1	不足以证明三位精度